

NuKAT Pre-Preliminary Design Review (pPDR)

December 12th, 2025

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pPDR Panelist Introduction

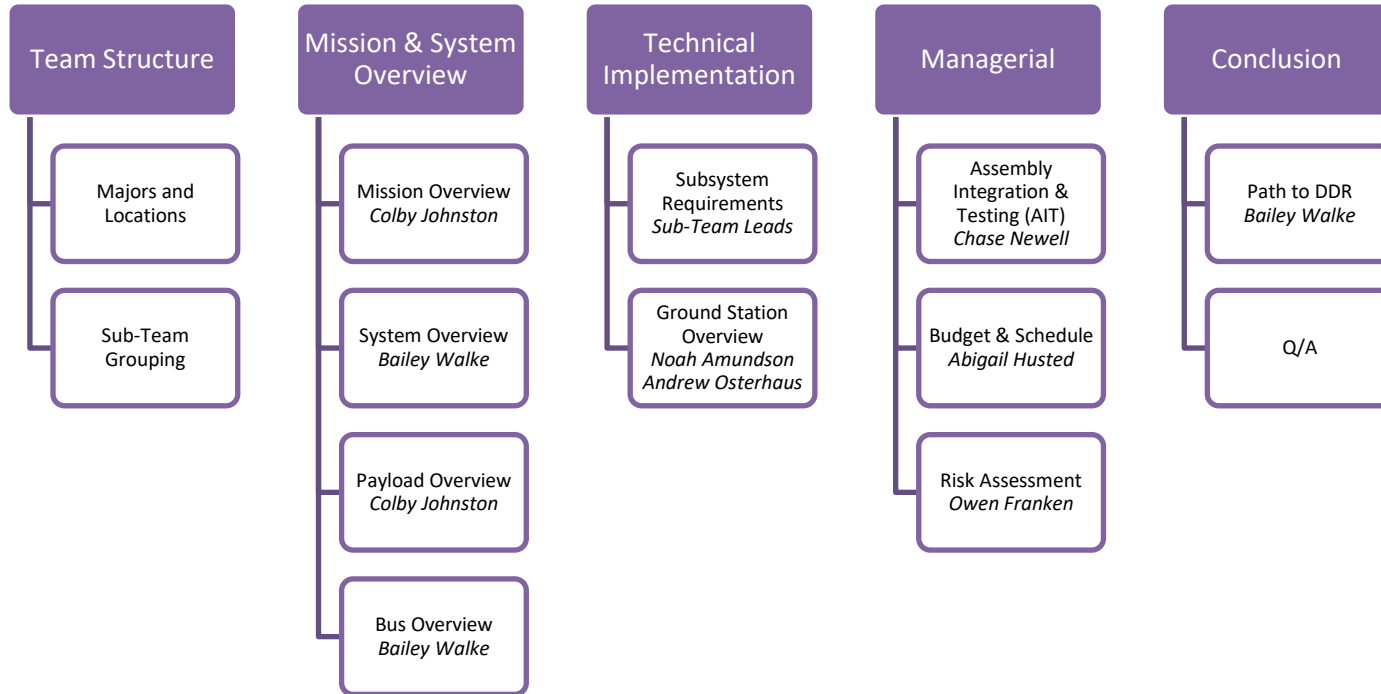
- Kyle Tidball
 - Engineer Specialist, Electronics Development Group, Textron Aviation
- Zach Metzinger
 - Engineer, METECS, Lunar Gateway Mission
 - Member, AMSAT Board of Directors
 - Volunteer, AMSAT GOLF Program
- Dr. Steven Bellinger
 - CEO, Radiation Detection Technologies (RDT)
 - Associate research faculty, K-State's Semiconductor Materials and Radiological Technologies (SMART) Laboratory
- Dr. Taylor Ochs
 - Nuclear Engineer, RDT
 - Award-winning research on the Dual-Sided MSND

pPDR Panelist Introduction

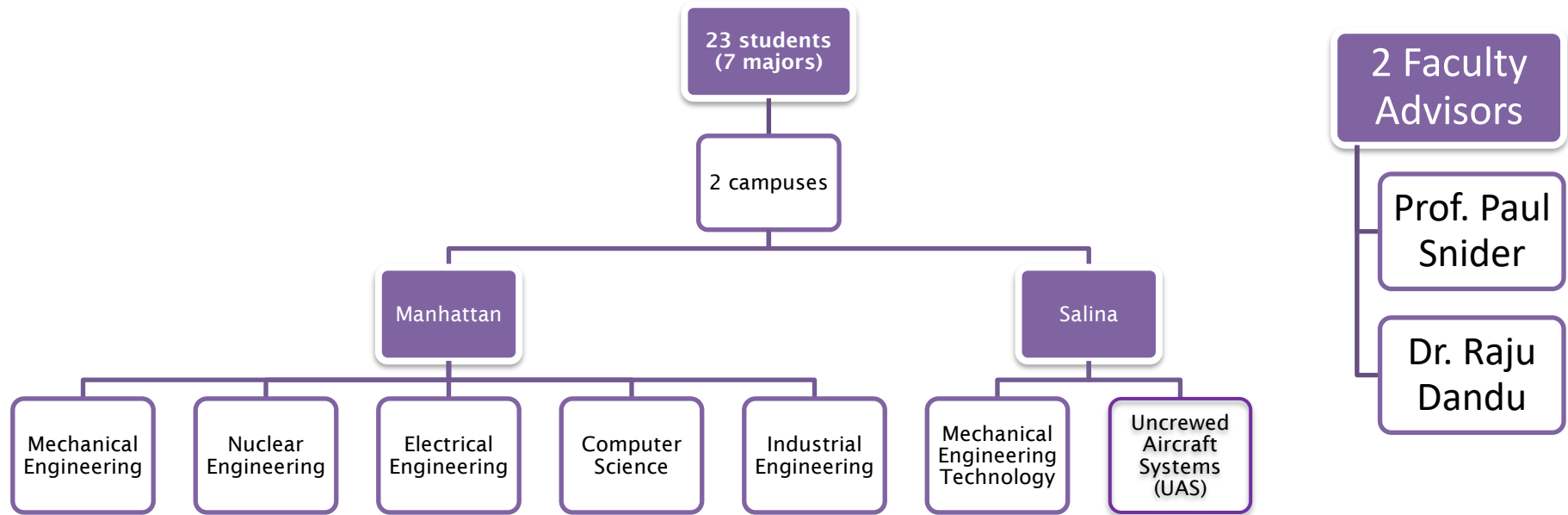
- Jeffrey Sheldon
 - Director, K-State's Sunderland Foundation Innovation Lab
 - FCC Trustee, K-State's Amateur Radio Club (W0QQQ)
- Ashleigh Mauler
 - M.S. student, K-State Nuclear Engineering
 - Research focus: Health Physics
- Danny Eckerberg
 - PhD student, K-State Nuclear Engineering
 - Research focus: Radiation Epidemiology



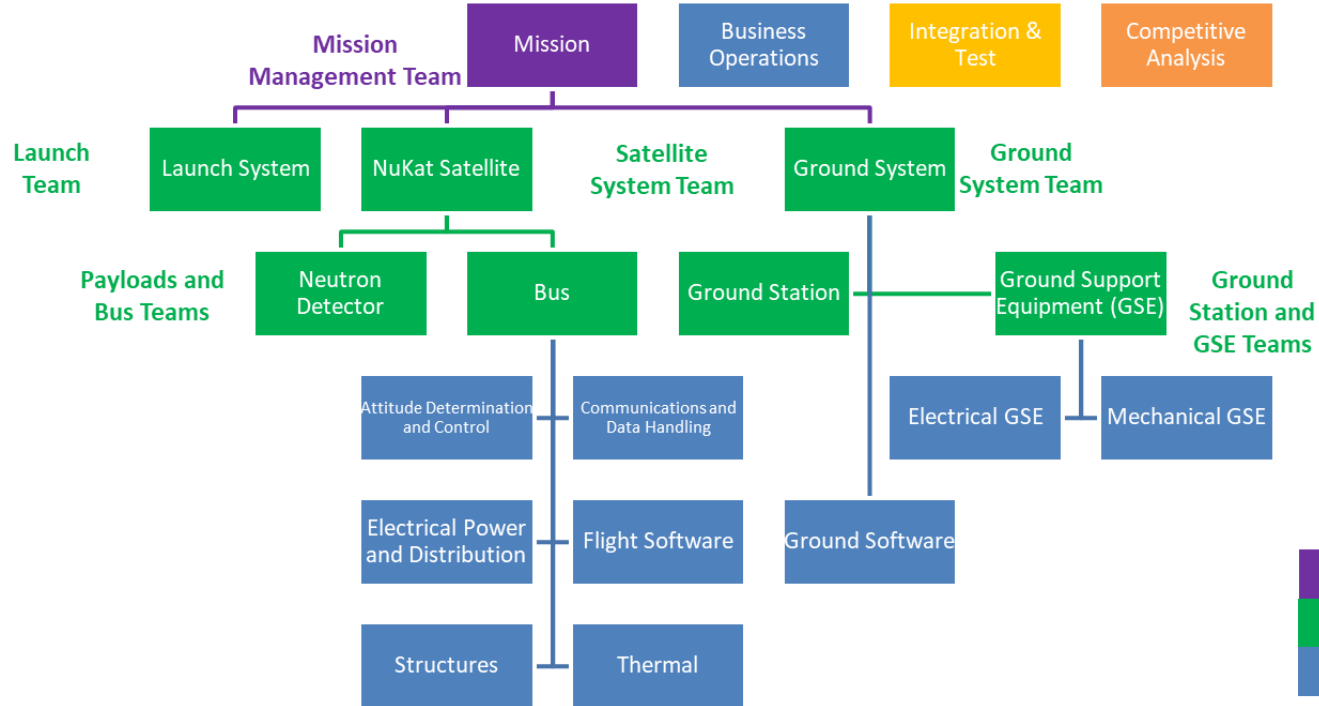
Agenda



Team Acknowledgement



Team Structure



Purpose of pPDR

Objective

Provide increased assurance that the proposed design approach and the manufacturing and test implementation plans will result in an acceptable product with minimal project risk.

Outcomes

- Confirmation of technical feasibility within budget and schedule
- Confirmation that design is on track to meet all requirements and to verify allocations
- Assessment of risks and mitigations
- Maturing an approved technical baseline

Review Focus Areas

- Mission objectives and rationale
- System/subsystem design and interfaces
- Orbit and environment
- Schedule, budget and risk management
- Readiness for preliminary design review



MISSION OVERVIEW

Colby Johnston – NuKAT Principal Investigator



Mission Background

Common Neutron Space Detectors	Downsides
$^3\text{He}/\text{BF}_3$ Proportional Gas	Pressure Vessel
Scintillators	Large Crystals/Photomultiplier tubes
Semiconductors (e.g., Si(Li))	Limited efficiency from planar geometry

- Microstructured Semiconductor Neutron Detectors (MSNDs)
- Original Domino heritage
 - NeRDI-1A and -1B missions
- New DSC Domino

Neutron
Detection
Metrics

- Flux
- Energy Spectrum
- Dose or Dose Rate

Mission Objectives

Mission Statement

Advancing space-based neutron detection via microstructured semiconductor neutron detectors (MSNDs) in low Earth orbit and validating the DSC Domino as a space-qualified detector platform.

1. Design, build and launch a CubeSat
2. DSC Domino space qualification
3. K-State CubeSat program pathfinder
4. STEM education, engagement, and outreach
5. Provide neutron counts, energy bins, and orbital location

Mission Requirements

Req. ID	Title	Requirement	Required Value	Expected Value	Status
NuKAT LEVEL 1 REQUIREMENTS: Quantitative Mission-Level					
NU-1.01	Neutron Detector Space Qualification	The neutron detector shall measure the thermal and fast neutron counts for comparison to accepted orbital readings and longevity qualification.	YES	YES	N/A
NU-1.02	Satellite Technology Demonstration	NuKAT shall demonstrate the effective use of the DSC Domino in space and build flight heritage for the detector	YES	YES	N/A
NU-1.03	Operational Integration and Success	NuKAT shall have a fully operational mission lifetime with the payload supported and integrated using a satellite bus system and ground station.	YES	YES	Not Compliant
NU-1.04	Operational Optimization	NuKAT shall optimize operational success and efficiency in terms of volume, power usage, data accuracy, and data processing in ensuring mission needs are met efficiently	YES	YES	Partial Compliance
NU-1.05	Low Cost Spacecraft	NuKAT component choices shall be evaluated for cost in addition to their needed/wanted technical capabilities	YES	YES	Compliant
NU-1.06	Launch Survivability	All components and systems of NuKAT shall be capable of surviving the conditions present in travel from K-State to the launch provider, while waiting integration, during integration with the launch vehicle, and during launch up until release into its final orbit.	YES	YES	Partial Compliance
NU-1.07	Space Environment Survivability	All components and systems of NuKAT shall be capable of surviving the space environment at the selected orbit and altitude.	YES	YES	Compliant
NU-1.08	Satellite Compliance	NuKAT shall be compliant with applicable guidelines from necessary governing bodies.	YES	YES	Compliant
NU-1.09	Satellite Guidelines	NuKAT should be designed to meet the requirements specified in Caly Poly CubeSat Design Specifications (CP-CDS-R14.1) and University NanoSat Program Satellite Development Guidelines (EXTDOC-315).	YES	YES	Partial Compliance
NU-1.10	Contamination and Hazardous Materials	NuKAT systems shall NOT use cadmium parts, cadmium-plated parts, zinc plating, mercury, and/or compounds containing mercury, pure tin or tin electroplate, and silicone rubber or room-temperature-vulcanizing (RTV) silicones with probability of transfer to Co-payload(s) or Launch Vehicle hardware.	YES	YES	Compliant

Minimum mission success:

- **Detector flight heritage** -- NASA TRL level 7
 - Averaged background-corrected channels (only 4 total data sets)
 - Reduced measurement frequency (60-second sample period)
- **Ground orbital-location correlation**
- **Reduced operational lifetime (3 months)**

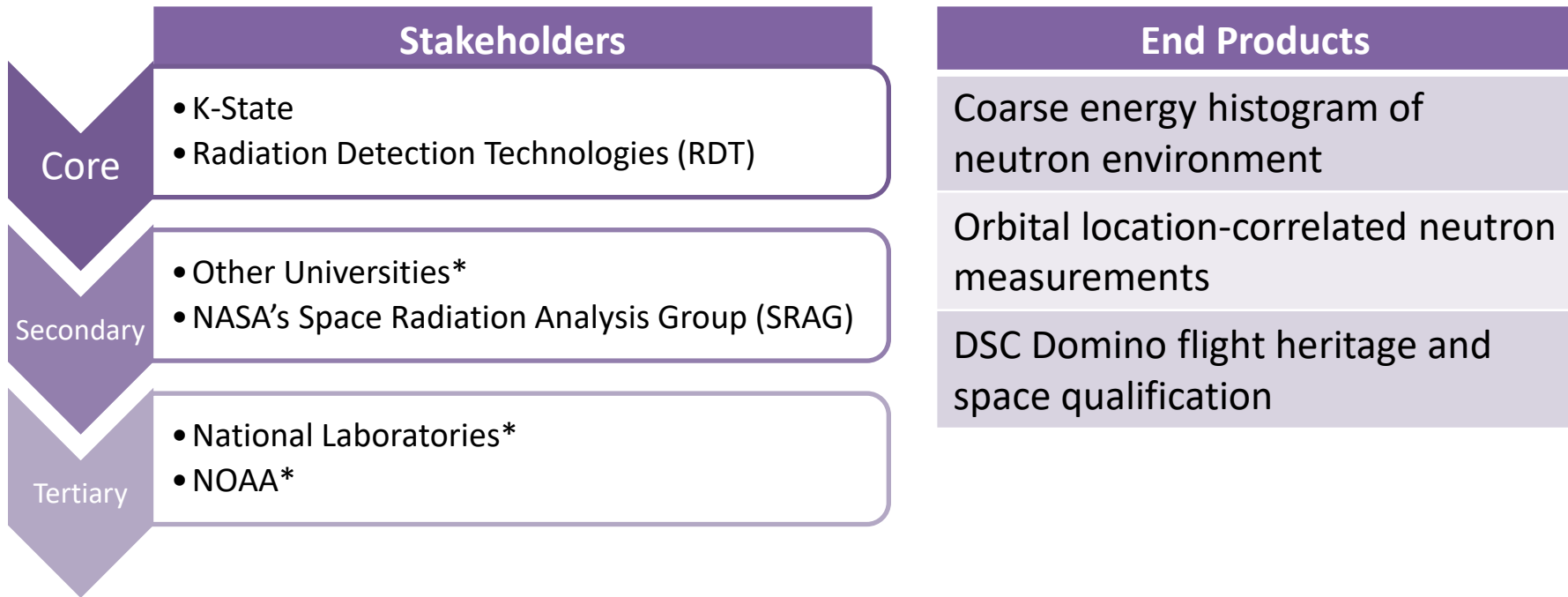
Complete mission success:

- **Detector space qualification** -- NASA TRL level 8
 - Background-corrected counts for each board (8 data sets)
 - Periodic full raw data downlink (16 data sets; each MSND)
 - Measurement frequency (30-second sample period)
- **Orbital-location from satellite systems**
- **Full operational lifetime (1 year)**

Mission Success

NuKAT (Neutron detector: K-State Analysis Tool) will achieve minimum mission success by building DSC Domino **flight heritage** and producing orbital location-correlated measurements over **three months**. NuKAT will achieve complete mission success from DSC Domino **space qualification** and satellite produced orbital location-correlated measurements over **one year**.

Mission Product



SYSTEM OVERVIEW

Bailey Walke – NuKat Chief Systems Engineer



System Requirements Cont.

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Level 2 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
2.1 Satellite Requirements					
SYS-2.01a	Orbital Parameters	NuKAT shall successfully launch into a low earth orbit at an altitude of 500 km and with an LTAN between 14:00 and 18:00	YES	YES	Compliant
SYS-2.01b	Deorbit Time	NuKAT shall be able to deorbit within 5 years of the conclusion of nominal mission operations.	YES	YES	Compliant
SYS-2.01c	Altitude Range	NuKAT shall not have an initial altitude lower than 350 km.	>350km	500km	Compliant
SYS-2.01d	Orbital Radiation	NuKAT shall be able to survive a radiation dosage of 417.25 krad.	YES	YES	Compliant
SYS-2.02	Altitude Trace	NuKAT altitude shall be known with ± 1 km	YES	YES	N/A
SYS-2.03	Location Trace	NuKAT latitude and longitude shall be known within "x" degree of certainty			Compliant
SYS-2.04	Time Trace	Detector data downlinked by NuKAT shall provide time and date data.	YES	YES	Compliant
SYS-2.05	Satellite Orientation Range	+/- 20 degrees of margin off Nadir	YES	YES	Compliant
SYS-2.06	Operational Modes	NuKAT shall utilize operational modes to optimize power usage.	YES	YES	Compliant
SYS-2.09	System Monitoring	NuKAT shall monitor all subsystems and payloads to ensure they are nominally operational at all modes of operation.	YES	YES	Compliant
SYS-2.10	Satellite Size and Weight	NuKAT shall be 2U in size and not exceed 4kg.	YES	YES, 2 kg	Compliant



System Requirements Cont.

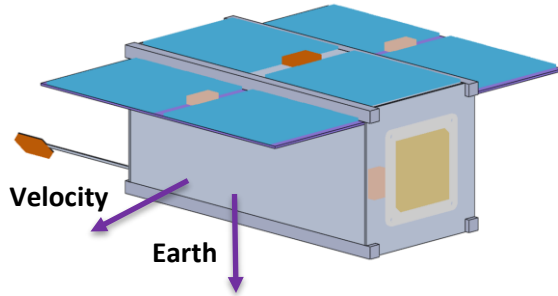
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Level 2 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
SYS-2.11	Satellite Center of Gravity Range	NuKAT shall have a center of gravity within range specified by CP-CDS-R14.1 Table 2.	YES	YES	Compliant
SYS-2.14	System Cost	NuKAT shall cost less than \$100,000 USD.	≤\$ "100,000"	\$"y"	Partial Compliance
SYS-2.15	Mission Lifetime	NuKAT shall be fully operational and supporting payload functions for at least 1 year.	>1 year	>"y" years	Compliant
SYS-2.16	Satellite Survivability	NuKAT shall be designed to withstand the launch and on-orbit environments of the launch vehicle without failure that results in damage of the launch vehicle and its contents or failure that causes injury to the ground handling crew.	YES	YES	Compliant
SYS-2.17	Data Interpretation	NuKAT's payload readings and locational/time trace readings shall be correctly correlated and packaged for downlinking.	YES	YES	Compliant
SYS-2.18	Data Timeliness	Interpreted data shall be provided to end user within 7 days.	< 7 days	Weekly	Compliant
SYS-2.24	Launch Vehicle Compliance	NuKAT System shall be compliant with selected launch provider dispenser system and regulations	YES	YES	Compliant



SATELLITE



Fast/thermal neutron counts

1 count every 30 seconds (**0.033Hz**); 300km ground distance between readings; **0.8 MB** of science data every day

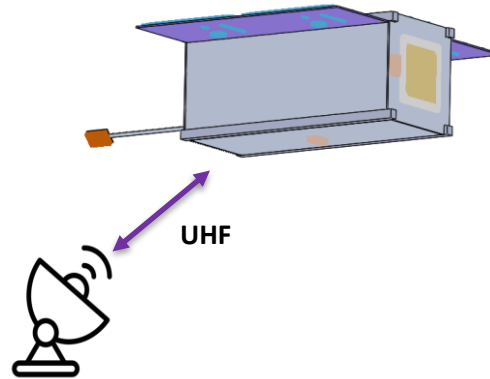
Science data tagging

Satellite orientation at count; data and time at count; latitude and longitude and count

Satellite health data

Critical component temperatures; battery level; operation mode log; data storage

SATELLITE/GROUND STATION



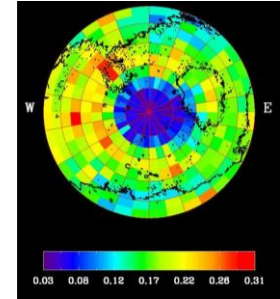
Data download

Downlinking **36 KB/min** (4.8 kbps) via UHF amateur radio; average contact time of around **20 minutes/day**; ground station contact every **1-7 hours**

Periodic command uplink

Manual mode command or specific data requests via UHF amateur radio

GROUND STATION



Science data post processing

Processing, validation and production of desired science data format

Science data distribution

Distributed monthly to stakeholders and interested parties

Satellite health monitoring

Reviewing health beacon packages to respond with uplink commands or software updates if necessary

Day In The Life

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TIME IN LIGHT

60 minutes in sunlight, 30 minutes in umbra every orbit.



GROUND STATION CONTACT

3-7 minutes per contact, revisiting every 1-7 orbits. Average of 20 minutes of contact per day.



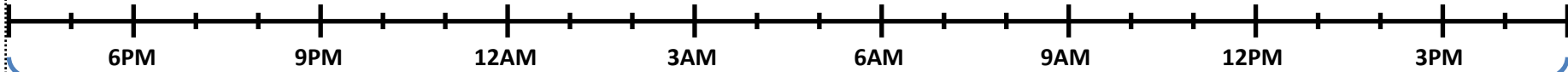
SOUTH ATLANTIC ANOMALY CONTACT

10-15 minutes per contact.



NORTH AND SOUTH POLE CONTACT

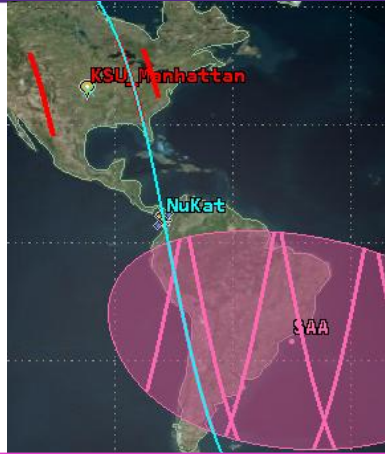
12 minutes per contact.



25 HOURS, 16 ORBITS

Day In The Life

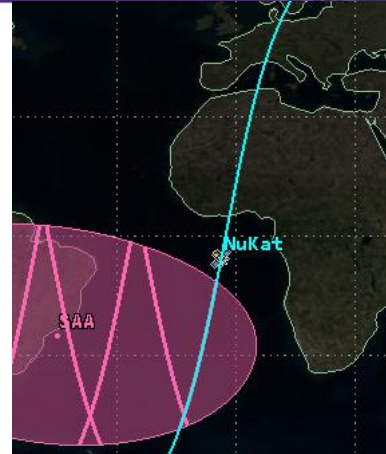
20



Ascending ground station and SAA contact in daylight



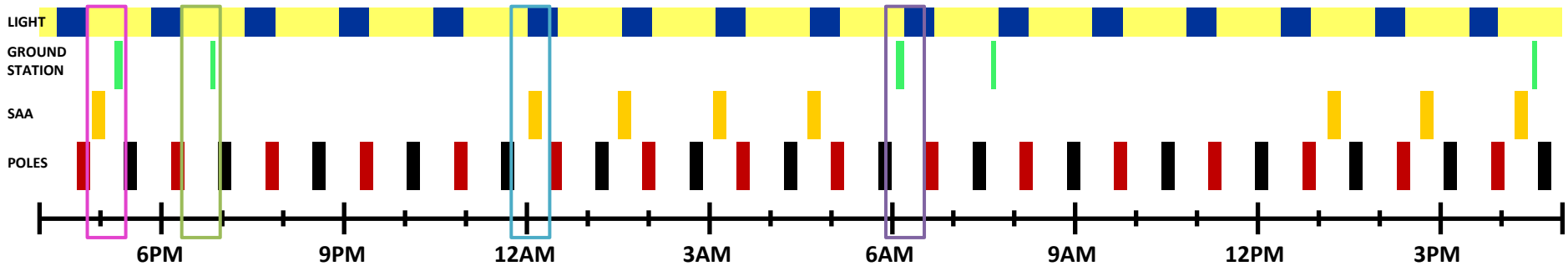
Ascending ground station contact in daylight



Descending SAA contact in umbra



Descending ground station contact in umbra



Concept of Operations

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Satellite returns
to nominal health

SCIENCE

Science payload and science data processing always ON. Transmitting, receiving and pointing operations can happen simultaneously and are triggered automatically.

Satellite falls out
of nominal health

Battery level falls
below 8 Whr (30%)

UPSET

(Detumble, EMI, Over/Under Voltage,
Over/Under Temperature)

Science payload and science data processing functions OFF. Prioritize power usage to ADC, EPD and THM systems, depending on type of upset.

Battery charge
reaches 40%

SURVIVAL

OBC monitors battery level, minimal thermal control to keep system within survivable temperature bounds. All other components and functions OFF.

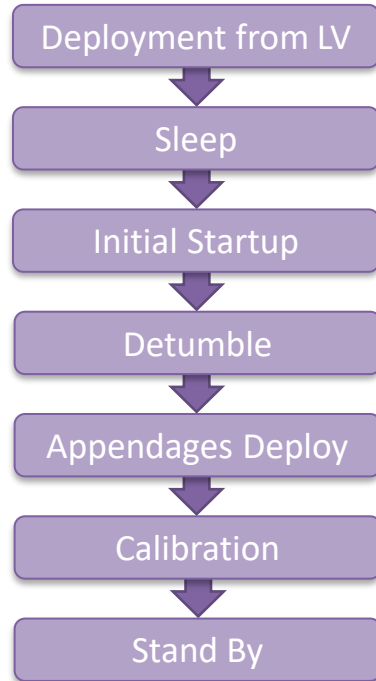
Battery level falls
below 8 Whr (30%)

Spacecraft Modes

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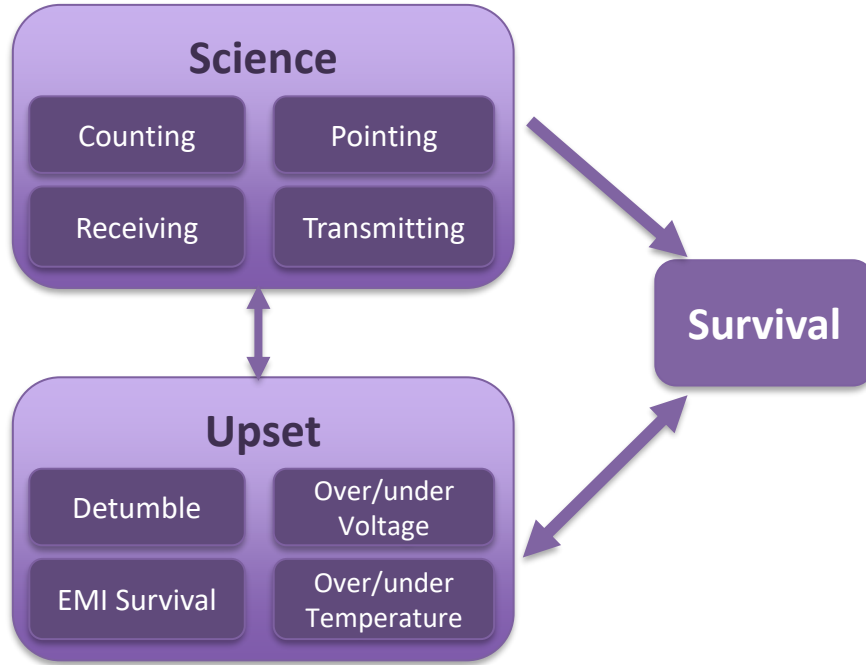
LAUNCH & INITIALIZATION

1 Month



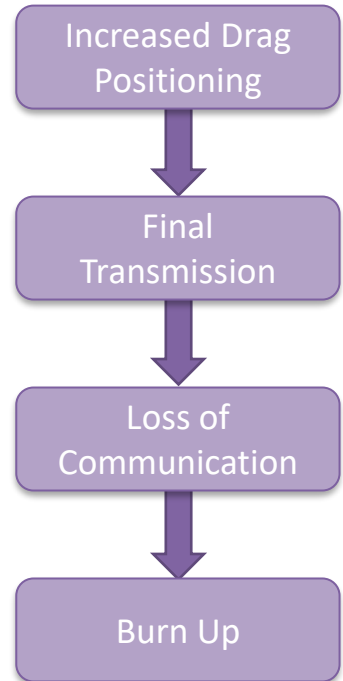
MISSION OPERATIONS

>12 Months



DECOMMISSIONING

6-18 Months



ORBIT & ENVIRONMENT

Igor Sheshukov, Cameron Weeks, Phillip Shirkey



Orbit

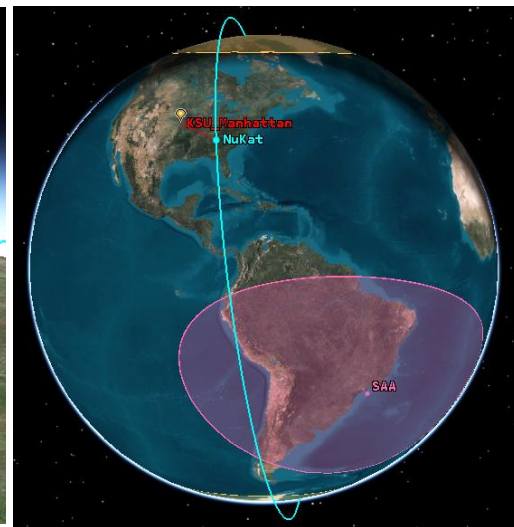
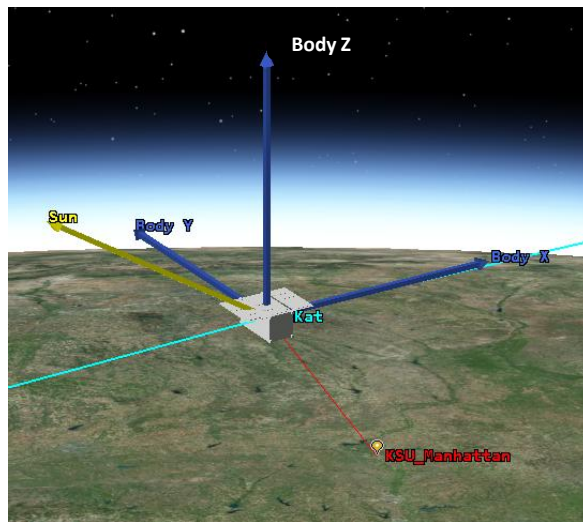
Points of Interest

Ground Stations	Science
KSU Manhattan	South Atlantic Anomaly (SAA)
KSU Salina	Earth Poles



Orbit

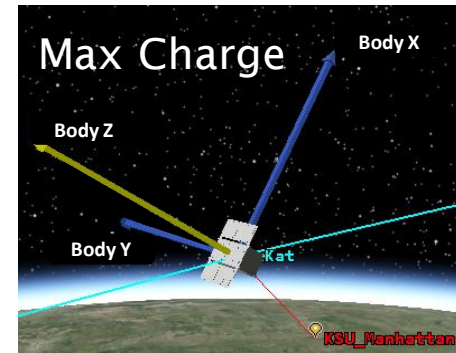
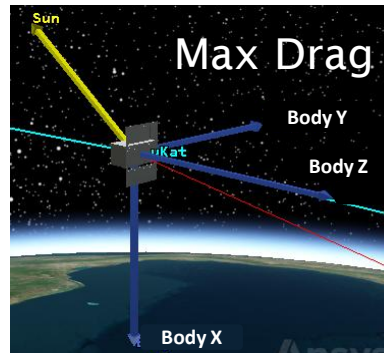
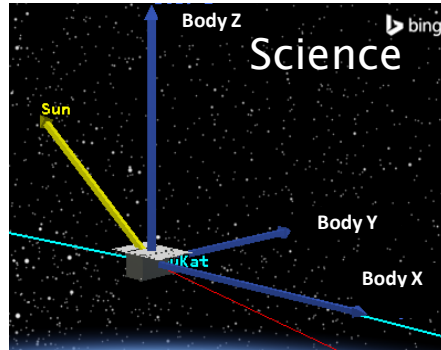
Type	Polar/Sun Synchronous
Altitude	~500km
Inclination	~97°±5°



Orbit

Reentry < 5 years
after loss-of-comms

Mode	Attitude/Orientation	Mission Life from 500km	Mission Life from 545km	Mission Life from 590km
Science	2U Face Forward	1-5 years	1-5 years	1-5 years
Max Drag	Solar Panel Face Forward	+ 0.9-0.3 years	+ 4.6-2.8 years	+ 10.7-10.3 years
Max Charge	Solar Panel Face Sun Normal	+ 1.3-0.4 years	+ 8.1-5.1 years	+ 12.0-11.6 years

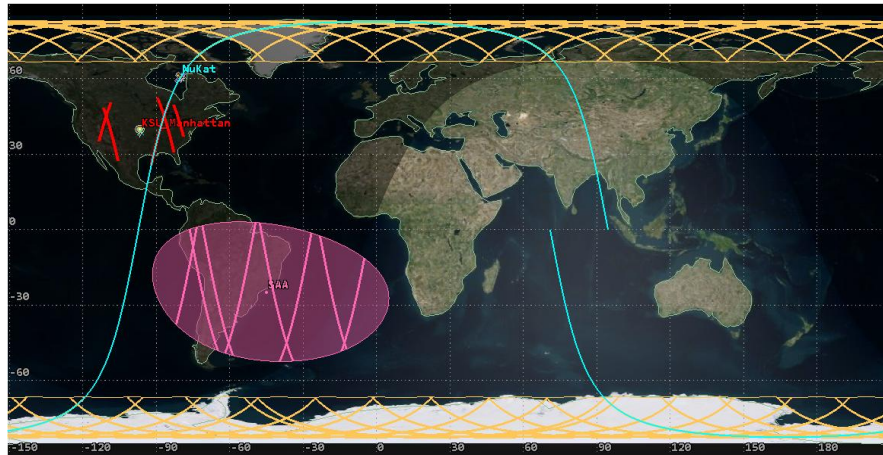
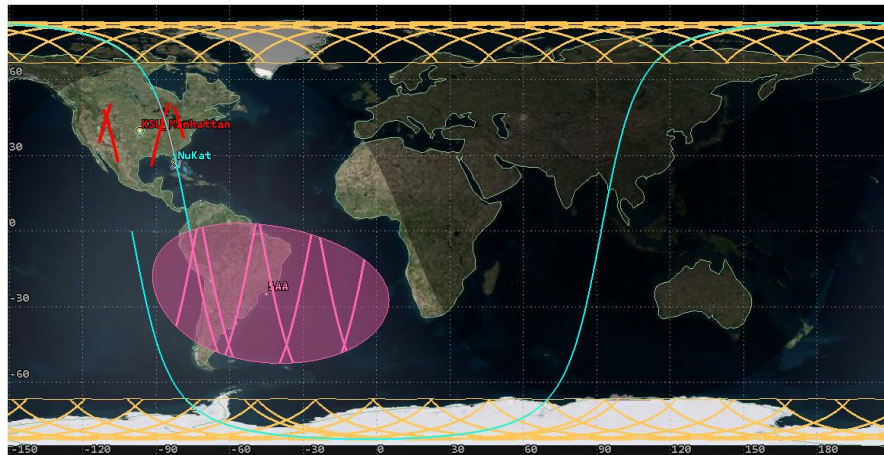


Left Figure: Attitude with 2U Face Forward / Science

Middle Figure: Attitude with Solar Panel Face Forward / Max Drag

Right Figure: Attitude with Solar Panel Face Sun Normal / Max Charge

Orbit



Access for NuKAT to:	KSU Manhattan	SAA	North Pole	South Pole
Avg. Contact Time	5.8min	11.3min	11.7min	11.7min
Avg. Revisit Time	7.5-13hrs	1.4-6.25hrs	1.4hrs	1.4hrs

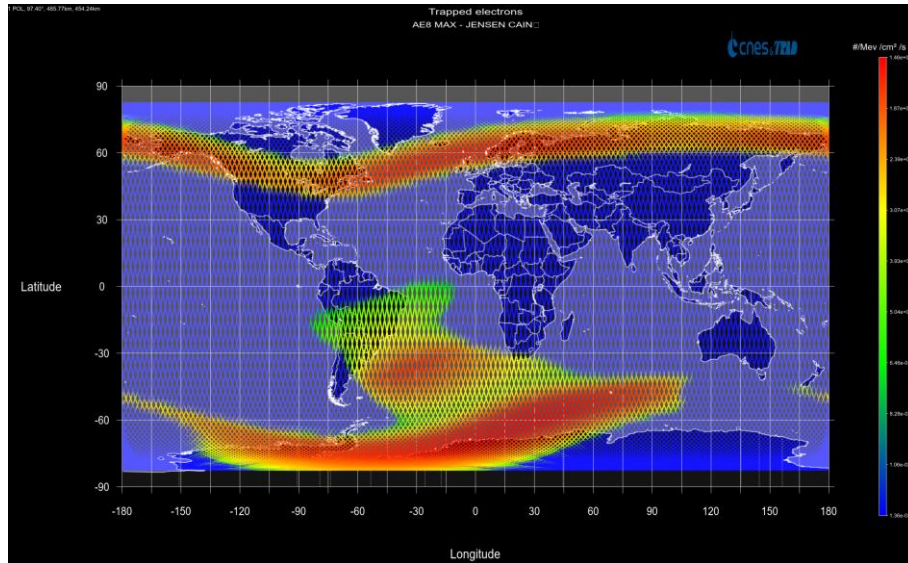
Space Environment

Mechanism	Effects	Source
Surface Charging	• Biasing of instrument readings • Power drains	• Plasma
Deep Dielectric Charging	• Biasing of instrument readings • Electrical discharges causing physical damage	• High energy electrons
Structure Impacts	• Structural Damage	• Orbital Debris
Drag	• Torques • Orbital Decay	• Thermosphere

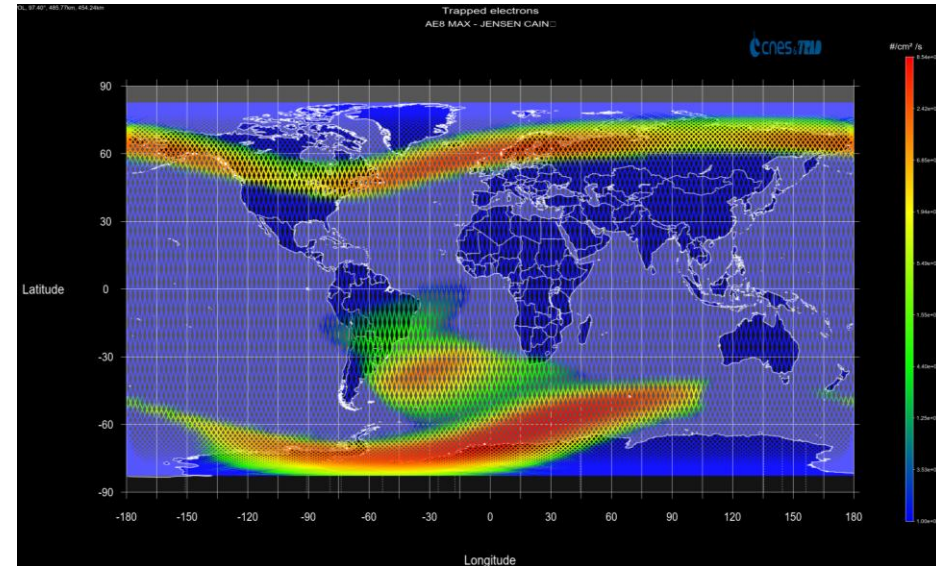
Space Environment

Mechanism	Effects	Source
Total Ionizing Dose	Degradation of electronics	- Trapped electrons & protons - Solar protons
Displacement Damage Dose	- Degradation of optical components and electronics - Degradation of solar panels	- Trapped electrons & protons - Solar protons - Neutrons
Single Event Effects	- Data corruption & noise on images - System shutdowns - Electronic component damage	- Trapped electrons & protons - Solar protons - Neutrons - Galactic cosmic rays
Surface Erosion	Degradation of components	- Particle radiation - Ultraviolet - Atomic oxygen

Radiation Environment



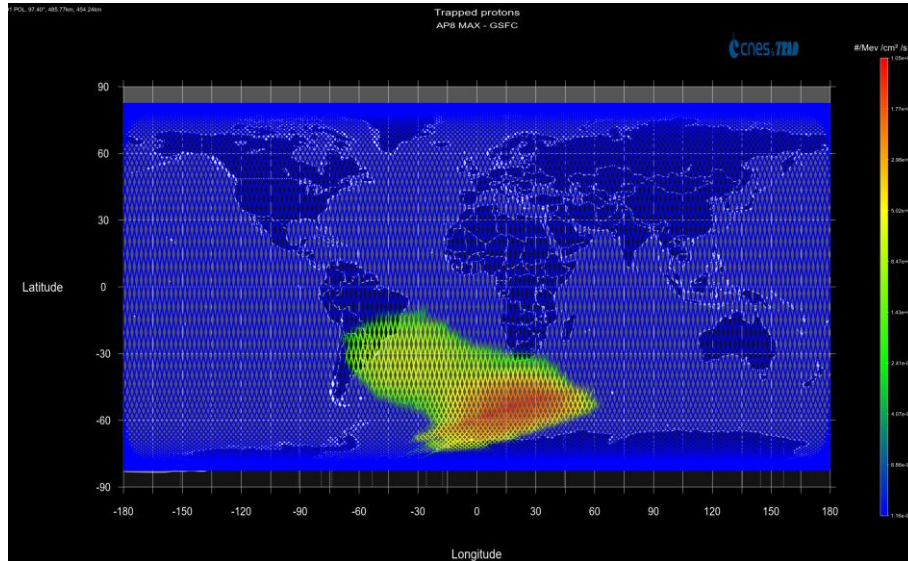
Trapped Electrons Differential Flux Map



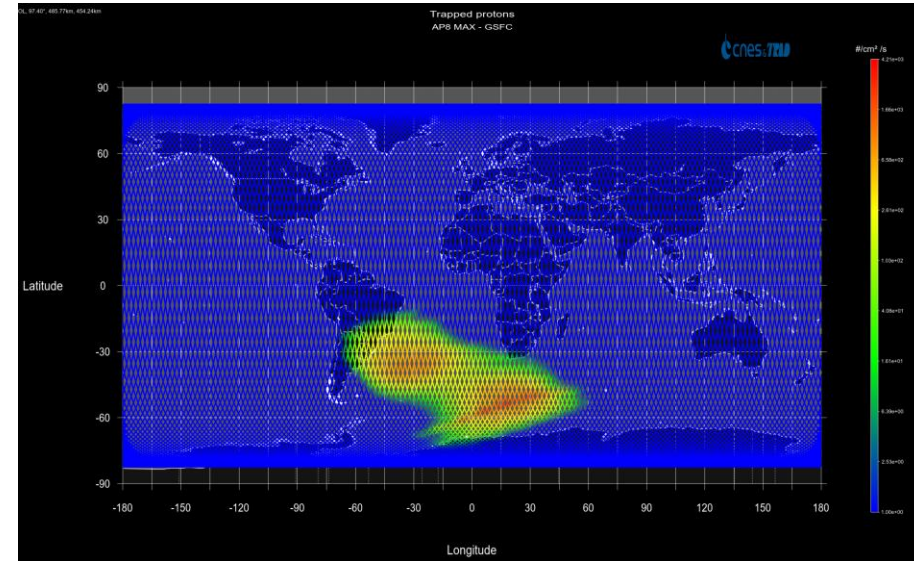
Trapped Electrons Integral Flux Map

Simulation Parameters: AE8 Max Jensen Cain 1960 at 500 km orbit

Radiation Environment



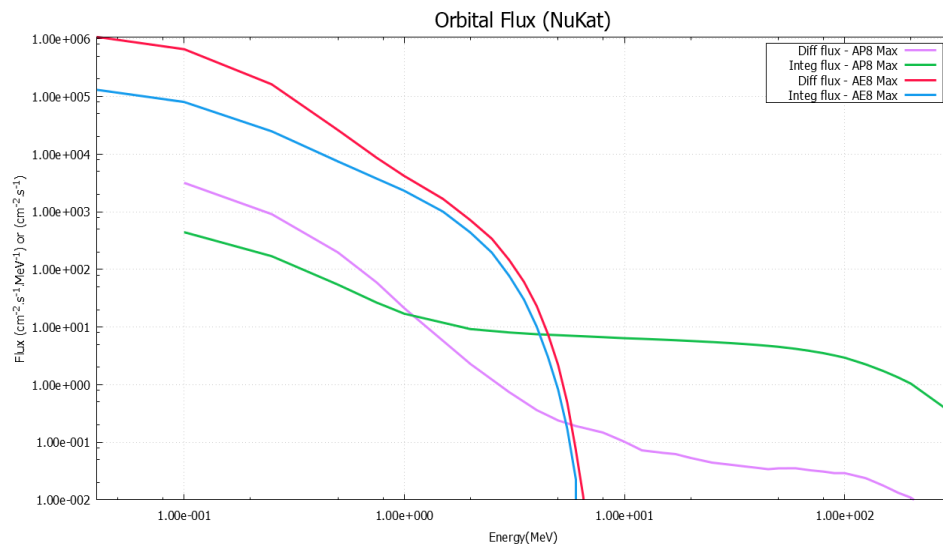
Trapped Protons Differential Flux Map



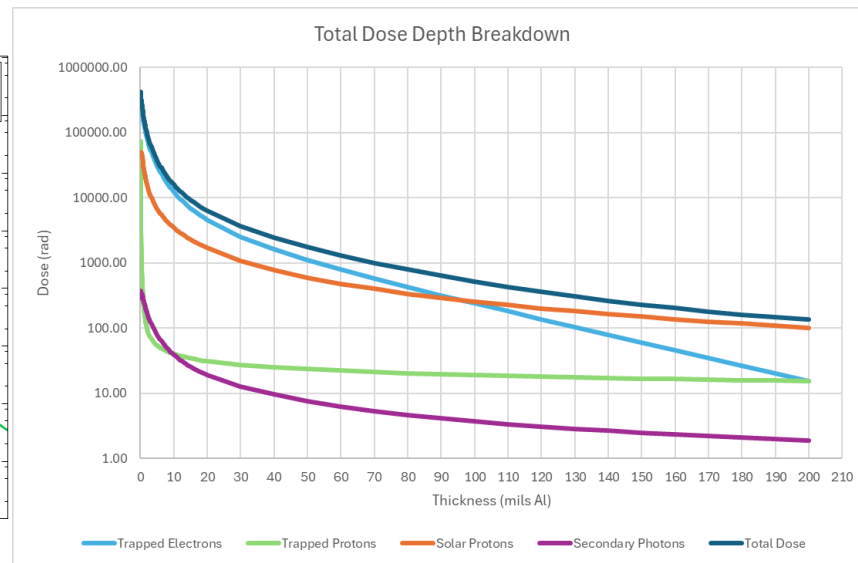
Trapped Protons Integral Flux Map

Simulation Parameters: AP8 Max GSFC 12/66 at 500 km orbit

Radiation Environment

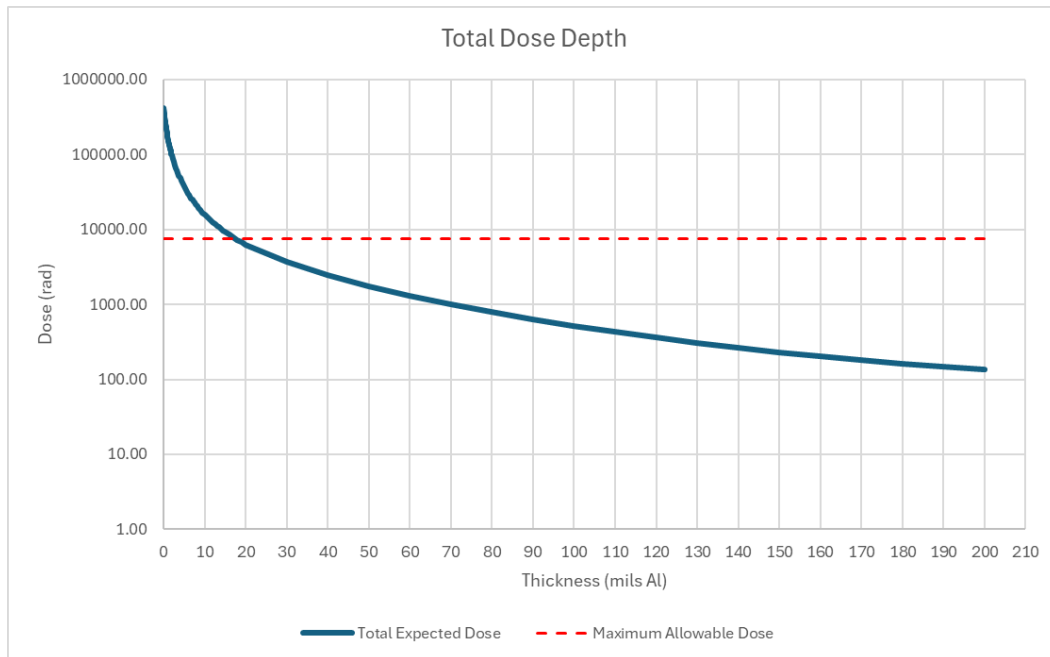


OMERE 5.9: From mission data files



OMERE 5.9: Semi-infinite slab shielding calculation with 2.7 g/cm^3 Aluminum

Radiation Environment

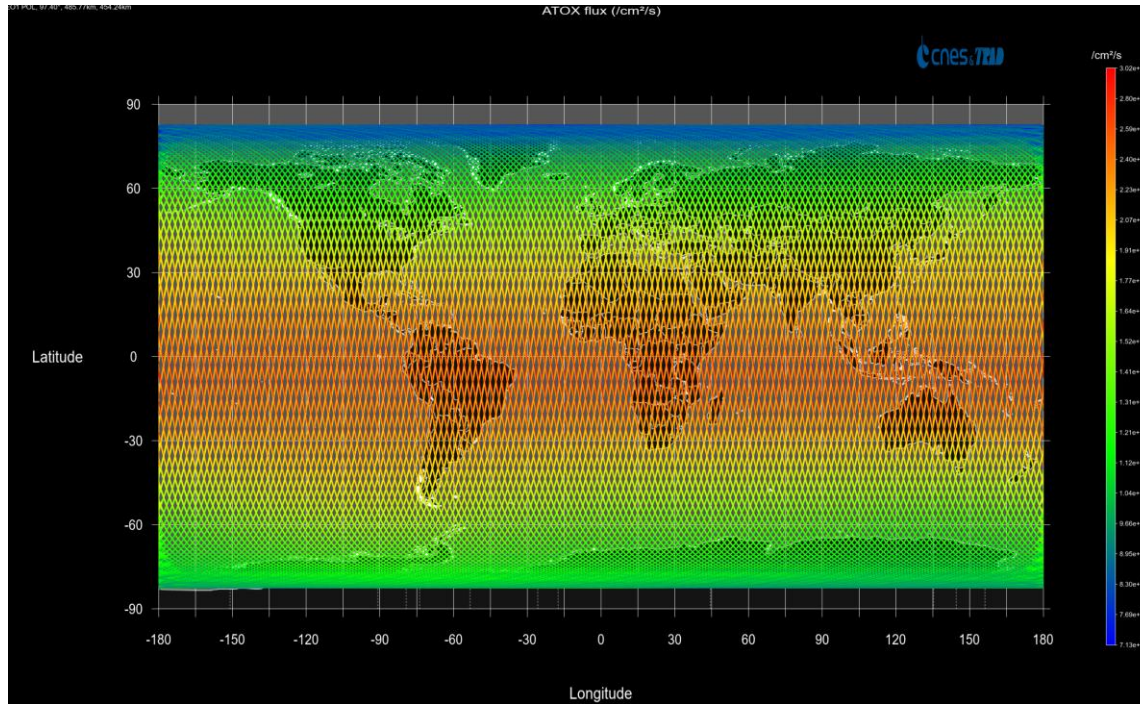


Shielding Thickness	Dose (500 km)
0.01 mils AL "no shield"	~417.25 krad
10.5 mils AL (0.2667 mm)	~14.85 krad
18.0 mils AL (0.4572 mm)	~7.27 krad
60 mils AL (1.524 mm)	~1.30 krad

NASA recommends for reliability to use a Factor of Safety of 2 for determining shielding

Current Max Allowable Dose is 15 krad
With a FOS of 2: 7.5 krad

Atmospheric Environment



500 km orbit

Maximum Flux:
3.02E14 atoms/(cm²/s)

Minimum Flux:
7.13E13 atoms/(cm²/s)

Launch Environment

- Launch Vehicle – SpaceX Transporter
 - Fundamental Frequencies
 - Primary lateral frequency $> 10\text{Hz}$
 - Primary axial frequency $> 25\text{Hz}$
 - Secondary structure minimum resonant frequencies $> 35\text{Hz}$
 - Expected Vibrations – 1g @ $> 30\text{Hz}$
 - Acceleration
 - Axial load 11g at Launch
 - Lateral load 4g at Launch, 7.5g Max

Launch Environment

- Integration Parameters
 - Electromagnetic Compatibility
 - Pending dispenser wiring schematics
 - Dispenser needs all access points on 1 long side
 - Ascent Venting
 - Payload outgassing and bus depressurization <4.5kPa/sec on ascent, test with TVAC chamber
 - Sensors must be oriented out of debris exit path
 - Rail Deployment

Launch System

- Tumbling rate: $<60^\circ/s$
- Volume: 2U, optional tuna can
- Mass: 4kg, optional 4.33+ with approval
- AIT compliance: covered in prior slide, requirements pending
- Cost: \$75,000-90,000
- Includes regulatory support

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.13 Launch System Requirements					
LNC-3.01	Deployment Tumble	The launch dispenser shall deploy NuKAT with a tumbling rate of less than 60 deg/s.	< 60 deg/s	< 30 deg/s	Not Compliant
LNC-3.02	Volume Compliance	LNC shall be capable of launching a 2U satellite into the desired orbit.	= 2U	= 2U	Compliant
LNC-3.03	Mass Compliance	LNC shall be capable of launching at least 2.66 kg into the desired orbit.	≥ 2.66 kg	≥ 4 kg	Compliant
LNC-3.04	Launch Provider Compliance	LNC shall provide STR with dispenser specifications and testing methods, to ensure compabitility with the provider's deployment system.	YES	YES	Not Compliant
LNC-3.05	Cost	LNC shall cost less than \$x.	< \$x	< \$x	N/A
4.13 Launch System Requirements					
LNC-4.01	Integration Compliance	All satellite components must comply with SpaceX Transporter and provider launch system requirements.	YES	YES	N/A

PAYLOAD OVERVIEW

Colby Johnston – NuKAT Principal Investigator

Payload Requirements

Level 2 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
2.2 Payload Requirements					
PL-2.01	Payload System Scope	The Payload System shall consist of at least one neutron detector.	YES/NO	YES	Compliant
PL-2.02	Neutron Detector Interface	PL shall interface to the bus mechanically, electrically and thermally.	YES/NO	YES	Compliant
PL-2.03	Neutron Energy Bins	Neutron detector moderation shall shift detection bins into thermal (0 - 1 eV), epithermal (1 - 500 eV), fast (500 eV+)	YES/NO	YES	Compliant
PL-2.04	Detector Efficiency	Neutron detector shall have peak detection efficiencies at 0.025 eV, 0.1 MeV, and 18 MeV	YES/NO	YES	Compliant
PL-2.05	Detector Orientation	Neutron detector shall have the MSNDs side of the DSC-Domino within 20-degrees of nadir while pointing in orbit and while primarily focusing on counting (e.g., not in a survival charging mode)	< 20-degrees	< 20-degrees	Compliant
PL-2.06	Detector Mounting	Neutron detector shall be mounted with the MSNDs facing parallel to the Nadir datum face within 5-degrees.	< 5-degrees	< 5-degrees	Partial Compliance
PL-2.07	Mission Lifetime	PL shall operate for at least 3 months with full success at 1 year.	≥ 3 months	≥ 1 year	N/A
PL-2.08	Detector Moderator	HDPE shall be used as the neutron moderator			Compliant

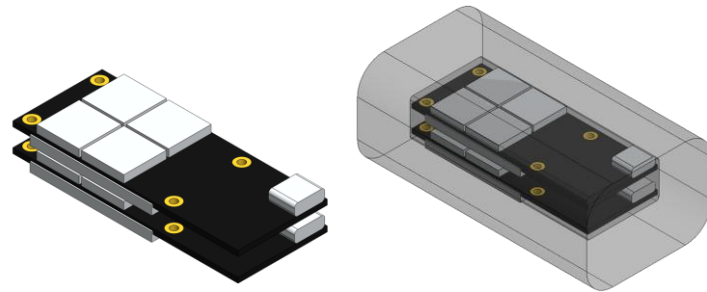
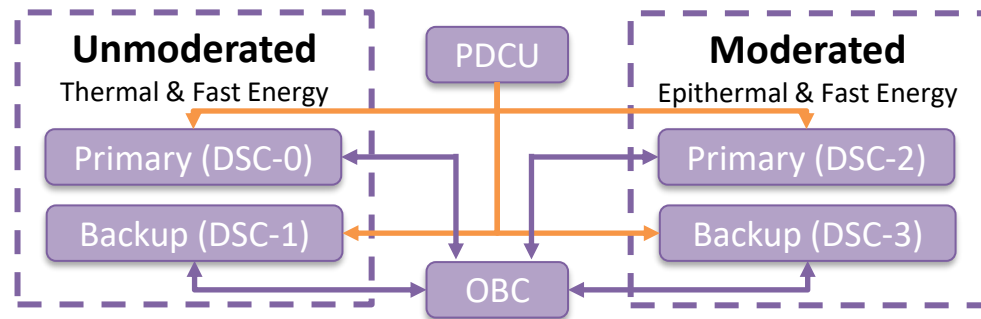
Payload Requirements

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.2 Payload Requirements					
PL-3.01	Neutron Detector Payload Accuracy	The neutron detector payload shall detect neutrons within "x"% of the accepted orbital neutron profile (after proper efficiency corrections from moderator choice/thickness and detector efficiency).	within "x"% of accepted value	within "y%" of accepted value	Not Compliant
PL-3.02	Neutron Detector Redundancy	Each main neutron detector shall have a backup detector for failure detection and space anomaly verification resulting in four total DSC Domino boards	Yes	Yes	Compliant
PL-3.03	Charged Particle Rejection (proton, beta, GCRs, gamma, etc.)	The reference MSNDs (or other detector, this needs to be determined with RDT) shall enable non-neutron particle rejection within "x"% accuracy	within "x"% of accepted value	within "y%" of accepted value	Not Compliant
PL-3.04	Neutron Detector Payload Power Consumption	The neutron detector payload shall have a power consumption of 65 mW or less per detector, with a 5 V voltage requirement, respectively.	< 65 mW	60 mW	Compliant
PL-3.05	Neutron Detector Payload Mass	The neutron detector payload shall have a mass of 0.31 kg or less (4 x 14.25 g for the detector and 246 g for the neutron moderator).	< 0.31 g	< 0.31 g	Compliant
PL-3.06	Neutron Detector Size	The neutron detector assembly (including moderator) shall occupy 0.3 U or less	< 0.3 U	< 0.3 U	Compliant
PL-3.07	Neutron Detector Thermal Limits	The neutron detector payload shall be kept between -20 °C and 55 °C during operation.	-20 °C < T < 50 °C	-20 °C < T < 50 °C	Partial Compliance
PL-3.08	Neutron Detector Data Format	The neutron detector data shall be collected in 32 bit unsigned integer format for each MSND (4 MSNDs per platform, 4 platforms)	= 32 bit uint	= 32 bit uint	Compliant
PL-3.09	Neutron Moderator Outgassing	The neutron detector moderator shall have a total mass loss coefficient below 1.0%. (or be encased such that the loss coefficient is below 1.0%)	< 1% TMLC	0.58% - NASA's Goddard Engineering and Technology Directorate	Compliant
PL-3.10	Neutron Moderator Size	The neutron moderator shall encase 2 of the detectors with a thickness of 10 mm on the "top" side of the board and 10 mm on the bottom side of the board.	= 10 mm +/- 1 mm	10 mm	Compliant

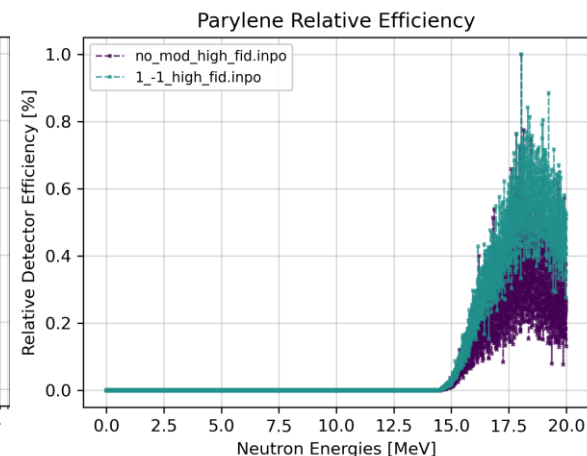
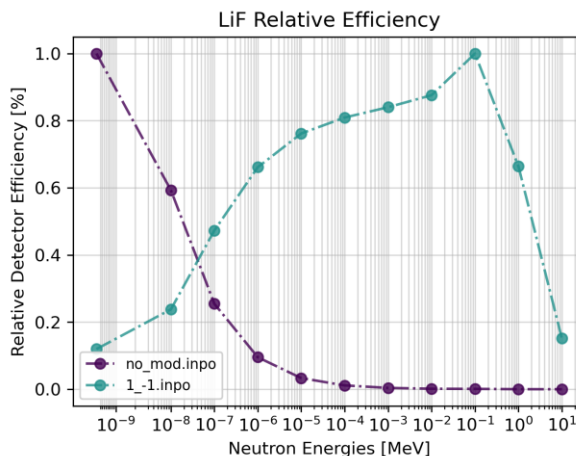
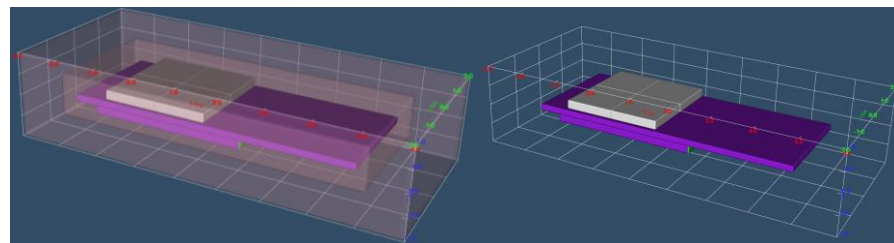
Payload Details

- Four detector packages
 - 2 primary and 2 backup
 - 4 MSNDs on each package
- SWaP (per package):
 - Size: 66.8 x 27.2 x 5.75 mm
 - Weight: 15 g
 - Power: 65 mW (across all operations)
 - Data package:
 - All 4 MSNDs: 16 bytes per count pull
 - Background-corrected (2 data points): 8 bytes per count pull
 - Temperature Sensor: 2 bytes per temp pull



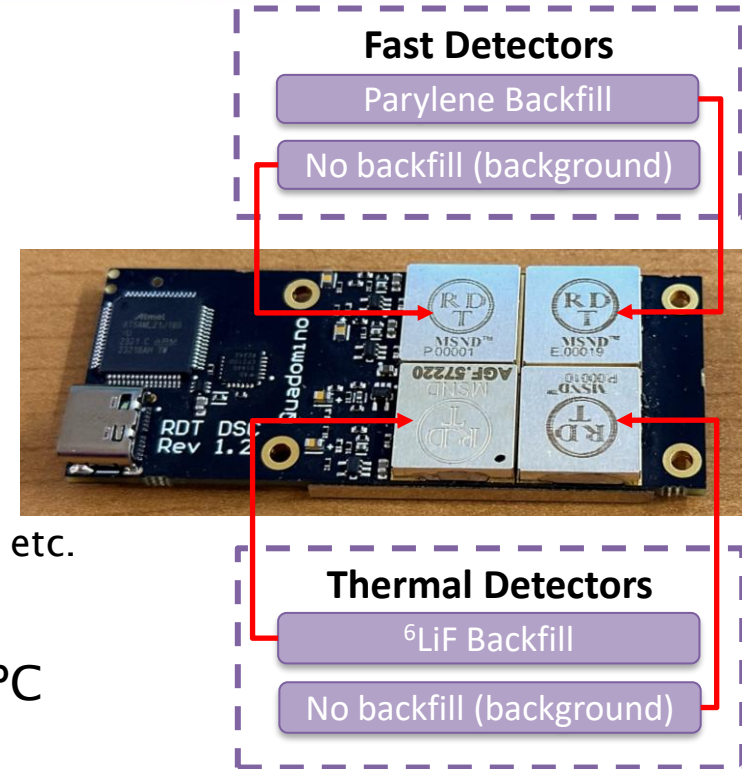
Payload Details

- Three coarse energy bins
 - MSND Backfill (^6LiF and Parylene)
 - Moderation (HDPE)
 - 10 mm thick
 - Efficiency Peaks
 - 0.025 eV (thermal)
 - 100 eV (epithermal)
 - 18 MeV (fast)



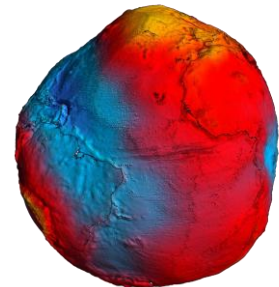
Payload Details

- Calibration and testing
 - Performed by supplier (RDT)
- Data validation
 - Onboard detector verification
 - Validation from previous missions
 - E.g., ISS detection data, NeRDI-1A and -1B, etc.
- Other payload specifications
 - Operating temperature: -20 °C – 55 °C
 - System interface: USB-C ports
 - Moderator Dimensions: 33.8 x 86.8 x 45.9 mm

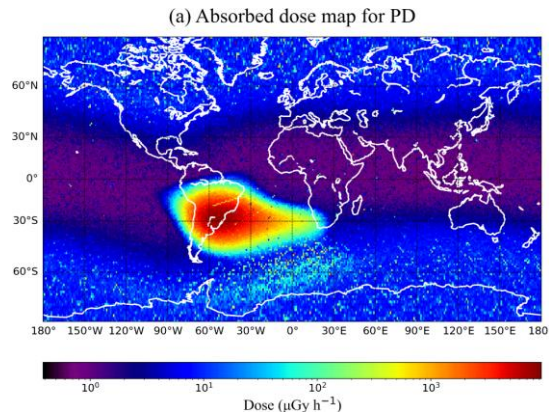


Science Data Use/Distribution

- Data Package:
 - Orbit-correlated raw data
 - Processed data at stakeholder request
- Periodic distribution via Listserv to stakeholders
- Interest areas
 - South Atlantic Anomaly
 - North and South Poles



National Geographic | *New Gravity Map Reveals Lumpy Earth*



Payload Readiness

The preliminary payload design is technically feasible, producible, and en route to compliance with our requirements.

BUS OVERVIEW

Bailey Walke – NuKat Chief Systems Engineer



Bus Requirements

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Level 2 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
2.3 Bus Requirements					
BUS-2.01	Bus System Purpose	NuKAT bus systems shall support payload needs, and integrate payload systems and bus subsystems.	YES	YES	Partial Compliance
BUS-2.02	Bus System Size	BUS subsystems shall take up no more than 1.5U of the satellite.	≤ 1.5 U	1 U	Compliant
BUS-2.03	Bus System Weight	BUS must weigh less than 3 kg.	≤ 3 kg	2 kg	Compliant
BUS-2.04	Bus System Power	BUS must provide 20 W of power.	$\leq$ "x" W	"y" W	Not Compliant
BUS-2.05	Bus System Thermal Range	BUS must be able to keep interior systems within a temperature range of 15 and 50 degrees Celsius.	YES	YES	N/A
BUS-2.06	Bus System Radiation Hardness	BUS must be able to endure 500 krad of radiation over the mission lifetime.	YES	YES	Compliant
BUS-2.07	Bus System Hosting	BUS subsystems shall not obfuscate payload systems or bus subsystems.	YES	YES	Compliant
BUS-2.09	Bus-Payload Integration	BUS shall provide data handling, ground communication ("x" kbps), attitude control, structure, and flight software to support payload needs.	YES	YES	Partial Compliance



Bus Requirements Cont.

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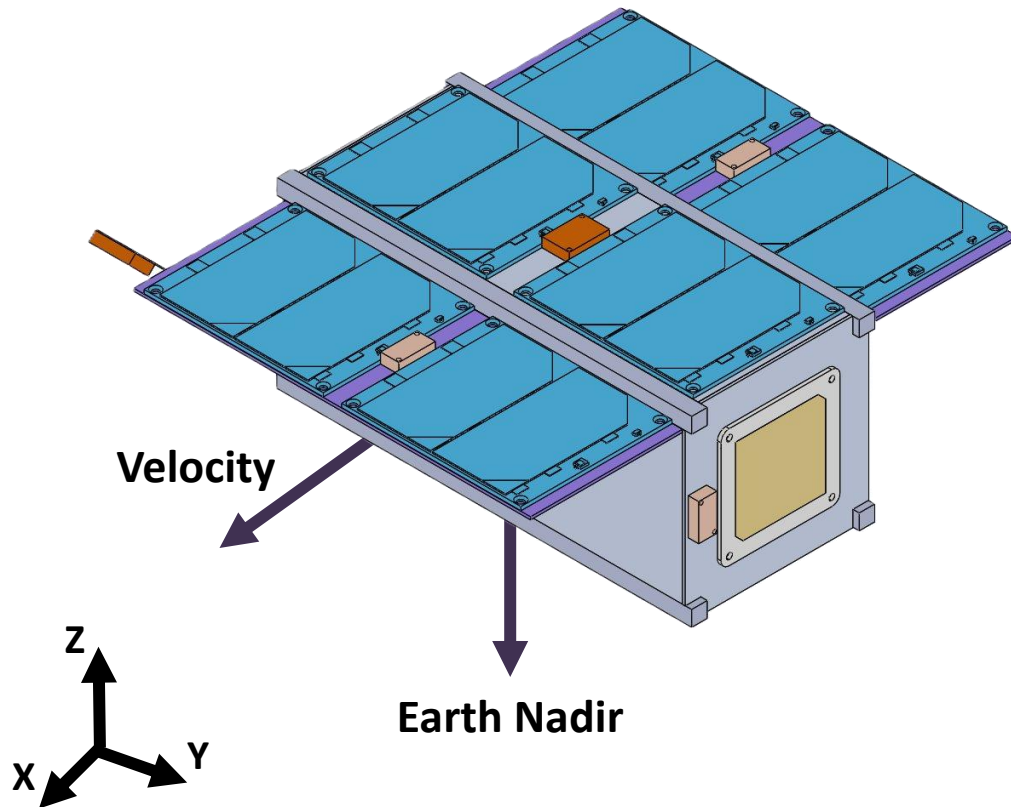
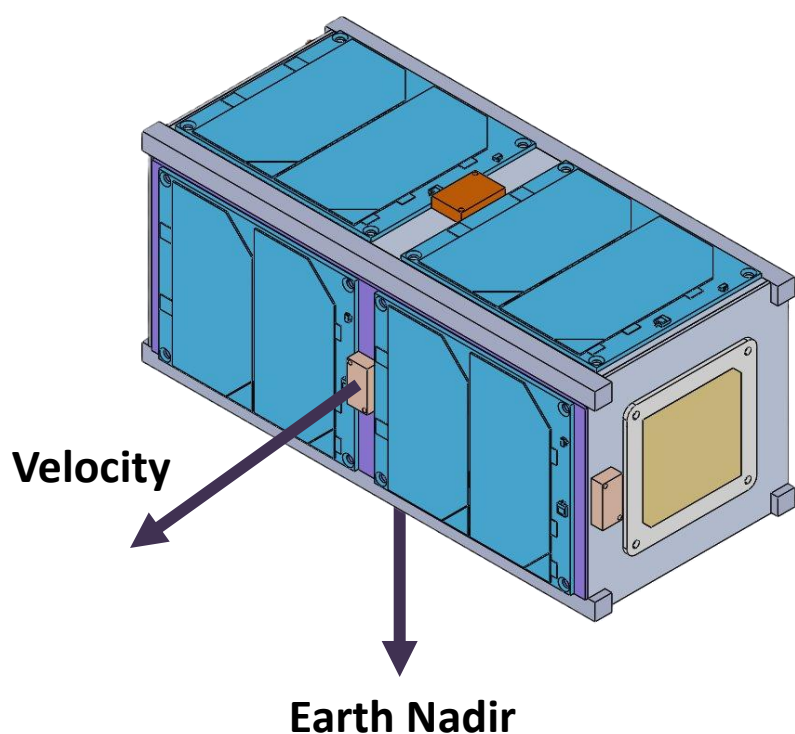
Level 2 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
BUS-2.10	Bus Preparation	BUS shall have means of testing, preparing, and protecting the satellite, it's systems, and it's components up until launch.	YES	YES	Partial Compliance
BUS-2.11	Space-Rated Components	BUS shall consist of components rated for space environments and proven to be space worthy for at least 12 months.	>12 months	24 months	Partial Compliance
BUS-2.13	Satellite Communication	BUS shall be able to communicate with the ground station. Communciation includes recieving uplinked commands and downlinking relevant science data and health beacons.	YES	YES	Not Compliant
BUS-2.14	Missed Pass Storage/Downlink	BUS shall be capable of storing and downlinking data for up to 7 days-worth of missed passes.	YES	YES	Partial Compliance
BUS-2.24	Payload Orientation	BUS shall ensure payload is in its correct orientation for data collection.	YES	YES	Compliant



Layout Overview

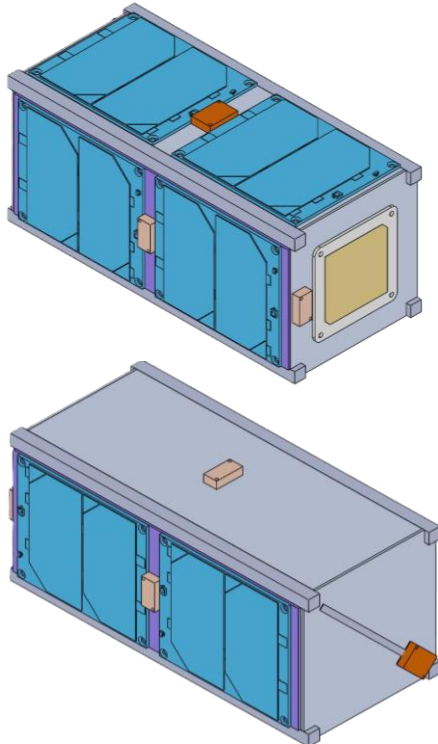
48



Satellite External View

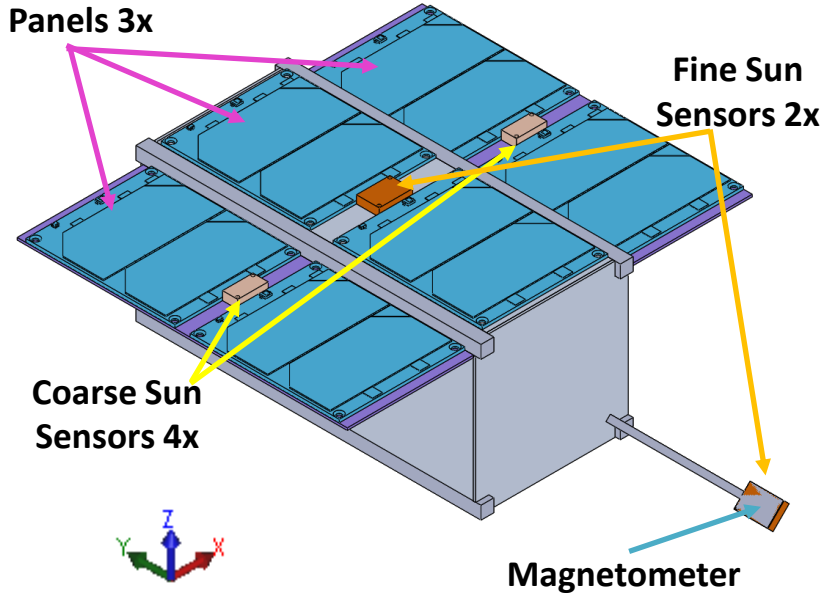
49

UNDEPLOYED



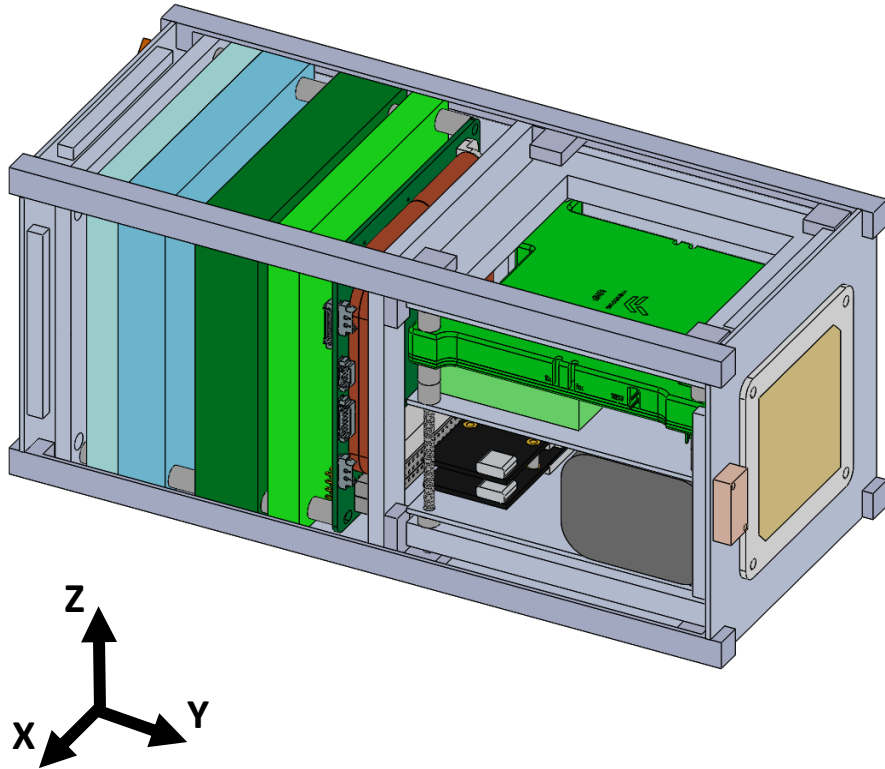
DEPLOYED

2U Solar Panels 3x



Satellite Overview (Internal)

50



PAYLOAD

- 4x RTD DSC Domino; 2x moderated, 2x unmoderated

ATTITUDE CONTROL

- *Attitude control:* 3x magnetorquers
- *Position monitoring accuracy:* 5 degrees

COMMUNICATIONS

- *Science and telemetry downlink, command uplink:* UHF amateur bands (4.8kbps)
- *Position monitoring:* OEM719 GPS/GNSS Receiver

POWER

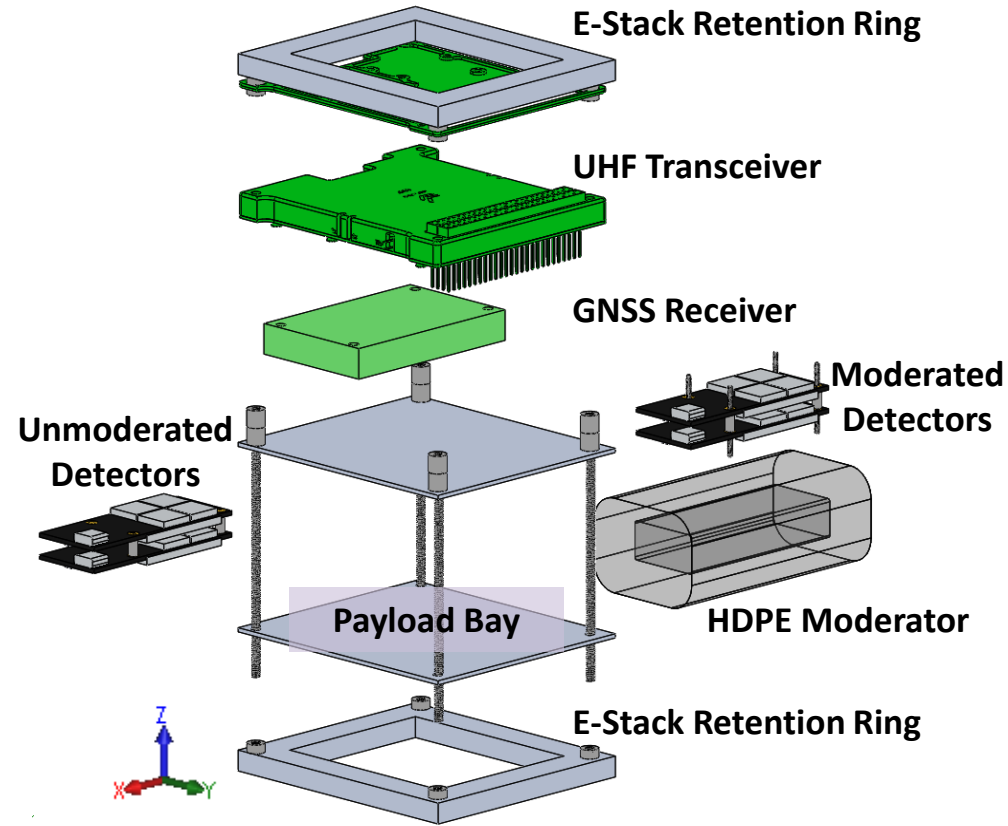
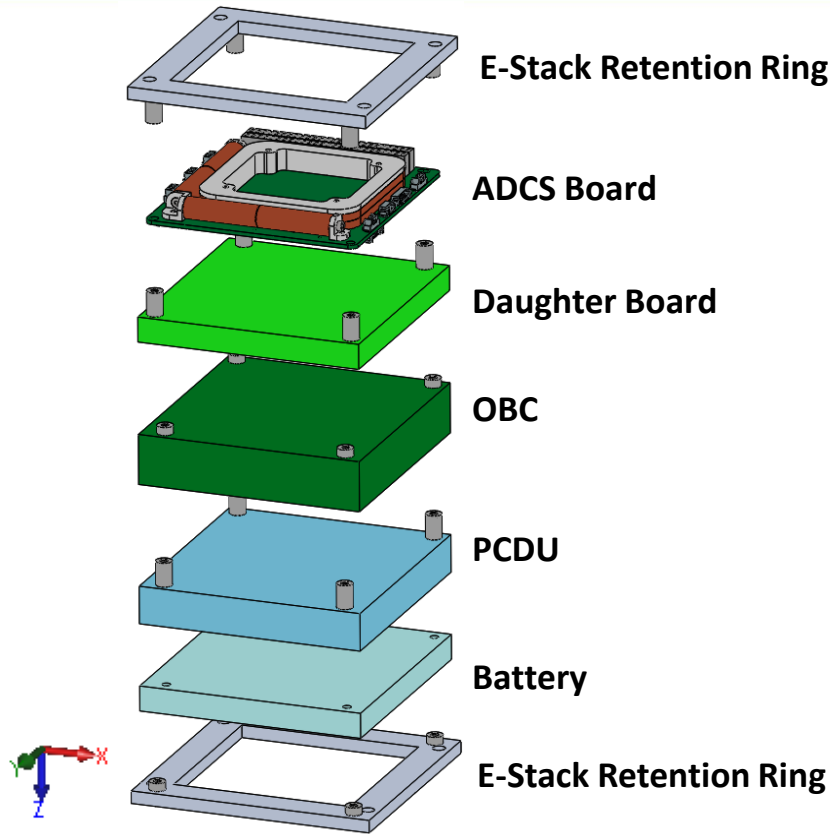
- 20 Whr battery
- *Power distribution:* DHV PCDU

COMMAND & DATA HANDLING

- NextAge NMC2 onboard computer
- *Data storage:* 512GB SD card

STRUCTURE

- 1.5mm-thick 6061 plating for radiation protection
- Rail design compliant with CalPoly spec

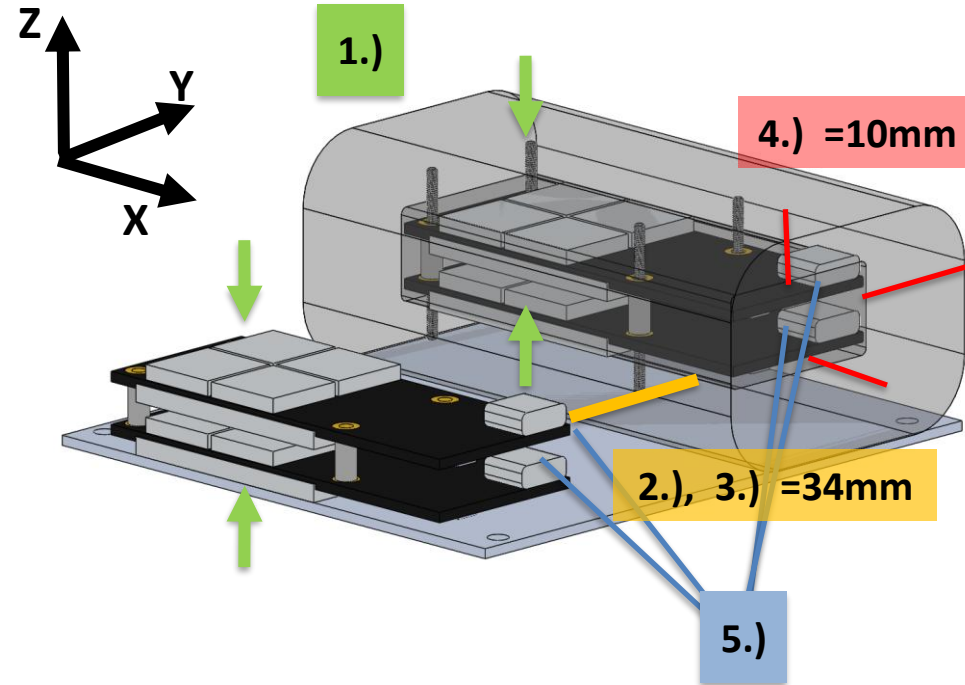


Payload Integration

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Requirements:

1. Platform normal are in line with earth normal (Z)
2. Moderated and unmoderated platforms must not overlap in Z direction
3. Moderated and unmoderated platforms must be spaced at least the width of one platform (26.5mm)
4. Moderator must provide 10mm of HDPE material on every side of the platforms
5. Each platform must interface with OBC via USB-C



Mass/Volume Budget

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Component	Model	Mass (g)	Volume (cm ³)	Critical Dimension (cm)
Detector Platform (4x)	RTD DSC Domino	9.50 (4)	8.30 (4)	0.6 (Z)
HDPE Moderator	Custom	212.15	222.84	3.4 (Z)
ADC Board	Tensor Tech MTQ-200M Board	150	155.66	2.3 (Z)
Coarse Sun Sensor (4x)	Tensor Tech CSS-10	0.50 (4)	1.40 (6)	0.6
Fine Sun Sensor (2x)	Tensor Tech FSS-15	3 (2)	1.721 (2)	2.0 (Y)
OBC	NextAge NMC2	85	138.24	1.5 (Y)
UHF Transceiver	EnduroSat UHF Transceiver	90.22	194.61	2.3 (Y)
UHF Antenna	TBD	TBD	TBD	TBD
GNSS Receiver	NovAtel OEM719	23	38.96	1.1 (Z)
GNSS Antenna	EnduroSat L1/E1 GNSS Antenna	17.1	21.84	0.2 (Y)



KANSAS STATE
UNIVERSITY

Carl R. Ice
College of Engineering

HPDE: High Density Polyethylene

ADC: Attitude Determination and Control

OBC: Onboard computer

UHF: Ultrahigh Frequency

GNSS: Global Navigation Satellite System

Bailey Walke

Mass/Volume Budget

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Component	Model	Mass (g)	Volume (cm ³)	Critical Dimension (cm)
PCDU	DHV PicoEPS Power Module	210	129.6	1.5 (Y)
Battery	DHV PicoEPS Battery Module	80	86.4	1.0 (Y)
Solar Panels	EnduroSat 1U	38 (6)	21.61 (6)	1.0 (Y)
Chassis Assembly	Custom	764.56	283.16	2.0 (Y)
Deployable Magnetometer	Custom	3.92	2.05	1.0 (Z-X)
Radiators/Insulators	TBD	TBD	TBD	TBD

Total Mass: 1.96kg

Total Critical Volume: 1,448cm³

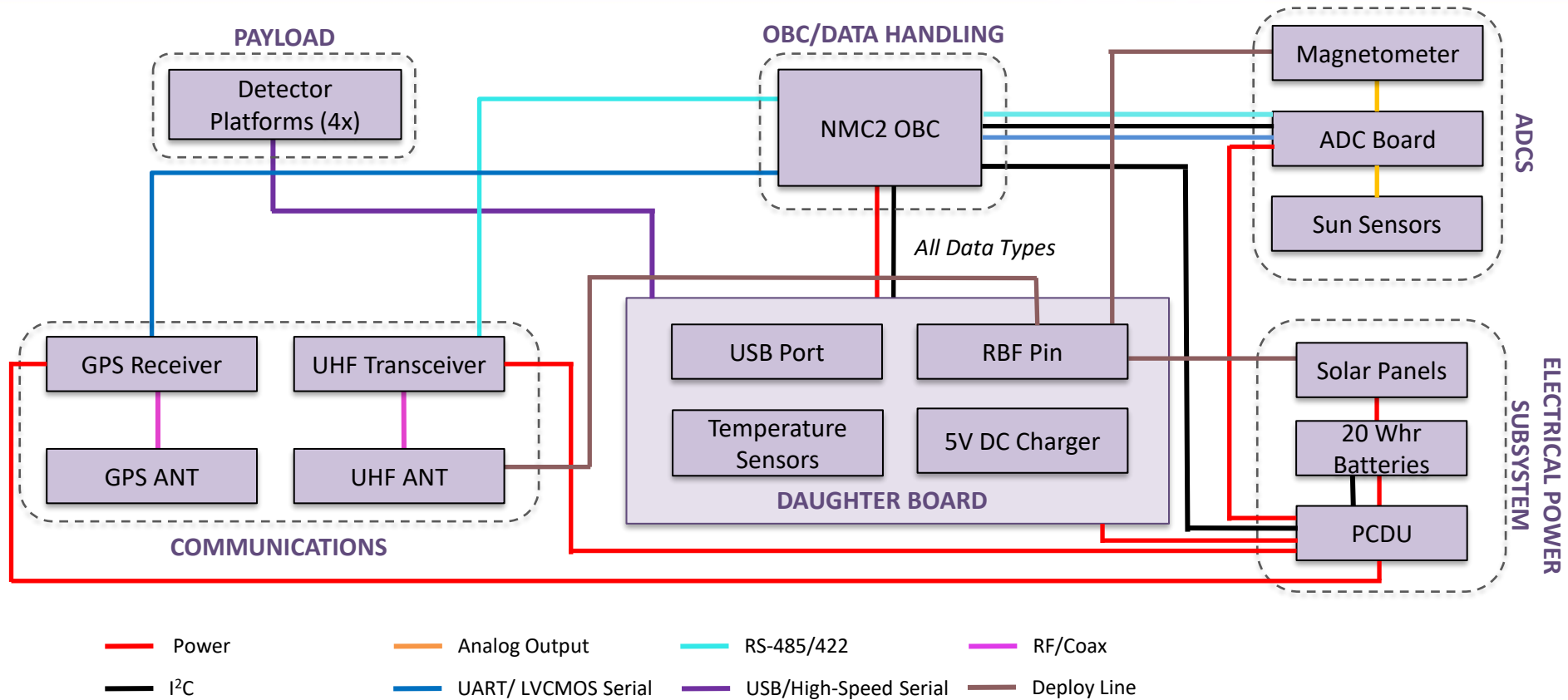
Total Mass +30% Margin: 2.55kg

Total Critical Volume +30% Margin: 1,882cm³



System Block Diagram

55



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Path Forward to Compliance

UHF antenna

Power positive

Data volume/
frequency

Bus Feasibility Statement

The Bus system partially meets requirements, fits our size allocation, and is on track to be feasible.

SATELLITE SUBSYSTEMS

Phillip Shirkey, Michael Bennett, Henry Bash,
Logan Hayward, Braden Adams, Sophia Fenton, Mitch Daues



Attitude Determination and Control

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
ADC-3.01	Core Capabilities	ADC shall allow coarse nadir tracking, sun tracking, ground station tracking, and PNT to support PLD mission and satellite operation.	YES	YES	Compliant
ADC-3.02a	NT Knowledge	ADC shall determine angle w.r.t. Earth normal vector, within 5 deg.	< 5 deg	< 5 deg	Compliant
ADC-3.02b	NT Pointing	ADC shall allow pointing to Earth normal vector, within 5 deg.	< 5 deg	< 5 deg	Compliant
ADC-3.03a	ST Knowledge	ADC shall determine angle w.r.t. Sun, within 10 deg.	< 10 deg	< 10 deg	Compliant
ADC-3.03b	ST Pointing	ADC shall allow pointing to Sun, within 10 deg.	< 10 deg	< 10 deg	Compliant
ADC-3.04a	GS Knowledge	ADC shall determine angle w.r.t. Ground Station, within "x" deg.	< "x" deg	< "y" deg	Partial Compliance
ADC-3.04b	GS Pointing	ADC shall allow pointing to Ground Station, within "x" deg.	< "x" deg	< "y" deg	Partial Compliance
ADC-3.05	PNT Knowledge	ADC shall collect PNT data for sensor calibration and GS tracking.	YES	YES	Compliant
ADC-3.06	Upset Capabilities	ADC shall allow sun tracking at low power and correct any deployment tumbling.	YES	YES	Compliant
ADC-3.07a	Safe Mode ST Knowledge	ADC shall determine angle w.r.t. Sun, within 10 deg under low power mode.	<10 deg	<10 deg	Compliant
ADC-3.07b	Safe Mode ST Pointing	ADC shall allow pointing to Sun, within 10 deg, under low power mode.	<10 deg	<10 deg	Compliant
ADC-3.08a	Detumbling Time	ADC shall complete detumble maneuver within 90 minutes.	< 90 min	< 90 min	Compliant
ADC-3.08b	Max Tumble Rate	ADC shall detumble the satellite from 60 deg/s or less.	< 60 deg/s	< 60 deg/s	Compliant

Attitude Determination and Control

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
ADC-3.09	Subsystem Requirements	ADC shall meet system engineering limits, SWaP-C, and resource budgets.	YES	YES	N/A
ADC-3.10a	Size	ADC shall require less than "x" U of space.	< "x" U	< "y" U	N/A
ADC-3.10b	Weight/Mass	ADC shall require less than "x" kg of mass.	< "x" kg	< "y" kg	N/A
ADC-3.10c	Power	ADC shall require less than "x" W of power.	< "x" W	< "y" W	N/A
ADC-3.10d	Cost	ADC hardware shall cost under "x" dollars to purchase.	< "x" dollars	< "y" dollars	N/A
ADC-3.11	Comms Budget	ADC shall require less than "x" bit/s of uplink and "y" bit/s of downlink data during operation.	< "x" bit/s UP and "y" bit/s DOWN	< "x1" bit/s UP and "y1" bit/s DOWN	N/A
ADC-3.12	Computing Budget	ADC shall require less than "x" bit/s of onboard data processing for telemetry and flight ops.	< "x" bit/s	< 0 bit/s (no OBC allocation required)	Partial Compliance

- GS pointing requirements not determined, because antenna configuration is not set.
- Subsystem SWaP-C and other allocations not determined, because system budgets are not set.
- Size: 0.2U Weight: 150g Power: 1.5-5W Cost: \$9600+

Attitude Determination and Control

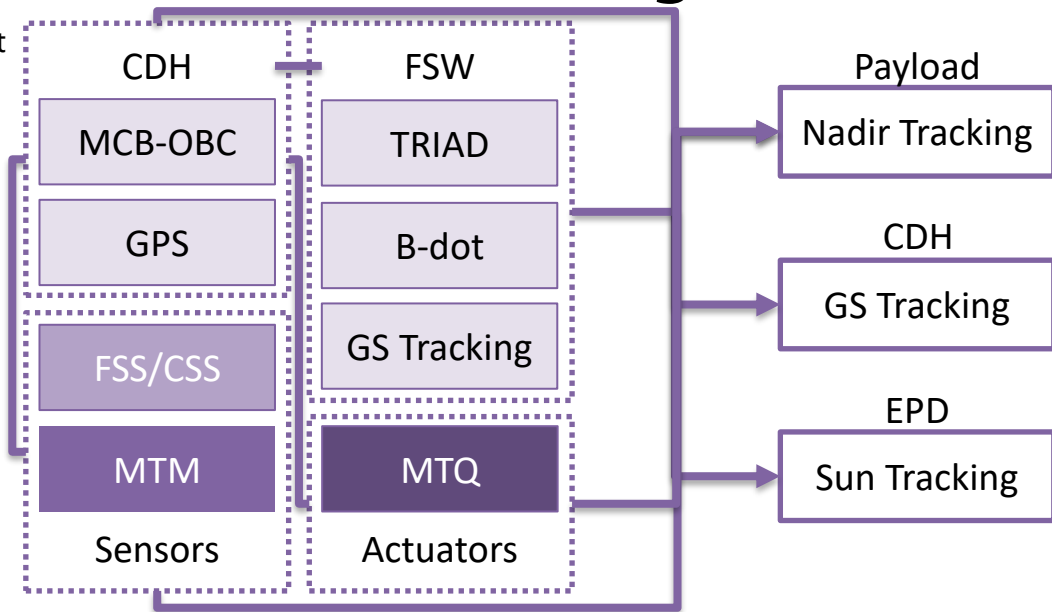
Level 4 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.3 Attitude Determination and Control Requirements					
ADC-4.01	Calibration Period	ADC shall be able to calibrate orientation and velocity sensors when required.	YES	YES	N/A
ADC-4.02a	Fine Sun Sensor Angular Accuracy	ADC shall use Fine Sun Sensors with an angular accuracy within 10 degrees.	<10 deg	<10 deg	Compliant
ADC-4.02b	Fine Sun Sensor Field of View	ADC shall use Fine Sun Sensors with a field of view of at least +/- 45 degrees.	>45 deg	>60 deg	Compliant
ADC-4.03a	Coarse Sun Sensor Angular Accuracy	ADC shall use Coarse Sun Sensors with an angular accuracy within 10 degrees.	<10 deg	<10 deg	Compliant
ADC-4.03b	Coarse Sun Sensor Field of View	ADC shall use Coarse Sun Sensors with a field of view of at least +/- 45 degrees.	>45 deg	>60 deg	Compliant
ADC-4.04	Magnetometer Angular Accuracy	ADC shall use Magnetometers with an angular accuracy within 5 degrees.	<5 deg	<5 deg	Compliant
ADC-4.05	Magnetorquer Angular Accuracy	ADC shall use Magnetorquers to make maneuvers accurate to less than 5 degrees.	<5 deg	<5 deg	Partial Compliance
ADC-4.06	FSW Interface with MCB	ADC Main Control Board must be able to receive commands, provide health & status, and push MTM data to PLD for ground re-normalization.	YES	YES	N/A
ADC-4.07	EPD Interface with MCB	ADC Main Control Board must be able to draw power from EPD.	YES	YES	N/A
ADC-4.08	CDH Interface with MCB	ADC Main Control Board must be able to pull GPS data from CDH to confirm position and time.	YES	YES	N/A
ADC-4.09	Miscellaneous Requirements	All ADC components shall fall within total SWaP-C for ADC system.	YES	YES	N/A
ADC-4.10a	Operational Temperature Range	ADC shall operate as designed between -20 and 55 degrees Celcius.	-20 to 55 deg C	-40 to 85 deg C	N/A
ADC-4.10b	Survivable Temperature Range	ADC shall endure any temperature between "x" and "y" degrees Celcius.	"x" to "y" deg C	"x" to "y" deg C	N/A
ADC-4.11	Radiation Hardness	ADC shall endure 24 krad of radiation during mission life.	< 24 krad	< 48 krad	N/A
ADC-4.12	Operational Lifetime	ADC shall operate as designed for 2 years.	< 2 years	< 5 years	N/A

Attitude Determination and Control

- ADC Subsystem
 - Power is distributed through MCB to sensors and actuators
 - MCB will control sensors and actuators to point at targets, commanded by master OBC
- CDH Interface
 - Link in: MCB requires position and time from GPS antenna, MCB must receive requests from OBC
 - Link out: MCB communicates telemetry and vitals to OBC
- EPD Interface
 - Link in: MCB requires 3.3V/5V connection, will push 3.3V to sensors and 5V to actuators
- FSW Interface
 - Link in: MCB takes commands from OBC to switch modes
 - Link out: MCB provides ADC H&S to OBC

Block Diagram



Attitude Determination and Control

- Hardware Interface
 - MCB and magnetorquers mounted to chassis
 - Sun sensors mounted to chassis/integrated w/ solar panels
 - Magnetometer deployed 10 cm away from EMI sources
 - Sun sensors must have unobstructed FOV (+/- 60 degrees from sensor normal)
 - All magnetorquers must be orthogonal w.r.t. each other
 - Location of leads, connections, etc. in supplier documentation
 - MCB and MTQ generate waste heat, should be thermally connected

Attitude Determination and Control

- Software Interface
 - MCB comes with detumble, sun pointing, and direct control modes
 - Nadir pointing, B-dot control, and GS pointing must be coded in-house
 - Interface: I²C, UART, and RS485/422
 - Connector: Molex Pico – Blade
 - Pointing will be done by MCB, but OBC will have mode authority

Attitude Determination and Control

- Magnetic environment
 - Components and assembled system must have residual magnetic field measured for MTM calibration and MTQ sizing
 - The World Magnetic Model (WMM) is used for preliminary sizing; as a reference, the local field in Manhattan now is around 40.33 μT
- MTQ preliminary sizing
 - Center of Gravity (CG): known distance from geometric center
 - Mass: known mass around 2.55 kg with margin
 - ADC placed as close to CG as possible
 - Known maximum magnetic dipole moment on each axis
 - Known moment of inertia on each axis
 - Nadir tracking, Sun tracking, and GS tracking are allowed (with margin for mode switch)
 - Rough time to detumble: 30 minutes or less
- MTQ final sizing will require a full map of magnetic field on orbital path over mission lifetime, to verify accurate pointing in weak field areas

Attitude Determination and Control

• Pointing Budget

Subsystem	Mode	Component	Target	Axis to Align and Frame of Axis	Accuracy (+,-)	Precision (+,-)	FOV (+,-)	Function
PLD	Science (Nadir Pointing)	Neutron Detector	Earth Normal	(+,-) Z Axis of Spacecraft to be aligned with Nadir of Earth	5 deg	5 deg	20 deg	Point for science mode; align to collect minimum 90% of expected particles
CDH	Up/Downlink (GS Pointing)	Comms Antenna	Ground Station	(+,-) X or Y Axis of Spacecraft to be aligned with GS, which one based on structure	UNKNOWN	UNKNOWN	UNKNOWN	Point for up/downlink; align to send minimum X% of packets
	Science, Calibration (Sun + Nadir)	GPS Receiver	GPS Satellites	Aligned with Equator during operation	5 deg	5 deg	30 deg	Position Determination
ADC	Science, Up/Downlink, Calibration	Magnetometer	Earth Normal	No alignment required; will calibrate on orbit to find relative direction of Earth Normal vector	5 deg	5 deg	360 deg	Attitude Determination
	Safe Mode (Sun Pointing)	Coarse Sun Sensor	Sun	Aligned with Sun vector during operation	10 deg	10 deg	45 deg	Attitude Determination, Sun Awareness
	Science, Up/Downlink, Calibration, Safe Mode	Fine Sun Sensor	Sun	Aligned with Sun vector during operation	10 deg	10 deg	45 deg	Attitude Determination
	ALL	Magnetorquer	N/A	Aligned internally, all 3 components must be orthogonal	5 deg	5 deg	N/A	Attitude Control; limit sensor resolution
EPD	Safe Mode	Solar Panels	Sun	(+) Z Axis of Spacecraft to be aligned with Sun	10 deg	10 deg	20 deg	Sun tracking to charge

Attitude Determination and Control

- Tech Specs: Tensor Tech partial data sheet
 - Still need test qualifications (thermal, rad, life)
- Flight Heritage
 - First integrated ADCS on orbit 2022
 - 100 ADCS products delivered by 2024
- Make/Buy Decisions
 - Make: CSS (save cost on non-critical sensors), code
 - Buy: MCB, MTQ, MTM, FSS (reliability)
 - Provider: Tensor Tech (lowest cost solution, integrates tracking software, low SWaP design)

Attitude Determination and Control

- Standards and Regulations

- Tensor Tech guidelines
- ISO 9001 manufacturing
- U.S. Export Controls & ITAR
- GEVS GSFC-STD-7000B
- ESCC Basic Spec. 22900

- Safety and AIT protocols

- Low hazard in testing
- Testing requirements:
 - HIL/SIL electronic testing for Sun Sensors, Magnetometers, Magnetorquers, all flight algorithms and pointing modes
 - TVAC testing for vented debris obstruction, hot/cold/warm operation
 - Full bus magnetic testing and correction (degaussing) for fixed & variable magnetic field effects, followed by air bearing tests

- Verification in Tensor Tech product data sheet

	QM	EM	FM
TID Radiation Test ¹	•	-	-
SEE Radiation Test ²	•	-	-
CPT	•	•	•
Sine Vibration (SV) Test	•	-	-
Random Vibration Test	•	-	•
Thermal Cycling Test ³	-	-	•
Thermal Vacuum Cycling Test	•	-	-

Attitude Determination and Control

The Attitude Determination and Control subsystem is technically feasible, meets our requirements, and is producible.

Electrical Power & Distribution

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.5 Electrical Power Distribution Requirements					
EPD-3.01	External Charging	The spacecraft shall have a means of charging its battery that does not require solar power.	--	--	Compliant
EPD-3.02	External Power Supply	The spacecraft shall have a means of externally powering the EPD system.	--	--	Not Compliant
EPD-3.03	Battery Level	The battery level shall remain above 50% during science/nominal mode, while remaining above 30% when upset mode occurs.	30%	50%	Not Compliant
EPD-3.04	Idle Mode Power	EPD shall be capable of powering all components at the calculated worst-case peak power draws during Idle Mode.	--	--	Not Compliant
EPD-3.05	Active Mode Power	EPD shall be capable of powering all components at the calculated worst-case peak power draws during Active Mode.	--	--	Not Compliant
EPD-3.06	Downlink Mode Power	EPD shall be capable of powering all components at the calculated worst-case peak power draws during Downlink Mode.	--	--	Not Compliant
EPD-3.07	Distribution and Control	EPD shall safely distribute and control all of the power generated.	--	--	Not Compliant
EPD-3.08	Power Conversion	EPD shall provide downstream power converters for different voltage loads.	--	--	Compliant
EPD-3.09	Bus Isolation	EPD shall provide bus isolation between subsystem loads.	--	--	Not Compliant
EPD-3.10	Bus Monitoring	EPD shall provide bus monitoring electronics such that each load may be accurately monitored for fault and efficiency.	--	--	Not Compliant
EPD-3.11	Protection	EPD shall protect itself from EMI, transients, bus faults, and load faults.	--	--	Not Compliant
EPD-3.12	Solar Safe/Arm Plug	EPD shall require a Safe/Arm plug for the solar array power generation.	--	--	Not Compliant
EPD-3.13	Battery Safe/Arm Plug	EPD shall require a Safe/Arm plug for the battery system.	--	--	Not Compliant
EPD-3.14	Remove Before Flight	EPD shall include Remove Before Flight locks, pins, and tags from the vehicle during ground testing and before launch.	--	--	Not Compliant

Electrical Power & Distribution

Road to Compliance

- EPD-3.01: There will be a USB-C for charging batteries during integration and testing.
- EPD-3.02: Physical implementations to occur after electronics are acquired and their integration is settled.
- EPD-3.04, 3.05, & 3.06: Further component information, load calculations, & budget analysis will be necessary for compliance to be proven.
- EPD-3.09 & 3.11: More design details necessary for isolation techniques and protection to be decided, then testing after integration will occur.
- EPD-3.12, 3.13, & 3.14: Physical implementations to occur after electronics are acquired and their integration is settled.

Electrical Power & Distribution

Level 4 Requirements:

NUKAT REQUIREMENTS VERIFICATION MATRIX

Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.5 Electrical Power Distribution Requirements					
EPD-4.01	Power Consumption	The PCDU shall not draw more than "x" Watts of power when operating.	2.6	TBD	Partial Compliance
EPD-4.02	Battery Size	The battery shall be of adequate size to provide power to systems while vehicle is in Earth's shadow .	TBD	20 Wh	Compliant
EPD-4.03	Solar Array Size	The affixed solar array shall conform to a "x"U vehicle size of "x".	2	2	Compliant
EPD-4.04	Solar Array Power	The affixed solar array shall provide nominally ~"x"W on sides of the vehicle.	10	24	Compliant
EPD-4.05	Lifetime	Component's expected life-span should exceed mission life expectancy of "x" months.	12	TBD	Not Compliant
EPD-4.06	Voltage Rails	The PCDU shall provide the necessary voltage levels for the busses.	3.3V, 5V, Vbat	3.3V, 5V, Vbat	Compliant

- EPD-4.01: Additional analysis of Active and Downlink modes required.
- EPD-4.02: Idle Load does not drain battery while in Penumbra.
- EPD-4.03 & EPD-4.04: Current Solar Array design is compliant and feasible.
- EPD-4.05: Longterm analysis of charge/discharge/thermal cycles required.
- EPD-4.06: PicoEPS provides necessary voltage rails.

Electrical Power & Distribution

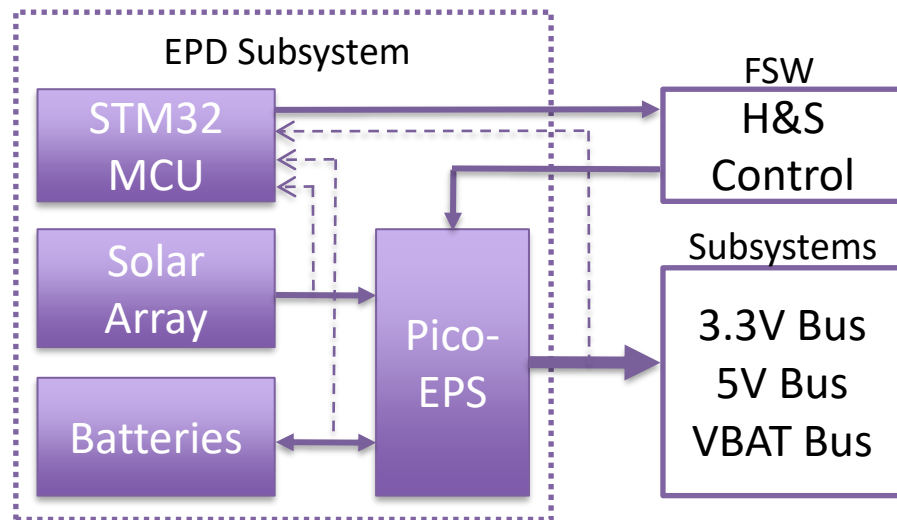
External Links

- Mechanical
 - Mounting
 - Keep-Out Zones
 - Exterior Solar Panels
 - Accessibility
 - Battery
 - Solar Panel Connection
- Thermal
 - Conduction
 - PCDU
 - Battery
 - Solar Panel Heat Radiation
- Command & Software
 - Command Protocol (I²C)
 - Enable/Disable
 - Telemetry Outputs
 - BMS
 - PCDU
- Electrical
 - Power Generation Input
 - External Testing Power Input
 - Voltage Rails
 - 3.3V
 - 5V
 - Battery Voltage
 - BMS Interface

Electrical Power & Distribution

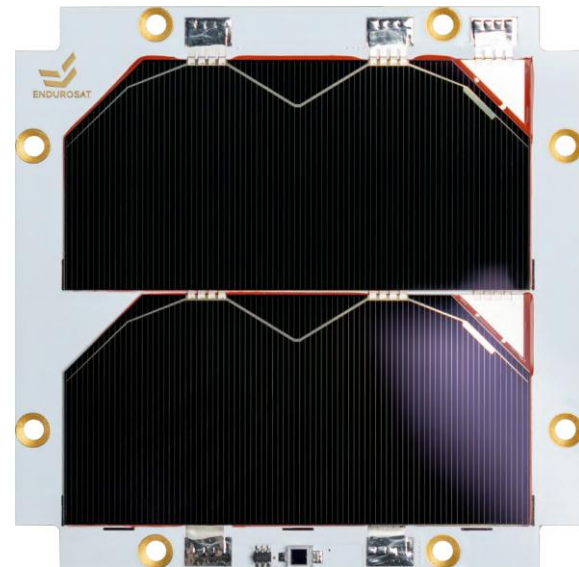
- **PicoEPS PCDU:**
 - Conditions and distributes power
- **Solar Array:**
 - Provides power generation
- **Batteries:**
 - Provides power storage
- **STM32 (Arduino):**
 - Provides the OBC with EPD Health & Status (H&S)

Block Diagram



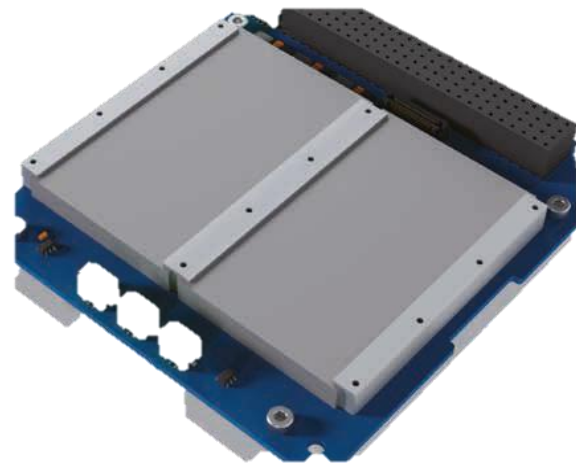
Electrical Power & Distribution

- Solar Panel - EnduroSat 1U
 - Includes: MTQ, Sun Sensor, & Temperature Sensor.
 - Reported Characteristics:
Max 2.4W, Max 0.5A, Max 4.8V.
 - PV-Cell Area: 60.36 cm²
 - Azur Space 3G30A
 - ~93.33% Degradation Factor at 125 Celsius
- Solar Array = 6 x 1U
 - Total PV-Cell Area = 0.036216 m²



Electrical Power & Distribution

- PicoEPS (TRL8)
 - Provides 3V3, 5V, 12V, & Unregulated Battery Voltage
 - 6 Panel Inputs
 - Voltage Range: 3V to 18V
 - Battery Capacity of 10Wh, 1400mAh
 - Optional battery packs for +10Wh, +1400mAh
 - Telemetry: Voltages, Currents, Battery Capacity, Temperatures, Status Registers & Error Flags



Electrical Power & Distribution

- PicoEPS Battery Module (TRL8)
 - Voltages: 7.4V Nominal, 8.4V Full Charge
 - Currents:
 - 3.04A Max Charge
 - 4A Max Discharge
 - Capacity of 10Wh, 1400mAh
 - Integrated BMS & Cell Balancing
 - Telemetry: Capacity Status & Aging Estimation
 - Temperature Control & Embedded Heaters



Electrical Power & Distribution

COTS Testing Standards:

- 1U EnduroSat Solar Panel Compliance:
 - ESA Standard ECSS E ST20-08C
 - ECSS-E-ST-10-03C
 - GEVS: GSFC-STD-7000A
 - Manufactured in ISO 14644 Class 6 Clean Room
- DHV PicoEPS Compliance:
 - GEVS: GSFC-STD-7000A
 - Manufactured in ISO 14644 Class 8 Clean Room

Electrical Power & Distribution

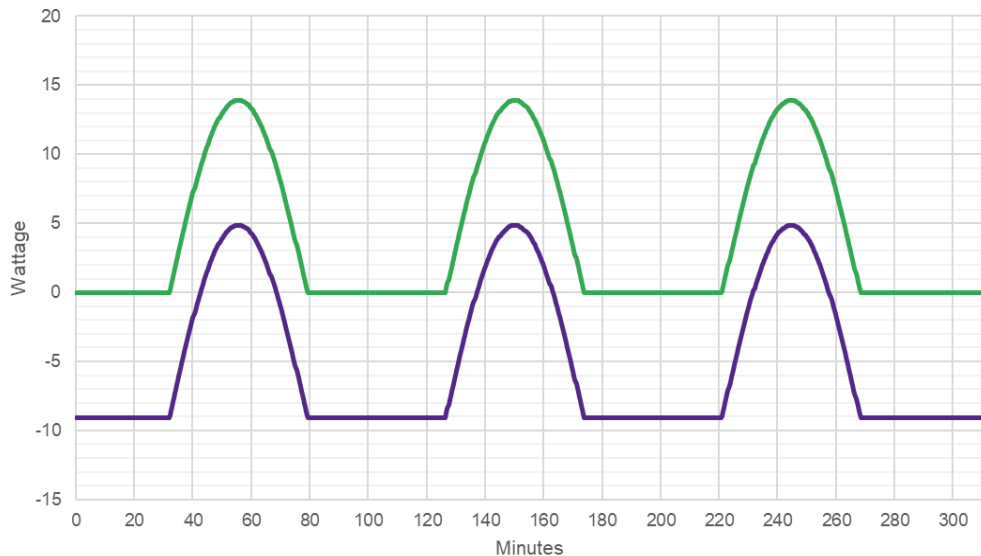
Preliminary Percentage Power Budget

Subsystem	% Allocated to Subsystem	Maximum Simultaneous Power Allocated to Subsystem (Watts)	Minimum Simultaneous Power Consumption in Idle Mode (Watts)	% of Known Operating Power Consumption
Payload	5%	0.6	0.25	3%
ADCS	10%	1.2	1.50	17%
C&DH	55%	6.5	5.30	59%
Thermal	10%	1.2	0.00	0%
Power	20%	2.4	2.00	22%
Structure	0%	0.0	0.00	0%
Total Consumption		11.9	9.05	
Max Generation			13.9	
Balance	100%	2.0	4.85	

Electrical Power & Distribution

Power Generation: ~4 Orbits, ~5 Hours

— Power Generation (W) — Power Balance (W)



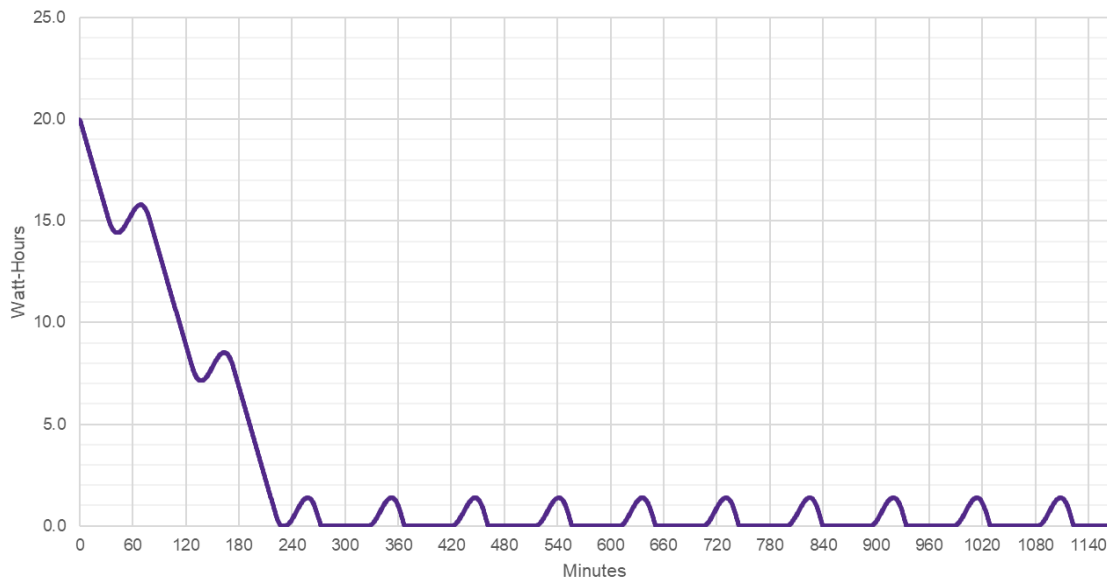
- Solar Array
 - Orientation: **Earth Nadir**
 - Size: 6U
- STK simulation uses PV-cell area to provide Effective Area
- Maximum Generation: 13.9 Watts
- Avg Generation, 3 Orbits: 8.48 Watts
- Minimal Power Consumption: 9.05 Watts
- Degradation Factor: 93.33%

$$P_{instant} = P_{incident} \cdot \text{Area}_{pv} \cdot \eta \cdot \cos(\theta_{pv}) \cdot \text{Degradation Factor}$$

$$P_{instant} = P_{incident} \cdot \text{Effective Area} \cdot \eta \cdot \text{Degradation Factor}$$

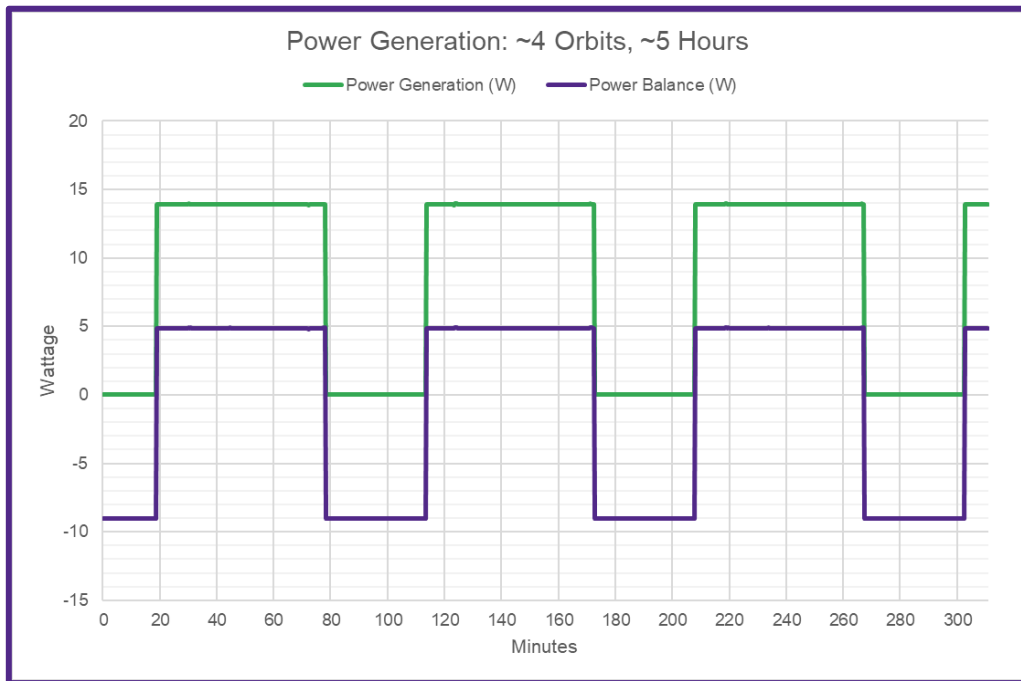
Electrical Power & Distribution

Battery Charge: 15 Orbits, ~19.5 Hours



- Solar Array
 - Orientation: **Earth Nadir**
 - Size: 6U
- Battery Capacity: 20 Wh
- Depth-of-Discharge: 100%
- Battery is dead after 227 minutes

Electrical Power & Distribution



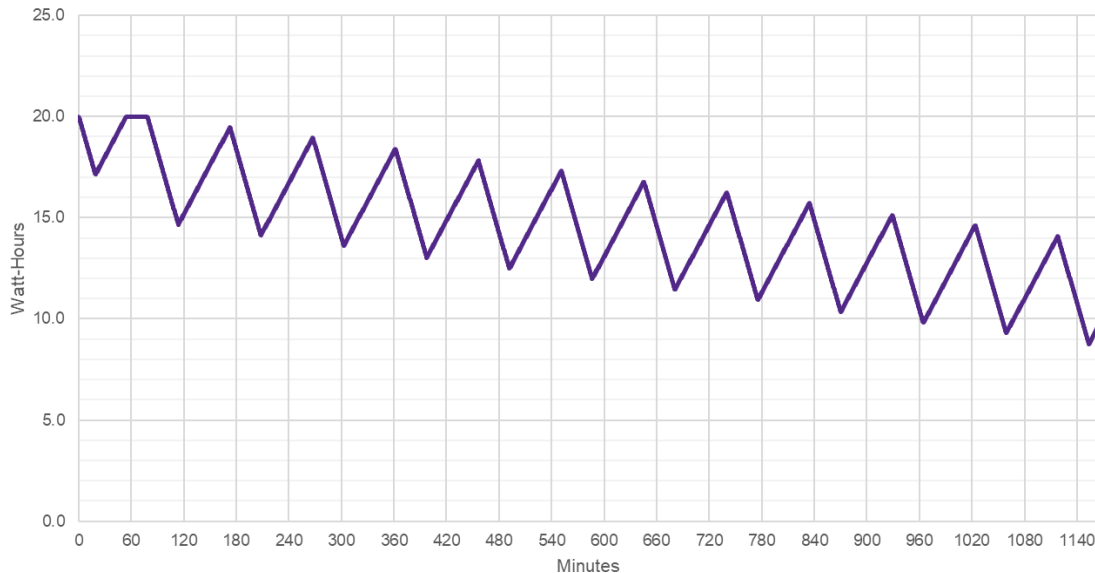
- Solar Array
 - Orientation: **Normal to Sun**
 - Size: 6U
- STK simulation uses PV-cell area to provide Effective Area
- Maximum Generation: 13.9 Watts
- Avg Generation, 3 Orbits: 8.48 Watts
- Minimal Power Consumption: 9.05 Watts
- Degradation Factor: 93.33%

$$P_{instant} = P_{incident} \cdot \text{Area}_{pv} \cdot \eta \cdot \cos(\theta_{pv}) \cdot \text{Degradation Factor}$$

$$P_{instant} = P_{incident} \cdot \text{Effective Area} \cdot \eta \cdot \text{Degradation Factor}$$

Electrical Power & Distribution

Battery Charge: 15 Orbits, ~19.5 Hours

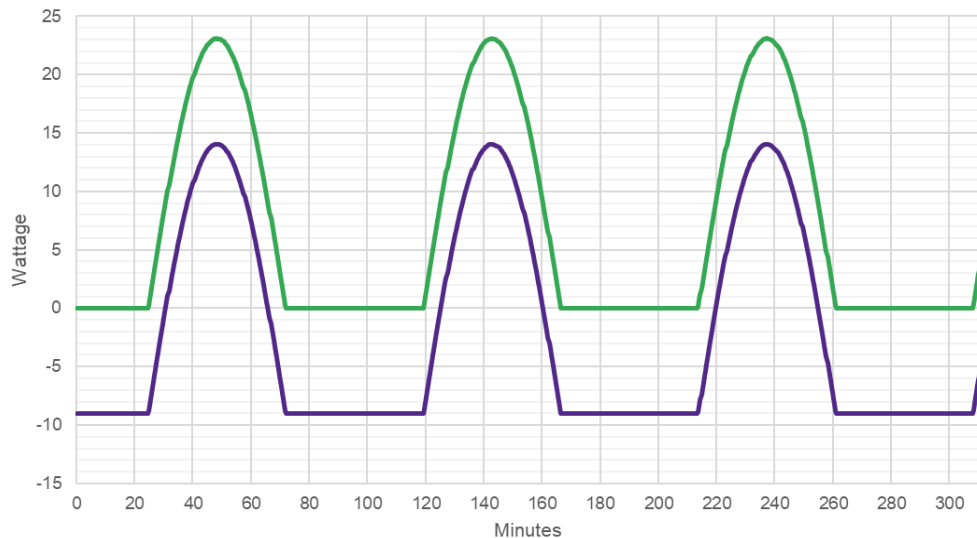


- Solar Array
 - Orientation: **Normal to Sun**
 - Size: 6U
- Battery Capacity: 20 Wh
- Battery survives for 25hrs, which is the limit of the STK data, but will likely reach 0Wh sometime after 25hrs.

Electrical Power & Distribution

Power Generation: ~4 Orbits, ~5 Hours

— Power Generation (W) — Power Balance (W)



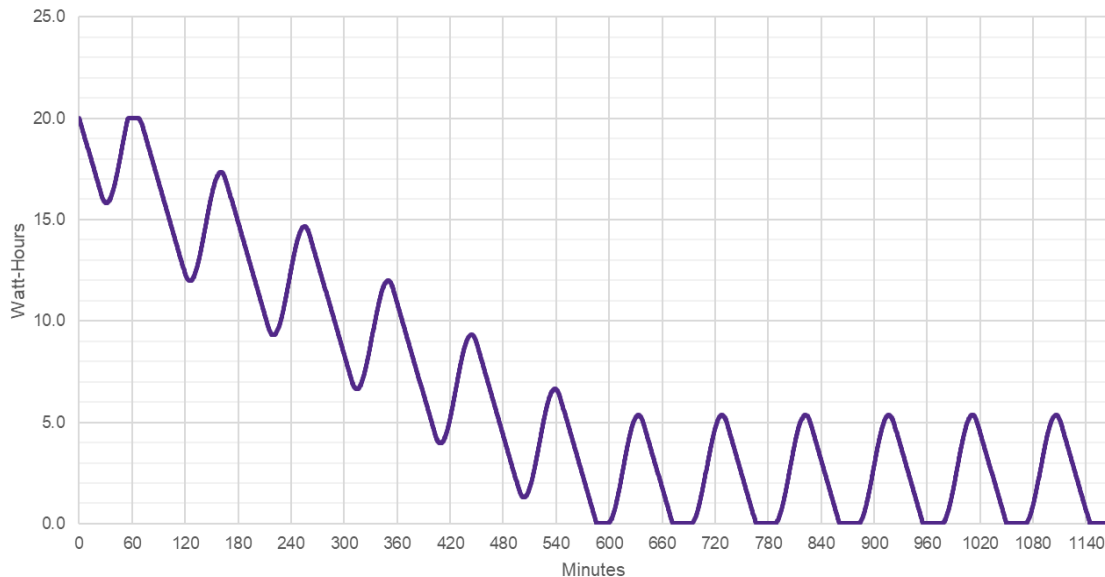
- Solar Array
 - Orientation: **Earth Nadir**
 - Size: 10U
- STK simulation uses PV-cell area to provide Effective Area
- Maximum Generation: 23 Watts
- Avg Generation, 3 Orbits: 14.5 Watts
- Minimal Power Consumption: 9.05 Watts
- Degradation Factor: 93.33%

$$P_{instant} = P_{incident} \cdot \text{Area}_{pv} \cdot \eta \cdot \cos(\theta_{pv}) \cdot \text{Degradation Factor}$$

$$P_{instant} = P_{incident} \cdot \text{Effective Area} \cdot \eta \cdot \text{Degradation Factor}$$

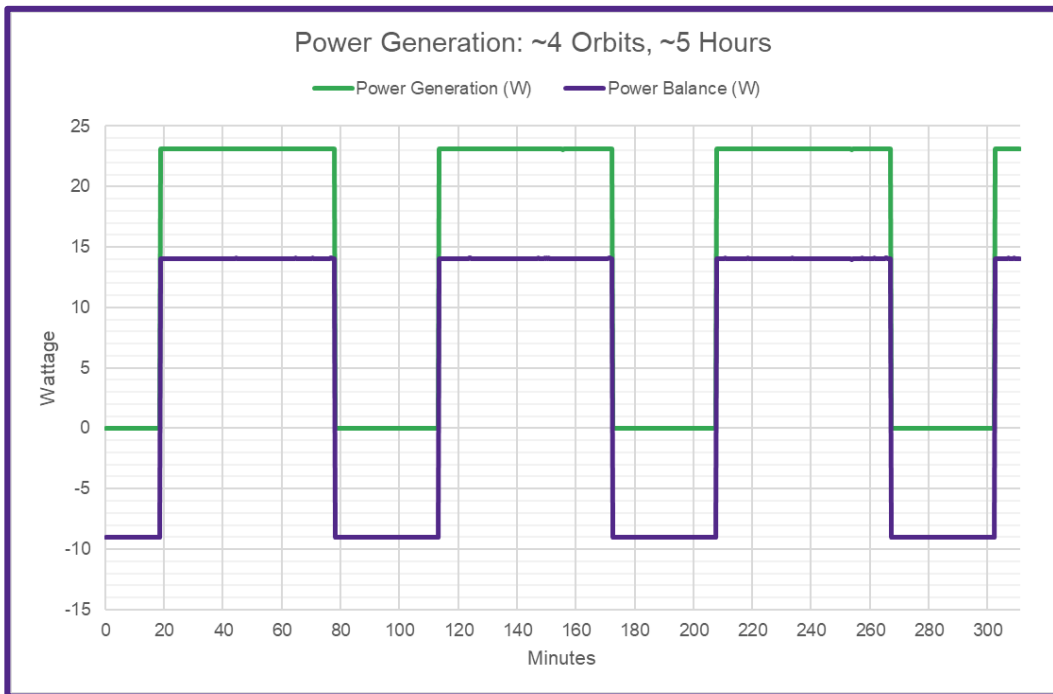
Electrical Power & Distribution

Battery Charge: 15 Orbits, ~19.5 Hours



- Solar Array
 - Orientation: **Earth Nadir**
 - Size: 10U
- Battery Capacity: 20 Wh
- Depth-of-Discharge: 100%
- Battery is dead after 585 minutes

Electrical Power & Distribution



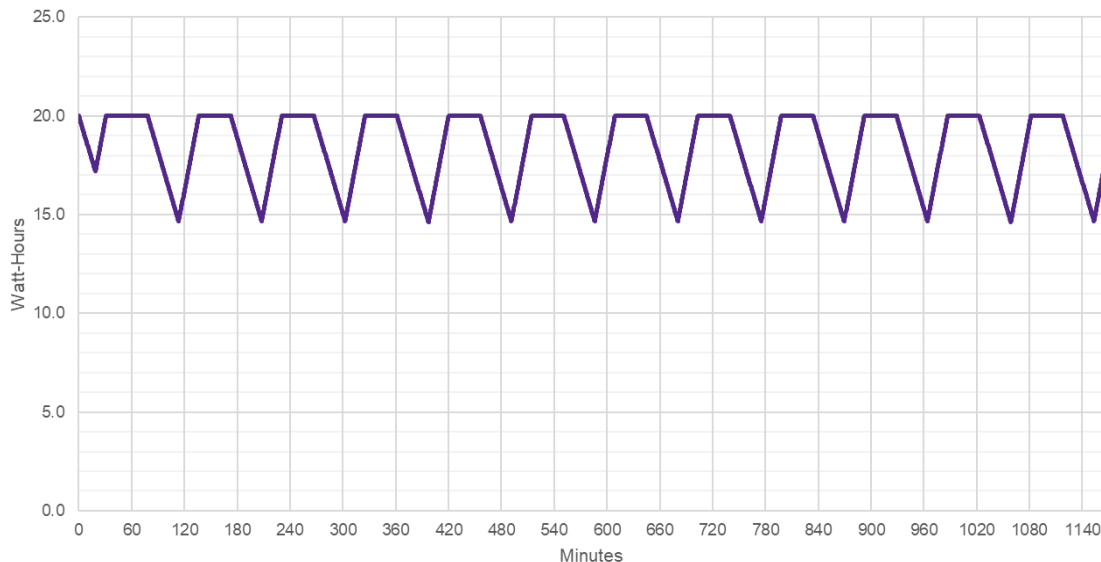
- Solar Array
 - Orientation: **Normal to Sun**
 - Size: 10U
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- Maximum Generation: 23 Watts
- Avg Generation, 3 Orbits: 14.5 Watts
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$$P_{instant} = P_{incident} \cdot \text{Area}_{pv} \cdot \eta \cdot \cos(\theta_{pv}) \cdot \text{Degradation Factor}$$

$$P_{instant} = P_{incident} \cdot \text{Effective Area} \cdot \eta \cdot \text{Degradation Factor}$$

Electrical Power & Distribution

Battery Charge: 15 Orbits, ~19.5 Hours



- Solar Array
 - Orientation: **Normal to Sun**
 - Size: 10U
- Battery Capacity: 20 Wh
- Lowest Charge: 14.62 Wh
- Average Charge: 18.42 Wh
- Depth-of-Discharge: 27%
- Battery cycle is stable.
- DoD is healthy percentage.

Electrical Power & Distribution

The preliminary design for EPD is not technically feasible, but is en route to meeting requirements, and the ability to produce design is not yet proven for NuKAT.

Moving forward, to balance the power budget, either power consumption must decrease, or power generation must increase.

Communications and Data Handling

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
CDH-3.01	C&DH + UHF TxRx Interface	The C&DH subsystem shall interface with the UHF transceiver via RS-485.	YES	YES	Compliant
CDH-3.02	C&DH + UHF Antenna Interface	The C&DH subsystem shall interface the UHF transceiver with the UHF antenna via MCX to SMP.	YES	YES	Compliant
CDH-3.03	UHF Operating Frequency	The C&DH UHF subsystem shall operate between 420-450 MHz range.	420-450 MHz	435-438 MHz	Compliant
CDH-3.04	C&DH + EPD Network Interface	The C&DH subsystem shall interface with the EPD via I2C.	YES	YES	Compliant
CDH-3.05	C&DH + Domino-2 Payload Network Interface	The C&DH subsystem shall interface with the Domino-2 payload via USB-C.	YES	YES	Compliant
CDH-3.06	C&DH + ADCS Network Interface	The C&DH subsystem shall interface with the ADCS via RS-485.	YES	YES	Compliant
CDH-3.07	C&DH + Thermal Network Interface	The C&DH subsystem shall interface with the thermal subsystem over [TBD communication protocol].	YES	TBD	Not Compliant

Path to compliance:

- Determine thermal subsystem interface

Communications and Data Handling

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
CDH-3.08	Communication Protocol	C&DH shall establish a data encryption, compression, and communication protocol w/ checksums and error correcting for transmission with ground station(s).	TBD	TBD	Not Compliant
CDH-3.10	Data Storage	The C&DH subsystem shall be capable of storing at least three * 213.12KB (the daily payload data generation) at a time in non-volatile memory over the mission lifetime.	>= 639.36 KB	320 GB	Compliant
CDH-3.11	Onboard Clock	The C&DH subsystem shall incorporate a clock that will be used to track RTC (Real Time).	YES	YES	Compliant
CDH-3.12	Downlink Data Bandwidth	The CDH subsystem shall have a data bandwidth and daily link time sufficient to downlink at least 3.25 times the expected daily payload data generation.	Bandwidth: 9600 bps Link time: 23 minutes	Bandwidth: 4800 bps Link Time: 23 minutes	Not Compliant
CDH-3.12	Link Margin	The C&DH subsystem will establish a link margin of 3 dB to ensure stable and robust communication.	>= 3dB	TBD	Not Compliant
CDH-3.13	Bit Error Rate (BER)	The C&DH subsystem will define a maximum allowable Bit Error Rate (BER) of [X] to ensure accurate communication.	TBD	TBD	Not Compliant
CDH-3.14	PNT Knowledge	GPS antenna shall collect latitude, longitude, & altitude w/i 500m, timestamped w/i 0.1s, for PLD mission and ADC tracking.	< 500 m & < 0.1 s	<= 50 m & <= 0.01 s	Compliant

Path to compliance:

- Determine RF communication protocols
- Design comms to close 9600 bps link
- Select antenna to complete link budget
- Determine acceptable Bit Error Rate

Communications and Data Handling

Level 4 Requirements:

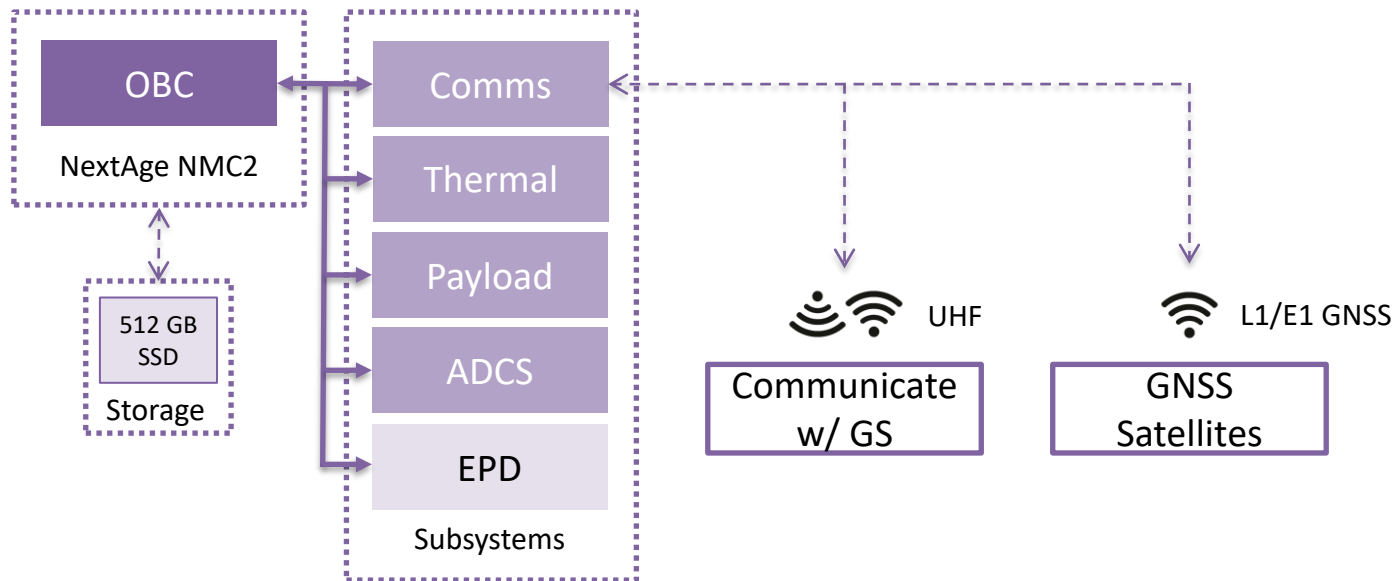
Req. ID	Title	Requirement	Required Value	Expected Value	Status
CDH-4.01	Satellite Antenna Gain	On-board satellite antenna will have a minimum gain of ["X"] dBi across its operational beamwidth.	TBD	TBD	Not Compliant
CDH-4.02	Satellite Antenna Beamwidth	On-board satellite antenna will have an operational beamwidth of ["X"] degrees.	TBD	TBD	Not Compliant
CDH-4.03	Link Budget Validation	The link budget shall demonstrate a link margin of at least 3 dB.	>= 3dB	TBD	Not Compliant
CDH-4.04	Transceiver Data Rate Cap (Tx)	The transceiver downlink data rate shall be bounded to ["X"] bps.	TBD	TBD	Not Compliant
CDH-4.05	Transceiver Data Rate Cap (Rx)	The transceiver uplink data rate shall be bounded to ["X"] bps.	TBD	TBD	Not Compliant
CDH-4.06	Modulation Implementation	The transceiver shall be capable of implementing ["X"] data modulation.	TBD	TBD	Not Compliant

Path to compliance:

- Determine values for requirements
- Continue developing link budgets as information arrives from suppliers and K-State ground station

Communications and Data Handling

Subsystem Block Diagram



Communications and Data Handling

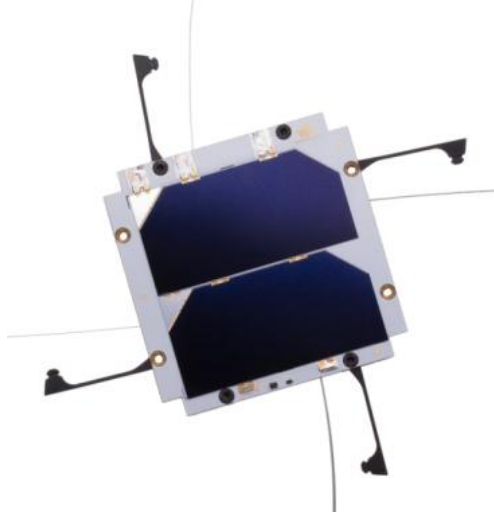
- EnduroSat UHF Transceiver II
 - Operating Temperature: -40 to 80° C
 - GMSK Supported
 - 1W transmit power
 - Mass: 94g



EnduroSat UHF
Transceiver II

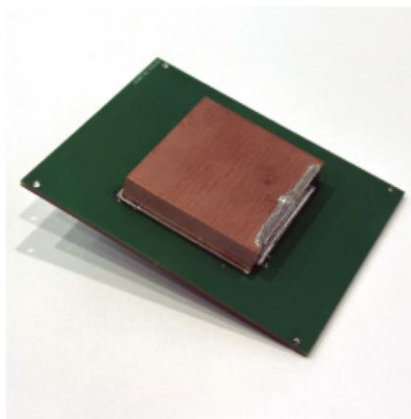
Communications and Data Handling

Antenna #1: EnduroSat
UHF Antenna III



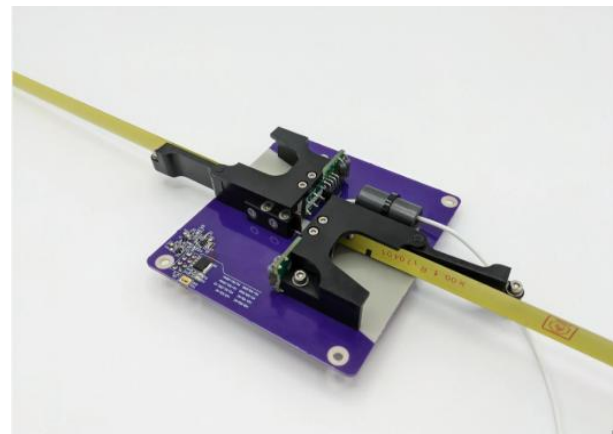
- + Easy interfacing, gain over beamwidth
- Significant attitude control required

Antenna #2: Kairo Space
UHF Patch Antenna



- + Mounts on bottom of satellite (little to no attitude control requirement)
- Less gain, linearly polarized

Antenna #3: Custom
(Image: OrcaSat UHF Antenna)



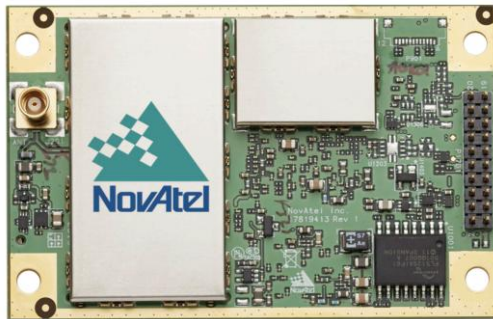
- + Customizability, cost
- Engineering challenge

Communications and Data Handling

- Modulation: GMSK (with Forward Error Correction if link allows)
- Bitrate: 9600 or 4800 bits/s
- Ground Station: K-State Radio Station (W0QQQ)
 - Power/transmission loss details pending for link budget calculations
- Preliminary (very rough) Calculations:

UPLINK		DOWNLINK	
Ground Station EIRP (rough estimated value)	25 dBW	Satellite Transmit Power	0 dBW
Free Space Path Loss	-176.5 dB	Antenna Gain (at half-power beam edge)	-3 dB
Antenna Gain (at half-power beam edge)	-3 dB	Free Space Path Loss	-176.5 dB
Received Power	-154.5 dBW	Ground Station Gain (estimated)	40 dB
Required Eb/N0 (GMSK, 10 ⁻⁵ BER)	10.6 dB	Received Power	-139.5
System Noise Temperature (at Satellite, estimated)	300K	Required Eb/N0 (GMSK, 10 ⁻⁵ BER)	10.6 dB
Channel Bandwidth	25 kHz	System Noise Temperature (at Ground, estimated)	700K
Bitrate	4800 bps	Channel Bandwidth	25 kHz
k * T _{sys}	-203.9 dB/Hz	Bitrate	4800 bps
Calculated Eb/N0	12.51 dB	k * T _{sys}	-200.2 dB/Hz
Link Margin	1.89 dB	Calculated Eb/N0	23.9 dB
		Link Margin	13.3 dB

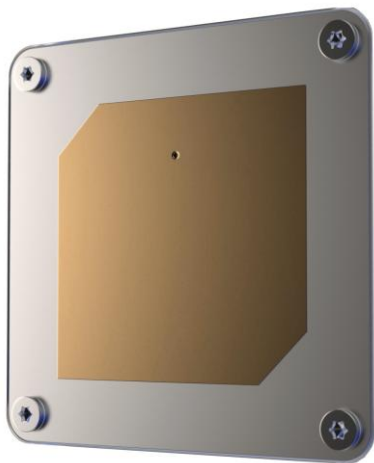
Communications and Data Handling



NovAtel OEM719
(GNSS Receiver)

- OBC Interface -> CAN
- Receiver to Antenna Interface -> MCX to SMP
- Operating Temperature: -40C - +85C
- Accuracy
 - Time: < 5 ns RMS
 - Velocity: < 0.03 m/s RMS
 - Lat/Long Positioning:
 - < 20m w/ Kalman Filter
 - < 50m w/o
- Capable of tracking many GNSS signals
 - GPS and Galileo included

Communications and Data Handling



EnduroSat GNSS L1/E1
(Passive Antenna)

- Antenna Interface -> SMP
- Operating Temperature -40C - +125C
- Operating Frequency -> L1/E1
 - L1 covers GPS
 - E1 covers Galileo
- Central Frequency = 1575.42 MHz
- RHCP
- HPBW = 88 deg
- Gain = 3.6 - 4dBi

Communications and Data Handling



GPS

- 6 Orbital planes
- 24 Satellites + Spare
- 55° Inclination Angle
- Altitude 20,200km



Galileo

- 3 Orbital planes
- 27 Satellites + 3 Spares
- 56° Inclination Angle
- Altitude 23,616km

GNSS next steps:

- Orbit simulations
- Link budget generation
- Further trade studies

Communications and Data Handling



NextAge NMC2
(+3 daughter boards)

- Designed with radiation best practices
 - Automotive grade parts
 - Component placement
 - Rigorous testing
- Operating Temperature: -40C - +85C
- Processor Information
 - 1 FPGA
 - 4 Risc V Application Cores
 - 1 Risc V Monitor Core

} Programmable via TriCortex and Verilog
} Programmable via C/C++ and Python
- 400+ GPIO Pins
- 500 GB SSD → 320 GB w/ ECC
- 16 GB DDR4 RAM

Communications and Data Handling

Data Per Domino Sample			
Field	Type	Bits	Bytes
Background Detector 1	uint32	32	4
Background Detector 2	uint32	32	4
Fast Detector	uint32	32	4
Thermal Detector	uint32	32	4
Temperature	float (signed)	32	4
Latitude	float (signed)	32	4
Longitude	float (signed)	32	4
Time (UNIX)	uint32	32	4
Altitude	float (signed)	32	4
Earth Normal Angle	int8	8	1
Total Data Per Sample	---	296	37

- Payload Data
 - 4 Detector Counts per Domino
 - Information for Data Validation
 - Temperature
 - Latitude
 - Longitude
 - Time
 - Altitude
 - Earth Normal Angle

Communications and Data Handling

Data Calculations

Assuming 23 minute link time	
Downlinked bits per day (9600 bps)	13,248,000
Downlinked KB per day (9600 bps)	1,656.00
Downlinked bits per day (4800 bps)	6,624,000
Downlinked KB per day (4800 bps)	828.00

Frequency (Hz)	Sample Period (s)	1 Domino (KB / day * 3.8 mrg)	4800 bps Compliant?	9600 bps Compliant?	2 Dominos (KB / day * 3.8 mrg)	4800 bps Compliant?	9600 bps Compliant?
0.017	60	202.46	Yes	Yes	404.93	Yes	Yes
0.033	30	404.93	Yes	Yes	809.86	Yes	Yes
0.05	20	607.39	Yes	Yes	1,214.78	No	Yes
0.1	10	1,214.78	No	Yes	2,429.57	No	No
0.2	5	2,429.57	No	No	4,859.14	No	No
0.5	2	6,073.92	No	No	12,147.84	No	No
1	1	12,147.84	No	No	24,295.68	No	No
2	0.5	24,295.68	No	No	48,591.36	No	No
5	0.2	60,739.20	No	No	121,478.40	No	No
10	0.1	121,478.40	No	No	242,956.80	No	No

*3.8 margin -> 30% general margin + 2.5 encoding margin multiplier

Communications and Data Handling

The preliminary design for CDH is moving towards technical feasibility, en route to comply with requirements, and is expected to be producible for NuKAT.

Flight Software

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.6 Flight Software Requirements					
FSW-3.01	Telemetry Collection	FSW shall collect telemetry from all hardware onboard NuKAT.			Compliant
FSW-3.02	Event Logging	FSW shall maintain a log of software events related to command execution, resets, and software errors.			Compliant
FSW-3.03	Ground Software Interface	FSW shall be able to receive, decrypt, and interpret commands from the Ground Station Software.			Compliant
FSW-3.04	Power Reset Actions	FSW shall immediately command NuKAT back to a normal state following a power reset.			Compliant
FSW-3.05	Health Data Transmission	FSW shall relay the instantaneous health status of NuKAT to the ground upon requesting command.			Compliant
FSW-3.06	Neutron Count Collection Time	FSW shall be capable of tracking what time and date each neutron count reading is collected.			Compliant
FSW-3.07	Location Tracking	FSW shall be capable of tracking NuKAT's physical position in orbit to appropriately associate neutron count readings.			Compliant
FSW-3.08	Operational Modes	FSW shall be capable of manipulating and tracking the operational modes for NuKAT and change between modes when necessary.			Compliant
FSW-3.09	Error Checking/Correction	FSW shall be capable of detecting and correcting errors caused by SEUs and lossy data transmission			Compliant

Flight Software

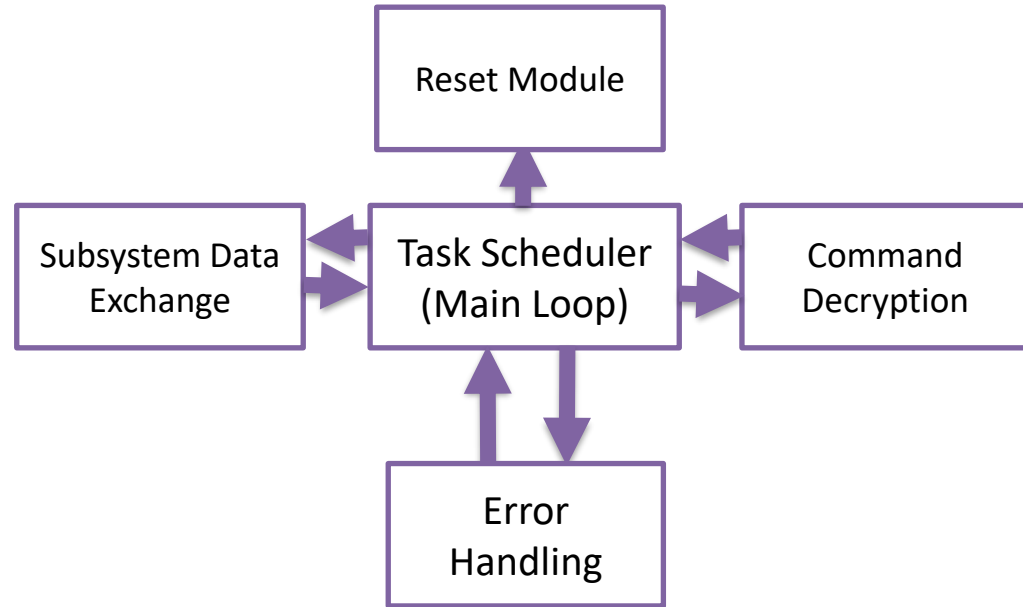
Level 4 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.6 Flight Software Requirements					
FSW-4.01	Onboard Computer Utilization	The flight software shall require under "x"% of full OBC Utilization.			Compliant
FSW-4.02	Software Code Count	The flight software should be written within the range of "X" and "Y" lines of code.			Compliant
FSW-4.03	Decryption Algorithm	The flight software shall implement an AES-256 decryption algorithm.			Compliant
FSW-4.04	Software Package	The flight software shall be implemented using the C, C++, Python, and TriCortex real-time AI languages.			Compliant
FSW-4.05	Software Structure	The flight software shall utilize NASA's core Flight System (cFS) framework.			Compliant

Flight Software

Block Diagram

- Interfaces handled by C&DH team with On-board Computer

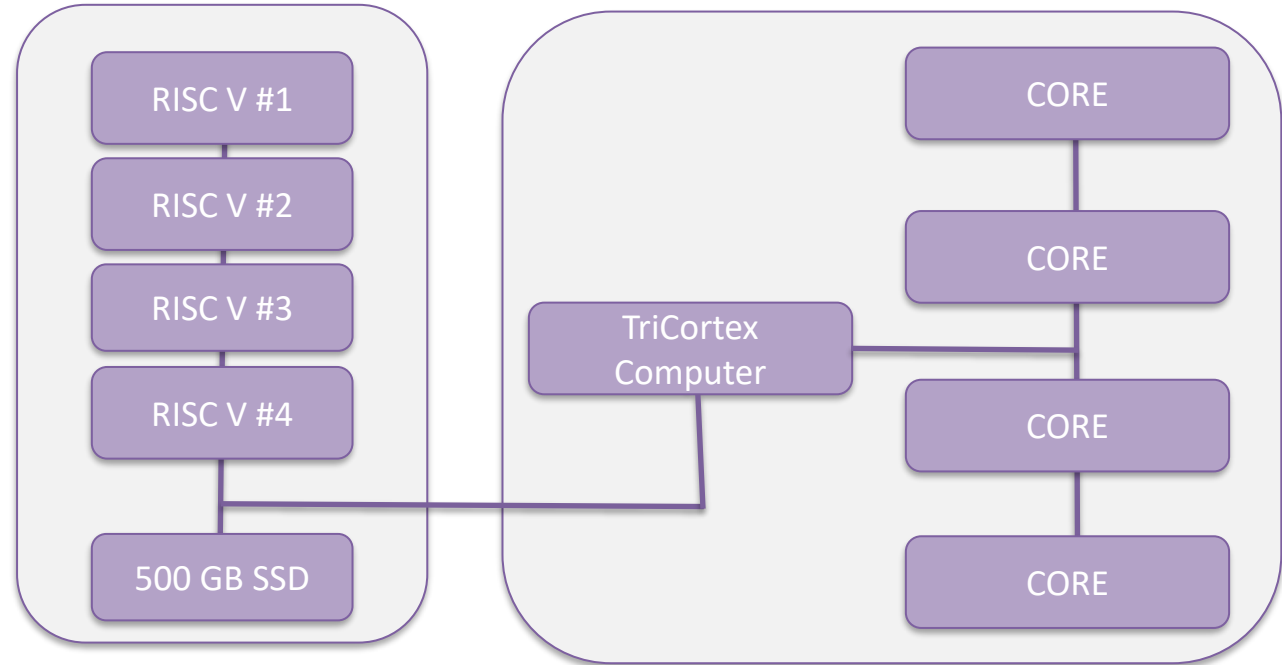


Flight Software

- Software implementation
 - C, C++
 - Python scripts
 - TriCortex real-time AI language from Nextage
- Utilization NASA's core Flight System (cFS) framework
- Early estimate of 10,000 lines of developed code

Software Architecture

- RISC V processors
 - core Flight System
- TriCortex Computer cores
 - Subsystem hardware interfacing
 - Co-processing
- Data exchanged through advanced high-performance bus (AHB)



Flight Software

NASA core Flight System (cFS)

- cFS enables easy reuse, rapid development, and portability
 - Dynamic runtime environment
 - Layered architecture
 - Component-based design
- 3 main components
 - Platform Support Package
 - Operating System Abstraction Layer
 - Core Flight Executive
- Used by 40+ small to large missions, including the Roman Space Telescope.

Flight Software

- Industry standards and compliance
 - NASA NPR 7150.2 – Software Engineering Requirements
 - NASA-STD-8739.8 – Software Assurance
 - NASA core Flight System

Feasibility Statement

From this, we believe that the Flight Software subsystem is technically feasible, meets our requirements, and is producible

Structures

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.7 Structure Requirements					
STR-3.01	Coordinate System	An X-Y-Z (X - velocity, Y- sun, Z- Earth) coordinate system for the NuKat shall be developed on a millimeter scale to provide a method for referencing any location	X-velocity, Y-sun, Z-earth	N/A	Compliant
STR-3.02	Satellite Dimensions	Satellite shall be 10cm by 227cm in the x-y and y-z face, and shall be 11.9cm by 11.9cm (maximum) in the x-z face within uncertainty +/- .01cm	227 cm (Y), 119 cm (X), 119 cm (Z)		Partial Compliance
STR-3.03	Access Port (USB)	The chassis shall include an interfacing port			Partial Compliance
STR-3.04	Deployment Sizing	NuKat shall conform to a 2U launch dispenser, using rails or tabs.	RAIL/TAB	RAIL	Compliant
STR-3.05	Deployment Rail Material	The NuKat rail surfaces that contact the NRCSD guide rails shall have a hardness equal to or greater than hard-anodized aluminum.	Rockwell C 65-70		Compliant
STR-3.06	Rail Texture	NuKat rail surfaces shall have a surface roughness of less than or equal to 1.6 micrometers.	1.6 micrometers		Compliant
STR-3.07	Exterior Dimensions	The rails of the NuKat shall be the only parts to touch the deployment mechanism (rails). The rails shall extend no less than 2 millimeters from the exterior faces of NuKat.	2 millimeters		Compliant
STR-3.08	Rail Alignment	The rails of NuKat shall be coplanar within +/- 100 micrometers	100 micrometers		Compliant
STR-3.09	Rail Dimensions	The rail dimensions shall conform to the deployer's specifications while simultaneously meeting the NuKat dimensions	100 mm x 100 mm		Compliant
STR-3.10	Rail Orientation	The rails shall run parallel to the Y-axis, one per corner of NuKat, which will allow the payload to slide along the rail interface of the deployer.	4 rails		Compliant
STR-3.11	Neutron Detector Mounting	Detectors shall be mounted securely with restrained movement and no obstructions to it collecting its data at its proper rate			Partial Compliance

Structures

Level 4 Requirements:

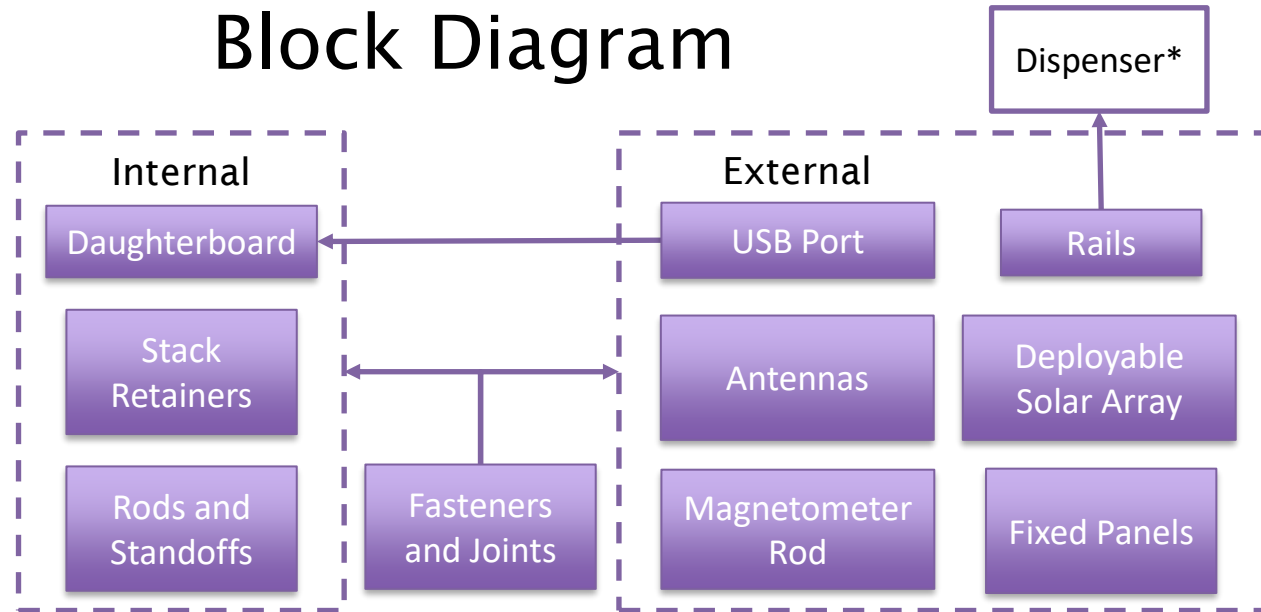
Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.7 Structure Requirements					
STR-4.01	Chassis Dimensions	The chassis shall be of maximum dimensions of 100mm X 100mm in the Z by Y plane, 227mm on the X axis	100mm x 100mm x 227mm		Compliant
STR-4.02	Chassis Material	The material of the chassis shall be aluminum 6061 with a density of 2.7 g/cm ³	2.7 g/cm		Compliant
STR-4.03	Rail Material and Hardness	The material of the rails shall have a hardness equal or greater than the hardness of hard anodized aluminum. The material will be aluminum 6061	Rockwell C 65-70		Compliant
STR-4.04	Rail Alignment	The rails shall be parallel to the Y axis within +/- 100 micrometers. Rails shall extrude no less than 2mm from the exterior surface of the satellite	+/- 100 micrometers		Compliant
STR-4.05	Exterior Interface	The chassis shall contain an external interface that is USB-C compatible	YES		Partial Compliance
STR-4.06	Payload Mounting	The payload shall be mounted normal to the Z axis with 'X' fasteners of 'type'			Partial Compliance
STR-4.07	Fasteners Material	Fasteners shall be of a 18-8 stainless steel	18-8 Stainless Steel		Compliant
STR-4.08	Fasteners Dimensions	Fasteners shall be in the range of M1-M3 bolt size and will fit with +/- .025mm standard tolerance	M1-M3		Compliant
STR-4.09	Radiation Shielding	Paneling shall consist of hard anodized aluminum 6061 of 1.5mm thick, being able to withstand 7 kRad	1.5mm		Compliant

Structures

Interfaces

- External
 - Rails contacting dispenser
 - External components with fixed and deployed mounting, paneling
 - USB-C port integrated
- Internal
 - M1-M3 fasteners for mounting
 - Retainers and rods for component mounting and installation

Block Diagram

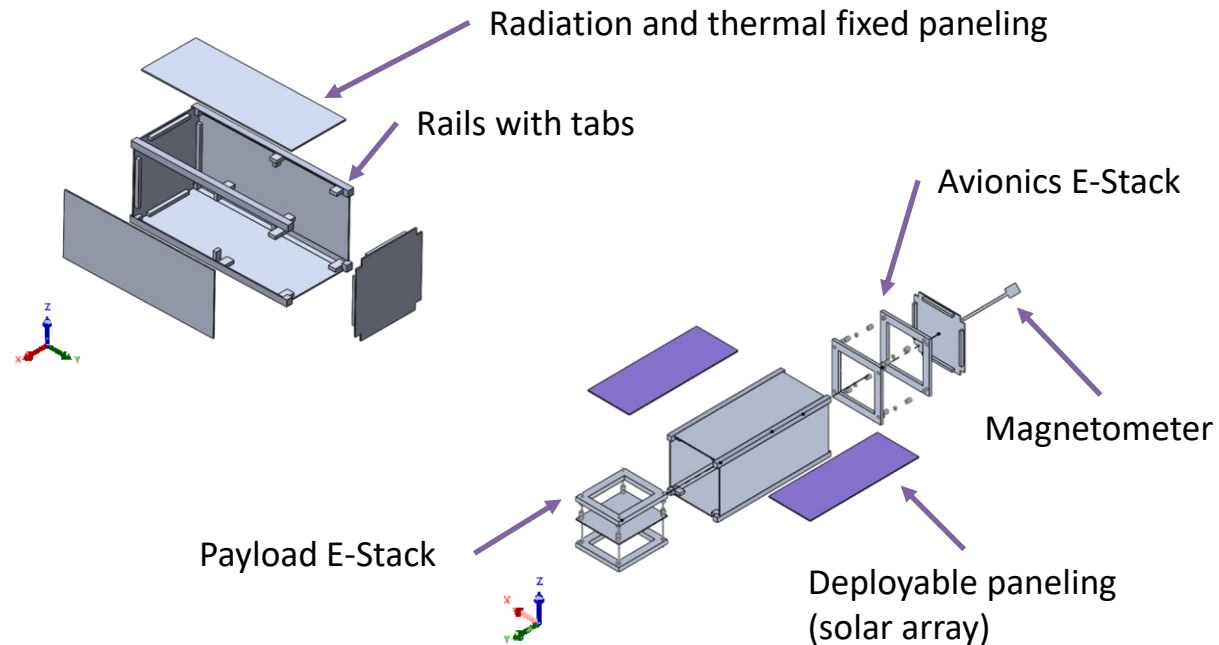


*Dispenser provided by launch provider

Structures

Configuration:

- Rails for deployment
- Aluminum paneling for thermal and radiation and external mounting
- Tabs manufactured for internal mounting
- Four internal retainers to contain Payload E-stack and Avionics E-stack



Structures

Component	Manufacturer	Number	Specifications	Mass(g)
Chassis	K-State TDI	1	6061 Aluminum	764.56
Magnetometer Arm	K-State TDI	1	6061 Aluminum	3.92
Threaded Rod	McMaster Carr	4	18-8 Stainless Steel 4-40	0.5 (4)
Fasteners	McMaster Carr		18-8 Stainless Steel (M1-M3)	~23
Standoffs	McMaster Carr		3/8", 1/4", 4-40	~25

Structures

Subsystem	Mass (g)
Payload	250.15
ADCS	159
CDH	215.32
EPD	518
STR	818.48
30% margin	588.28
Estimated total	2549.23 g

MOI Matrix: (high level)

	$x \text{ (g*cm}^2\text{)}$	$y \text{ (g*cm}^2\text{)}$	$z \text{ (g*cm}^2\text{)}$
x	99075.16	157.11	19.63
y	157.11	37541.05	-1226.24
z	19.63	-1226.24	100167.84

Center of Mass: (high level)

$$x = -0.01 \text{ cm}$$

$$y = 11.06 \text{ cm}$$

$$z = 0.07 \text{ cm}$$

Structures

- Manufacturing standards
 - We will confirm that K-State TDI is able to produce our custom components in compliance with ISO-14644 Class 8 cleanliness standard
- Launch Regulations
 - The biggest regulator will be our launch provider and their standards for vibration, material hardness, dimensions, toughness and radiation hardness

Feasibility Statement

Based on our current design and information, we believe that this is technically feasible. It will require a customized design of the chassis components, magnetometer and potentially UHF antenna, but they are manufacturable.

Thermal

Level 3 Requirements:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.8 Thermal Requirements					
THM-3.01	Thermal Control System Volume	The subsystem (heaters and heat strips/insulation) shall not take up more than 10.5% of the NuKAT's total volume	10.50%	8.41%+straps	Compliant
THM-3.02	Thermal Control System Mass	The physical mass of the thermal subsystem (heaters, heat strips, insulation) shall not exceed 100 grams	100g	75.4g	Compliant
THM-3.03	Thermal Control System Energy Usage	Energy usage per orbit shall not exceed x Wh. This includes active thermal regulation (RTDs, control system for heater) inside of EPS and resistive heaters.	>2.1 W simultaneously	TBD	Partial Compliance
THM-3.04	Thermal Control System	The satellite shall be maintained within -20 to 50 C temperature range (hard cutoff range) based upon the operational requirements of each (narrowed in op range)	-20-50	TBD	Partial Compliance
THM-3.05	Thermal Control - Radiator	The radiator shall maintain a temperature within a -20 to 40 C range (passive system, heat strips)	-20-40	TBD	Partial Compliance
THM-3.06	Thermal Control - Payload	the thermal control system shall maintain the payload temperature with a -20 to 55 C range	-20-55	TBD	Partial Compliance
THM-3.07	Thermal Control System Measurement	Temperature shall be recorded within x intervals of each operating component within the cubesat and relayed to controls	1 min	TBD	Compliant
THM-3.08	Thermal Control System Derivative	Temperature of each component should not change with respect to time at a rate greater than x degrees per x time interval. Accomplished with balance of control systems, radiation, and heat generation	TBD	TBD	Partial Compliance

Thermal

Level 4 Requirements:

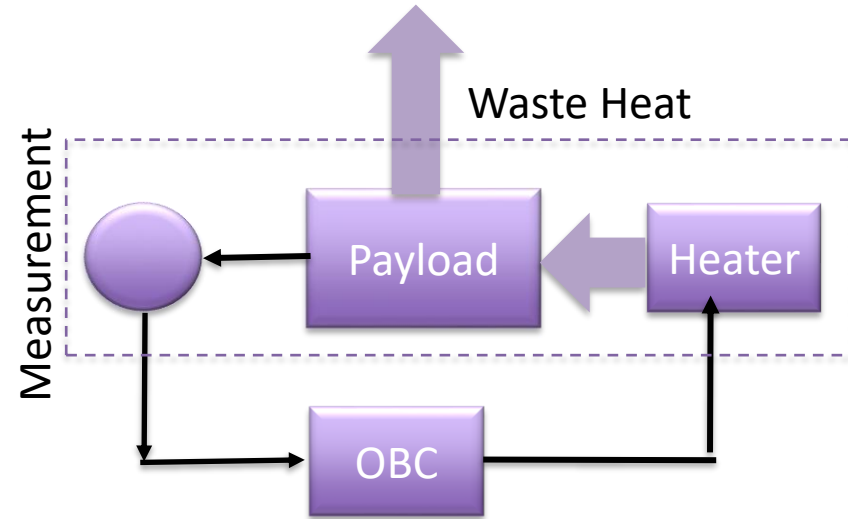
Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.8 Thermal Requirements					
THM-4.01	Thermistor Operating Temperature	The Thermistors shall operate with a temperature range of "X °C" to "Y °C"	-20-55	TBD	Partial Compliance
THM-4.02	Thermistor Power Consumption	The thermistors shall operate with a power consumption no greater than "X" W	>2 W	TBD	Partial Compliance
THM-4.03	Thermistor Reading Interval	The thermistors shall provide readings within a time interval of "X" seconds	60	TBD	Partial Compliance
THM-4.04	Thermistor Reading Accuracy	the thermistor reading shall be accuracate within +/-"X" %	2%	TBD	Partial Compliance
THM-4.05	Thermistor Mounting Method	the thermistors shall be mounted to ensure proper operation	yes	yes	Compliant
THM-4.06	Thermistor Quantity	there shall be a sufficient number of thermistors to monitor critical systems	TBD	TBD	Not Compliant
THM-4.07	Thermistor/OBC communication	the thermistor shall be able to communicate with the OBC	yes	yes	Compliant
THM-4.08	Heater Operating Temperature	The heaters shall be operable within a termperature range of "X °C" to "Y °C"	-20-55	TBD	N/A
THM-4.09	Heater Power Consumption	the heaters shall consume no greater than "X" W	TBD	TBD	N/A
THM-4.10	Heater Response Time	the heaters shall be cabable of providing sufficient heat within a set time	TBD	TBD	N/A
THM-4.11	Heater Mounting Method	the heaters shall be mounted to ensure proper operation	yes	TBD	N/A
THM-4.12	Heater size	the heaters shall be of adaqueate size to provide sufficient heat to components	TBD	TBD	N/A

Thermal

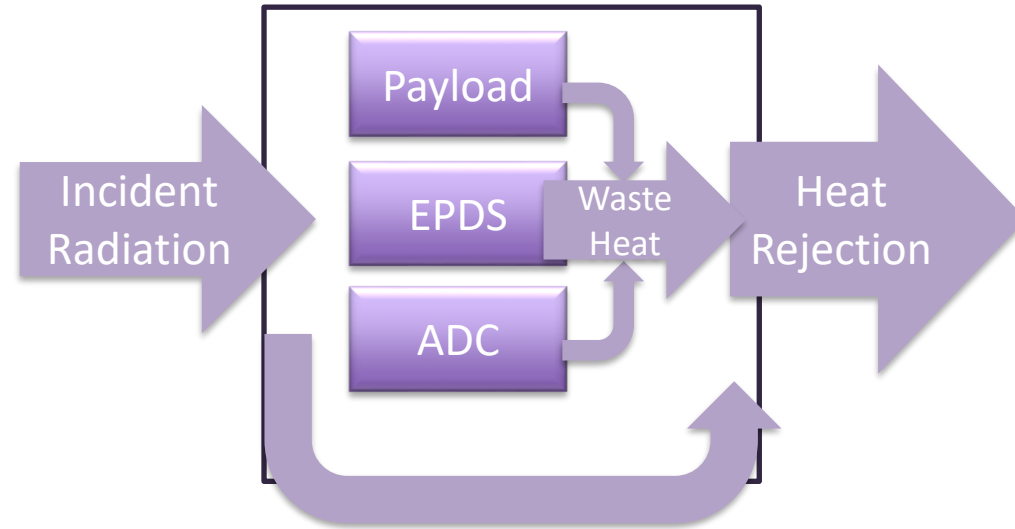
- Internal
 - Heat sinks to radiators
 - Components to heat sinks
- External
 - Any component with waste heat (EPDS, Comms, etc)
 - Structure:
 - Anchoring
 - EPDS:
 - Heater Power
 - FSW/C&DH:
 - Measurement processing, Heater on/off

Thermal

Component Level

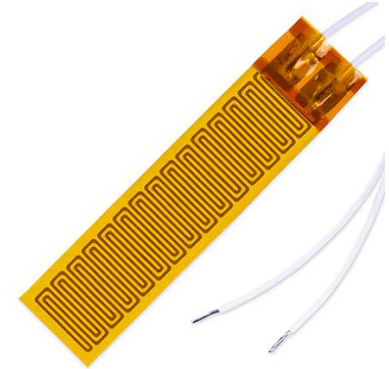


Satellite Level



Thermal

- Thermistors for temperature measurement
- Heaters and other components TBD



Device	Collins 0118MM	TE 44900	Variohm 311P18
Op Temps (C)	-269-400	-55-90	-55-90
Flight Qualified	✓	✓	✓
Interchangeability	±1.0%	±0.1 C	±1.0 C

Thermal

- Standards:
 - MIL-PRF-23648
 - MIL-T-23648
 - ASTM-e595
 - ESA 4006
- Export Controls:
 - Expect no issues

Thermal

This is technically feasible and producible,
but does currently not meet requirements

GROUND SYSTEM

Noah Amundson, Andrew Osterhaus



Ground System

Level 2 Compliances:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
2.4 Ground Station Requirements					
GS-2.01	Ground Station Communication	Ground station shall be able to communicate with a NuKAT. NuKAT shall operate between (420-450 MHz) range which falls under the UHF band requirement.	430-440 MHz	420-450 MHz	Compliant
GS-2.02	Tracking	NuKAT shall be tracked by the ground station within a 1-5% pointing accuracy.	YES	YES	Partial Compliance
GS-2.03	Mission Data	NuKAT shall interface with mission data users.	YES	YES	Partial Compliance

Ground System

Level 3 Compliances:

Req. ID	Title	Requirement	Rationale	Parent Requirement	Status
GSH-3.01	Location	The ground system shall have a primary and secondary location that allows for clear communication at TBD time intervals. Primary location shall be located within campus boundaries.	Primary location shall be KSU radio station. A secondary location is needed in case of equipment failure or power loss.	GS-2.01	Compliant
GSH-3.02	Contact Time	The ground system shall be aware of the satellite's location to achieve communication for 5.8 minutes.	Contact time needs to allow for required data to be transmitted and recieved.	GS-2.02	Partial Compliance
GSH-3.03	Security	The ground station shall be in a secure location with restricted access.	Ground station hardware can be sensitive equipment that should only be accessed by informed and trained personell.	GSH-3.06	Compliant
GSH-3.04	Power	The ground station shall be able to provide "x" amount of power to components.	Certain components may have specific power requirements to efficiently and properly function.		Partial Compliance
GSH-3.05	Licensing	The ground station shall have licesning to operate UHF communication bands and meet FCC requirements.	Experimental radio license is common for universities. Capable of supporting amateur radio station.	CDH-3.01, CDH-3.02	Partial Compliance
GSH-3.06	Training	Personell directly operating the ground station shall have expert level training on its systems.	Operation of the ground station will be needed.		Partial Compliance
GSH-3.07	Operational Lifetime	The ground station shall have an operational lifetime that meets or exceeds mission requirements.	NuKAT has no way to deliver data or be monitored if the ground station is inoperable.	PHAT-1.05	Partial Compliance

Ground Station Hardware

Level 4 Compliances:

Req. ID	Title	Requirement	Rationale	Parent Requirement	Status
GSH-4.01	FCC Specific Liscensing	Students planning on working the ground station must pass the FCC exam offered through Jeff Sheldon at KSU Manhattan.	Students will not be able to help operate the ground station without passing the FCC exam.	GSH-3.05	Partial Compliance
GSH-4.02	Satellite Antenna and Gain	Ground station antenna must be able to comunicate and operate in UHF bands between 430-440 MHz and with an antenna gain between 10-15 dBI.	Antenna is needed to communicate with satellite and antenna gain will affect the link budget.	GS-2.01	Compliant
GSH-4.03	SDR Transceiver	Must have UHF transmit frequencies within "x" MHz and recieve frequencies within "x" MHz.	Faciliate communications to and from the satellite.	GS-2.01	Partial Compliance
GSH-4.04	Rotator	Rotator will be needed for tracking NuKat in orbit.	Assists antenna in tracking the satellite in space.	GS-2.01	Partial Compliance
GSH-4.05	Contoller	TBD if controller will be needed.	Contains the software needed to track the satellite through space.	GS-2.02	Partial Compliance
GSH-4.06	Amplifier	Select the correct low-noise amplifier that meets the mission requirements.	A low-noise amplifier is required to take a low signal and increase its power. This is esppcially crucial as signals are traveling through space.	GS-2.02	Partial Compliance
GSH-4.07	Satellite Antenna Beamwidth	Ground station antenna will have an operational beamwidth of X degrees	Size of beamwidth affects attitude control.	GS-2.01	Partial Compliance
GSH-4.08	Computer	TBD if computer will be needed.	Computer will be needed to view the data through the software.	GS-2.03	Partial Compliance
GSH-4.09	Cabling	Cabling will be needed to connect hardware together.	Hardware will need to be connected together and with computer.	GS-2.03	Partial Compliance
GSH-4.10	Liscensing	Need to obtain FCC, IARU, FAA, and ITU.	Liscensing is needed to be able to launch satellite.	GSH-3.05	Partial Compliance
GSH-4.11	Ground Station Storage	Expected storage needed for 6 years is between 1.01×10^{10} bits 2.03×10^{10} bits (4800-9600 BPS)	Need to be able to hold enough storage at the ground station for the lifetime of the mission.	GS-2.03	Partial Compliance
GSH-4.12	Command Encryptor	Needed to ensure data isn't lost or destroyed during the encryption process.	Protects data from being lost or destroyed.	GS-2.03	Partial Compliance

Ground Station Hardware

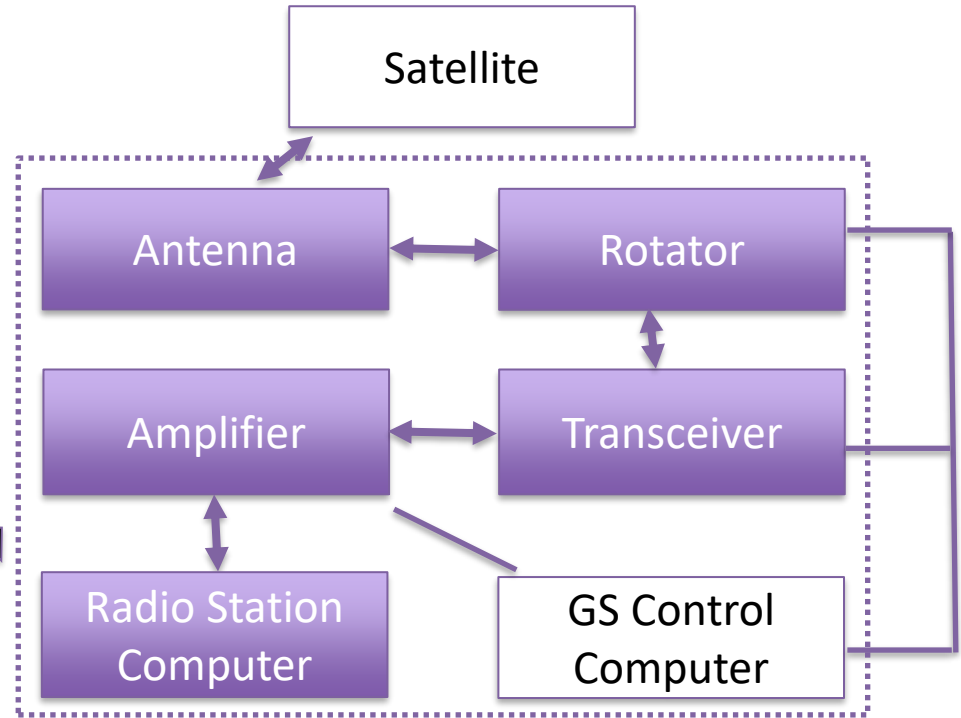
External Interfaces

- Operators
- Ground Station

Internal Interfaces

- Database

Stored at KSU
Radio Station



Ground Station Licensing

Licensing

- Federal Communications Commission (FCC)
- International Amateur Radio Union (IARU)
- Federal Aviation Administration (FAA)
- International Telecommunication Union (ITU)

Ground Station Software

Level 3 Compliances:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
3.11 Ground Software Requirements					
GSW-3.01	Network	The ground software shall establish a network connection between what is hosting it and the ground station. TBD if the ground station has network sockets	Yes/No	Yes	Partial Compliance
GSW-3.02	File Transfer	The ground software shall reliably and securely be able to transfer information to the satellite.	Yes/No	Yes	Partial Compliance
GSW-3.03	Communication Protocol	The ground software shall establish an encryption protocol for uplinking commands during communications. Need of a compression protocol is TBD on data estimates	Yes/No	Yes	Compliant
GSW-3.04	Data Processing	The ground software shall be able to process and display data in a meaningful manner	Yes/No	Yes	Compliant
GSW-3.05	Storage/Data Pipeline	The ground software shall store and retrieve data from a database	Yes/No	Yes	Compliant
GSW-3.06	Distribution	The ground software shall distribute process data to users.	Yes/No	Yes	Compliant
GSW-3.07	Alert System	The ground software shall alert a user or admin when necessary	Yes/No	Yes	Compliant



Ground Station Software

Level 4 Compliances:

Req. ID	Title	Requirement	Required Value	Expected Value	Status
4.11 Ground Software Requirements					
GSW-4.01	Host	The ground station software shall be hosted on a server in the Kansas State Network	Yes/No	Yes	Not Compliant
GSW-4.02	Software Package	The ground station software will be based on the open-source software Cosmos and additional functionality will be added on as needed.	Yes/No	Yes	Compliant
GSW-4.03	InfluxDB	The ground station software shall use InfluxDB as the database for storing telemetry data and neutron counts	Yes/No	Yes	Compliant
GSW-4.04	AES-256	The ground station software shall use Advanced Encryption Standard to encrypt uplink commands	Yes/No	Yes	Compliant
GSW-4.05	Data Distribution	The ground station software shall utilize the Ruby Mail package to send out data on request.	Yes/No	Yes	Compliant
GSW-4.06	Alert System	The ground station software shall build an alert system into COSMOS that will alert an admin or registered individual to anomalies or extreme events.	Yes/No	Yes	Compliant

Ground Station Software

Non-Compliant:

- Host
 - We are not currently cleared to host our software on the K-State network
- Mitigation
 - Formal request from our team
 - We will need to stress the importance of our mission, the value for students, and the value of the data we will return
 - Student-led team
 - Research
 - Interaction with Radio Club

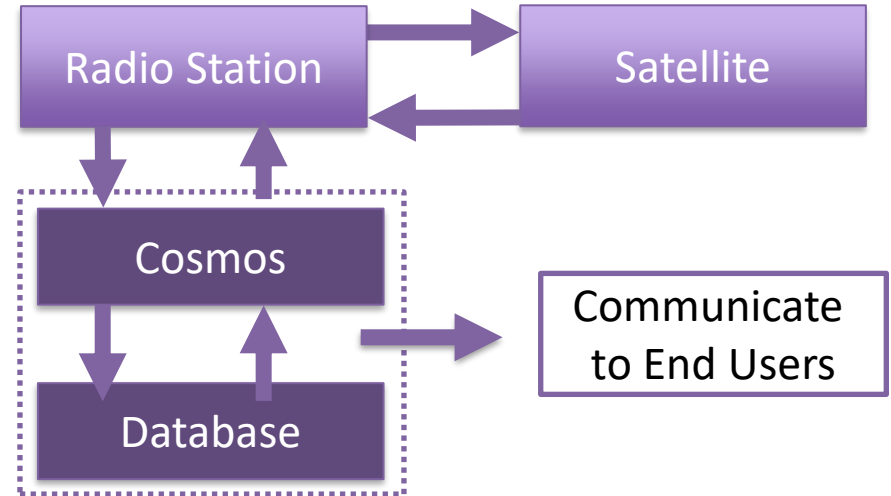
Ground Station Software

External Interfaces

- Operators
 - Those who will be operating COSMOS through commands
- End Users
 - Those who will be requesting the data that we collect
- Connection to the ground station
 - This will be the network connection to the radio station
- Uplink/Downlink
 - This is the information that we will be sending and receiving from the radio station

Internal Interfaces

- Database
 - Connection between COSMOS and the



Ground Station Software

There are difficulties ahead in gaining access to host our software on the Kansas State network, but we believe that the Ground Station Software is technically feasible.

ASSEMBLY, INTEGRATION & TEST (AI&T)



Assembly, Integration & Test

Req. ID	Title	Requirement	Required Value	Expected Value	Status
2.5 Ground Support Equipment Requirements					
GSE-2.01	Testing	Ground support equipment will be able to accurately test the power, communication, thermal, and mechanical capabilities of the satellite			Partial Compliance
3.14 Electrical Ground Support Requirements					
EGSE-3.01	Location	The assembly, integration and testing facility shall have a secure location on campus.			Compliant
EGSE-3.02	Security	The facility shall have access restricted to designated team personnel.			Compliant
EGSE-3.03	Lab Instruments	The facility shall provide electrical testing equipment not limited to a multimeter, power supply, function generator, o-scope, data acquisition system, etc.			Compliant
EGSE-3.04	Proper Grounding	The facility station shall provide a grounded workspace.			Compliant
EGSE-3.05	Computing Capabilities	The facility shall supply computing and networking resources.			Compliant
4.14 Electrical Ground Support Requirements					
EGSE-4.01	Climate	Facility temperature should stay at $68^{\circ}\text{F} \pm 5^{\circ}\text{F}$ and relative humidity of $40\% \pm 10\%$			Partial Compliance
EGSE-4.02	Power Delivery	Facility shall provide 120V regulated power for all equipment.			Compliant
EGSE-4.03	Electrostatic Discharge Prevention	All assembly and handling surfaces must have grounded ESD mats.			Compliant
EGSE-4.04	Lab Instruments	The facility shall provide calibrated electrical instruments.			Compliant
EGSE-4.05	Computer Requirements	The facility shall hold computers with all necessary software and tools needed.			Compliant
EGSE-4.06	Computer Signal Emulation	The facility shall provide necessary software/hardware needed to emulate signal input.			Compliant
EGSE-4.07	Antenna Interface Testing	The facility shall provide the capability to interface with antenna procedures and log results.			Compliant

Verification Methods

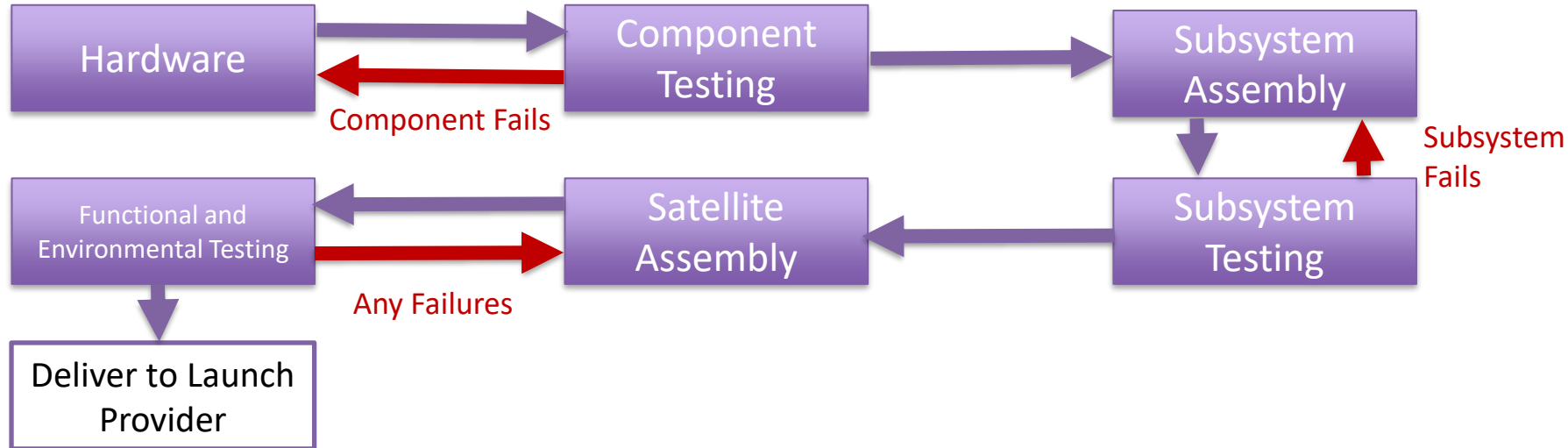
- RVM Verification Methods
 - Grouped by Analysis, Test, Inspect, Demonstrate (ATID)

Mission Parameter (Requirements)	Type	Description
Neutron Detector Payload Accuracy	D	Demonstrate that DSC Domino neutron detector is accurate within "x" percent
Core Capabilities (Nadir Tracking)	T	Test that Attitude Determination system can track the Nadir datum
Operational Lifetime (Attitude Determination)	A	Analyze ADCS in environmental conditions of space for "x" years.
Neutron Detector Mounting Angle	I	Inspect the detector mounts and ensure correct orientation.

- Current status has 249 of 290 verification methods complete
- Expected completion December 17th, 2025

AI&T Flow

- Flow diagram of AI&T sequence showing major assembly and verification activities



AI&T Facility

- Assembly Area:
 - Dedicated area to satellite assembly
- Testing Stations:
 - Workstations for electrical and mechanical verification
- Storage:
 - Secure storage for hardware



Ground Support Equipment

- Electrical Support

- Lab Equipment: Multimeter, oscilloscope, power supply, function generator.
- Signals Emulation: EGSE hardware and software to emulate outputs and provide test sequences for flight software.
- Data Acquisition: EGSE hardware and software to capture and log signals and telemetry from all subsystems.

- Mechanical Support

- Tools: Calipers, micrometer, mass scale, surface roughness gauges, hardness tester.
- Process (Outsourced): Shock testing, vibration testing, thermal testing, vacuum testing.



BUDGET & SCHEDULE

Abigail Husted



Budget – Subteams

- Individual part orders will be tracked for budget control
- Values come from Subteam estimates and limited quotes
- Broad category estimates based on NASA CubeSat 101/201 and Arizona State University CubeSat 3U Budget Spreadsheet

Subteam	Cost
PLD	\$5,000
EPDS	~\$11,300
ADCS	\$10,000
C&DH	~\$30,716
FSW	~\$6,000
STR	~\$4,000
THM	~\$1,000

Budget – Hardware & System

Other Hardware	Cost
Harnessing & Connectors	~\$1,500
Spares	~\$2,400

System and Infrastructure	Cost
Contingency (~20%)	\$33,788.8
Integration & Testing	~\$12,000
Ground Station	~\$4,028
Launch System	~\$76,000

~\$197,733.8 total

Schedule

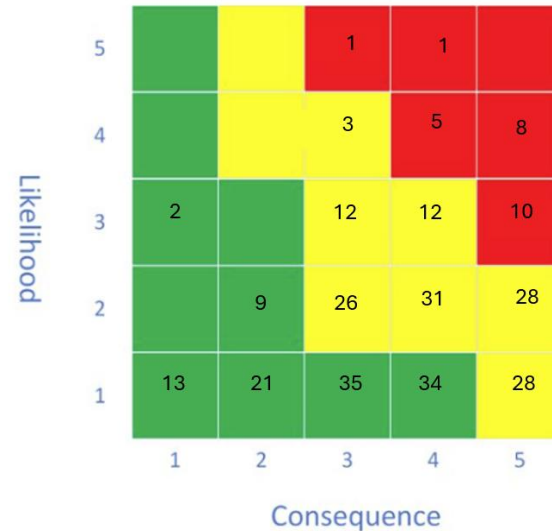
- **Jan-Feb 2026:** subsystem design and interfaces, component selection, update CAD file, preliminary production plan
- **Mar-May 2026:** Integration drawings, interface control docs, baseline system design, prepare for and deliver CDR
- **Jun-Aug 2026:** Order long lead items, build engineering model, unit level functional testing, flight software development
- **Sep-Nov 2026:** Assemble flight structure/harness, integrate sub systems, execute thermal/vibration/EMI/EMC testing
- **L-6M:** MICD Delivery, **L-3M:** Payload environment/func test reports, **L-2M:** All licenses/approvals received, **L-45D:** Payload delivery for integration with launch provider

RISK ASSESSMENT

Risk Assessment Strategy

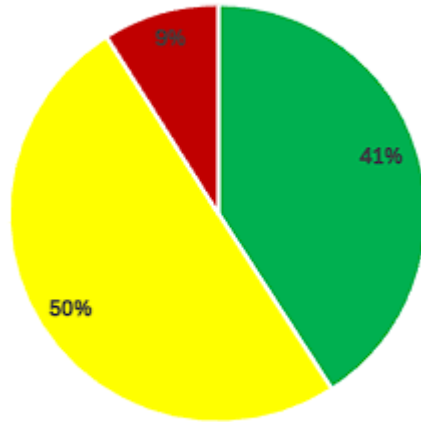
- Over 275 risks identified
- Each risk comes from a system requirement.
- Focus on red level risks
- Working with sub teams to identify new potential risks
- Creating potential mitigation methods

5x5 Matrix Number Breakdown of RVM



Risk Assessment Strategy

Risk Ratings

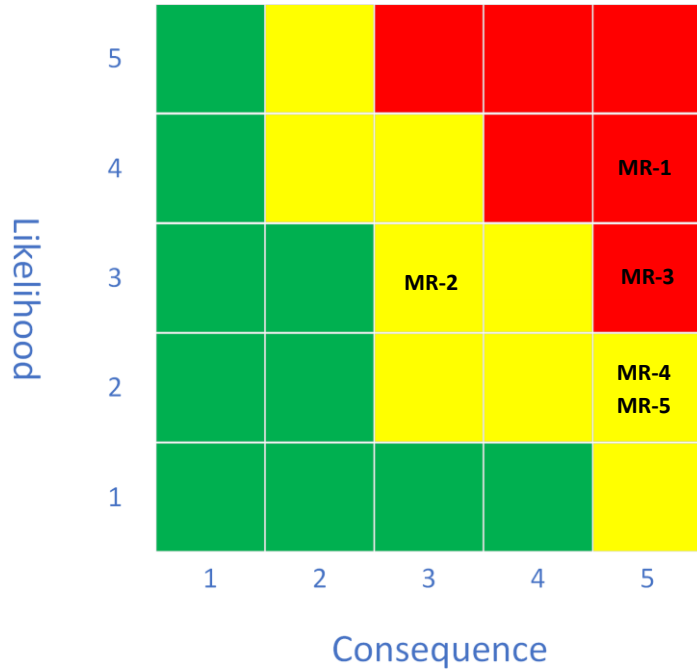


■ Green ■ Yellow ■ Red

	Frequency	Percent
Green	114	40.86%
Yellow	140	50.18%
Red	25	8.96%
Total	279	100.00%

Mission-Level Risks (Level 1)

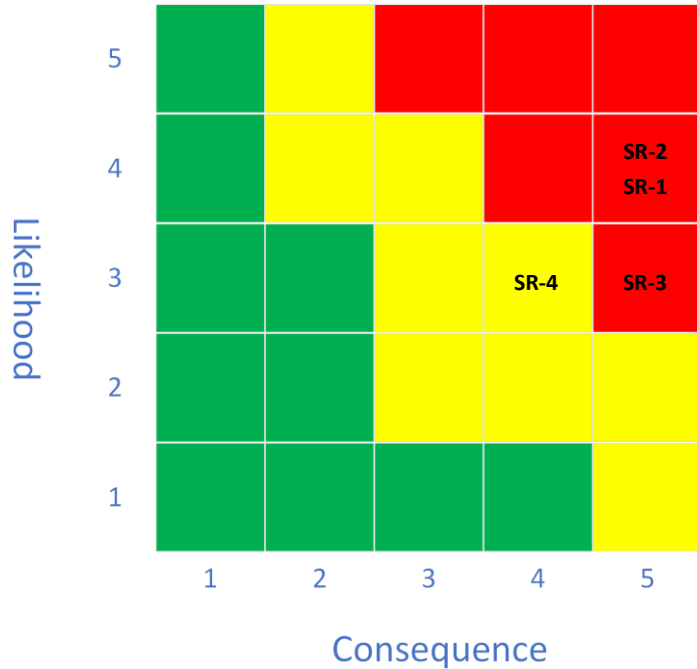
149



ID	Risk	Mitigation Strategy	Approach
MR-1	Satellite + payload may not remain fully integrated and operational for the minimum mission lifetime. (NU-1.03)	Validate bus-payload integration early, run end-to-end mission simulations, and test satellite/GS functionality in hardware-in-the-loop.	M
MR-2	Neutron detector may fail space qualification or produce invalid thermal/fast neutron data. (NU-1.01)	Perform vacuum, vibration, and thermal qualification; characterize detector response pre-flight; analyze data consistency with orbital expectations.	A
MR-3	Components may not survive radiation, vacuum, outgassing, and temperature extremes in orbit. (NU-1.07)	Build a full environmental budget (thermal, radiation, vacuum), select rad-tolerant components, and validate via TVAC + radiation simulation.	M
MR-4	NuKAT is unable to demonstrate the effective use of the DSC Domino in space and build flight heritage for the detector. (NU-1.02)	Create proof and work to create vigorous testing to ensure use of DSC Domino will be effective in space.	A
MR-5	Hardware may not survive launch vibration, shaking, shock, and transportation to launch site. (NU-1.06)	Perform launch-profile vibration, shock, and environmental tests early; strengthen mounts and fasteners; verify survival margins.	M

System-Level Risks (Level 2)

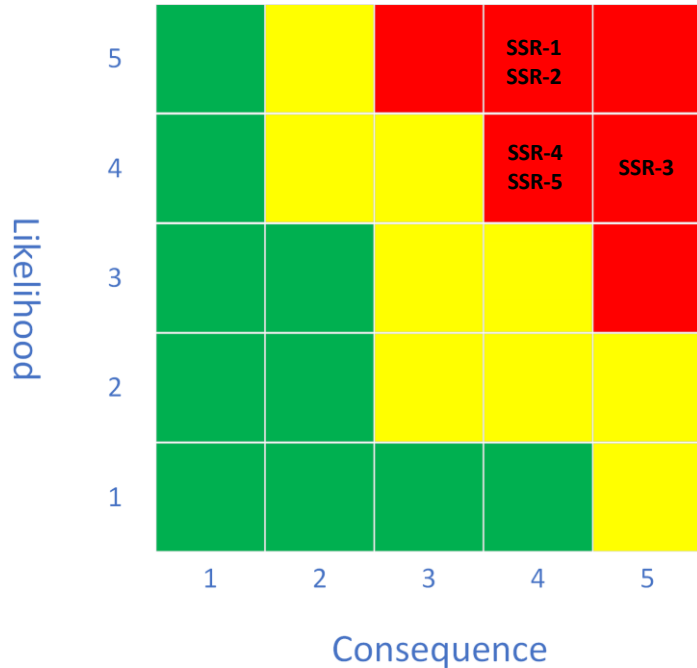
150



ID	Risk	Mitigation Strategy	Approach
SR-1	Orbital radiation environment may exceed design margins and damage on-orbit hardware. (SYS-2.01d)	Perform detailed orbital radiation analysis, size shielding and rad-hard parts to total-dose + margin and verify with radiation environment simulations.	M
SR-2	High-energy proton/electron environment may cause SEUs and upsets in critical systems. (SYS-2.01e)	Model expected electromagnetic environment, select tolerant components, add error-correcting / watchdog schemes, and validate with HW-in-the-loop fault-injection tests.	M
SR-3	Satellite may not structurally survive launch and on-orbit environments. (SYS-2.16)	Use FEA/vibro-acoustic analysis, then run vibe/shock/thermal-vac tests on structure and key assemblies with adequate safety margins.	M
SR-4	NuKAT is unable to be fully operational and supporting payload functions for at least 1 year. (SYS-2.15)	Work to ensure rigorous verification and testing of satellite parts and potential life cycle, as well as working to remove potential failure points that could result in a lifetime of less than 1 year.	M

Subsystem-Level Risks (Level 3)

151



ID	Risk	Mitigation Strategy	Approach
SSR-1	Ground station compatibility TBD may prevent C&DH from transferring required telemetry. (SYS-3.06)	Finalize ground station infrastructure early, test all C&DH message formats with GS hardware.	M
SSR-2	Neutron detectors may exceed thermal limits (-20 to 55°C) during operation. (PL-3.07)	Conduct thermal modeling + TVAC testing; size heaters/insulation to meet worst-case cold and hot scenarios.	M
SSR-3	C&DH UHF subsystem may not reliably operate within the 420– 450 MHz requirement. (CDH- 3.03)	Select compliant transceiver hardware; validate frequency performance through RF chamber testing.	M
SSR-4	Encryption/compression protocol may not meet reliability requirements for uplink/downlink. (CDH-3.08)	Define protocol early; perform end-to- end data transfer tests with error injection.	M
SSR-5	C&DH may not consistently send required telemetry or software updates to the FSW. (SYS-3.03)	Validate interface definitions; run integrated hardware-in-the-loop tests to confirm data flows.	M

CONCLUSION

Bailey Walke

Path to Preliminary Design Review

Focus Area	PDR Goal
Orbit & ConOps	Detail operations timeline, apply for RF license
Requirements	Manage system and subsystem requirements; ensure all are measurable and verifiable in the RVM.
Architecture & Interfaces	Manage top-level architecture and interfaces.
Detail Subsystem Designs	Detail CAD, structural and thermal models/simulations, mass, power, data and link budgets
Risk Reduction	Manage system-level risks, reduce and mitigate subsystem risks
Verification & Test Planning	Develop functional performance and environmental test plan per GEVS and reserve required facilities

NuKAT Preliminary Design Review to be presented next semester



Upcoming Trade Studies/Solutions

- UHF Antenna
 - Custom design
 - Modified COTS parts
 - Modified ConOps
- Power Positive
 - Solar panel reconfiguration
 - Modified ConOps
 - Adjusting component power draw
- Science data volume/frequency
 - Data compression and/or reduction
 - Different antenna options
 - Additional ground stations

PDR Board Feedback

- What mission and design definition content is missing?
- What mission and design definition needs more development?
- What mission and design definition trade studies are needed?
- Feedback on path to compliance/noncompliances

BACKUP

ADCS Critical Tech

- Fine Sun Sensor 1-6x
- Coarse Sun Sensor 6x
- Triaxial Magnetometer
- Navigation Satellite Receiver
- Magnetorquer 3x
- Onboard Computer
- TRIAD Angle Algorithm
- B-Dot Spin Algorithm
- GS Tracking Algorithm
- General Flight Algorithm
- Upset Recovery Algorithms

COTS or Custom

- Custom - Cost, Precision
- Custom - Cost, Precision
- COTS - Precision
- COTS - Precision
- COTS - Precision
- COTS/Custom - Cost, Precision
- Custom - Cost
- Custom - Cost
- Custom - Cost
- Custom - Cost
- Custom - Cost

EPD: PV-Cell Degradation Calculation

Using available data from the DHV Solar Panels, which use the same PV Cells as the EnduroSat 1U, the **Azur Space 3G30A**, to approximate the solar panel degradation.

$$\text{Degradation Factor} = \frac{P_{EoL}}{P_{BoL}}$$

Operating Temperature (Celsius)	Beginning-of-Life Power (W)	End-of-Life Power (W)	Degradation Factor
-50	2.48	2.41	97.18%
28	2.07	1.98	95.65%
125	1.5	1.4	93.33%

93.33% is used to provide the largest safety margin for extra power.

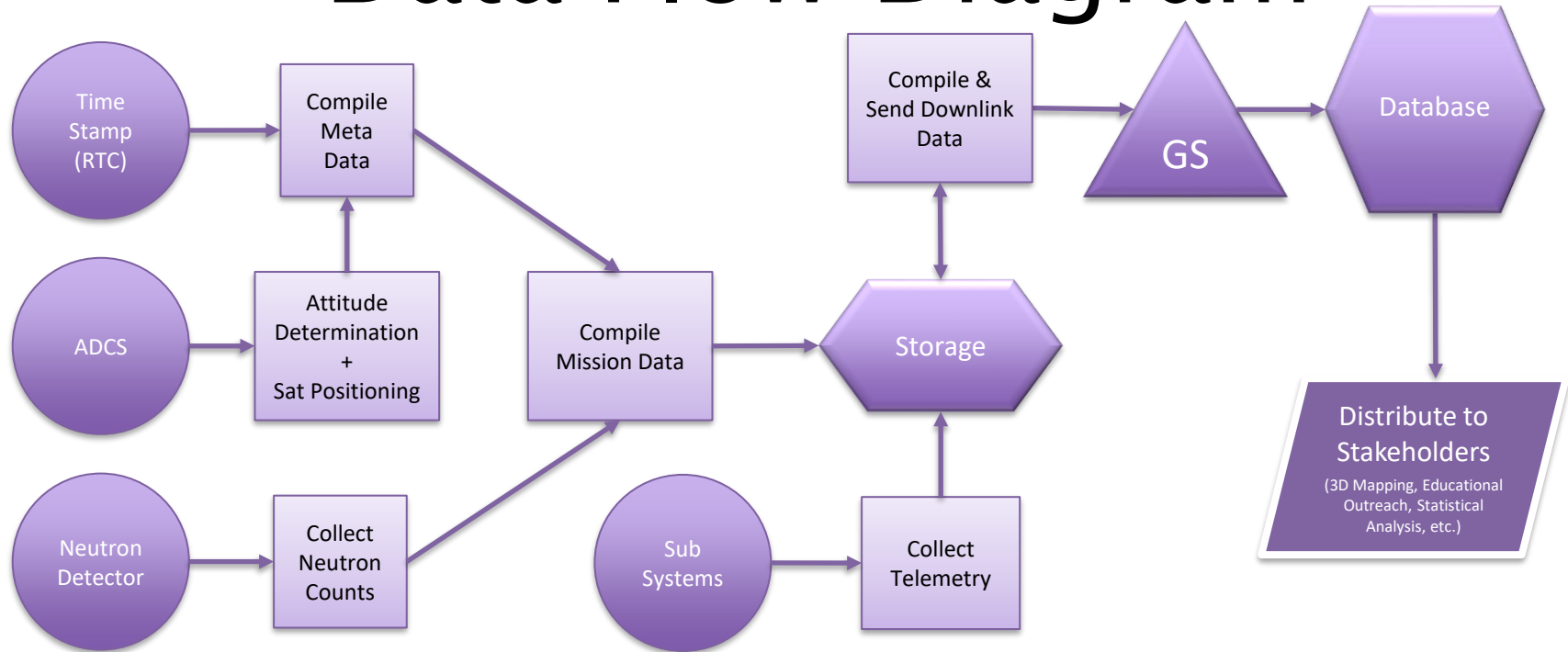
EPD: Reference for Preliminary Power Allocations

Subsystem	Minimum Power Consumption	% of Operating Power for Small Spacecraft	% of Operating Power for Medium to Large Spacecraft
Payload	20 – 50 W	40	40 – 80
Propulsion	0 W	0	0 – 5
Attitude Control	0 W	15	5 -10
Communications	15 W	5	5 -10
Command and Data Handling	5 W	5	5 -10
Thermal	0 W	5	0 – 5
Power	10 – 30 W	30	5 – 25
Structure	0 W	0	0

"New SMAD's Table 10-9" F. Zhu, *A Guide to CubeSat Mission and Bus Design*. Honolulu, HI: Pressbooks, 2022.

Available: <https://pressbooks-dev.oer.hawaii.edu/epet302/>

Data Flow Diagram



Communications and Data Handling

Subsystem	Component	Throughput Estimation	CPU Utilization	Non-volatile Memory	Volatile Memory	Update Frequency	Timing Accuracy
Payload	DSC Domino	High	Medium	High	Medium	Medium/High	Low
Structures and Mechanisms	Deployment Mechanisms	--	Low	--	--	--	Strict
Thermal Control	Temperature Sensors	Medium	Low	--	--	--	Low
	Radiators	--	--	--	--	--	Low
Power	PCDU	Medium	Low/Medium	Medium	Low	High	Medium
Telemetry and Control	UHF Transmitter	High	Medium	High	--	--	--
	UHF Receiver	Medium	Low	Low	--	--	--
	GNSS Receiver	Medium	Low	Low	--	--	--
Command and Data Handling	On-Board Computer	High	High	High	High	High	Strict
Attitude Determination and Control	Sun Sensors	Medium	High	Low	--	--	--
	ADC Board	Low	Low	Low	Medium	Medium	Strict
	Magnetometer	Low	Low	Low	Low	Low	Low

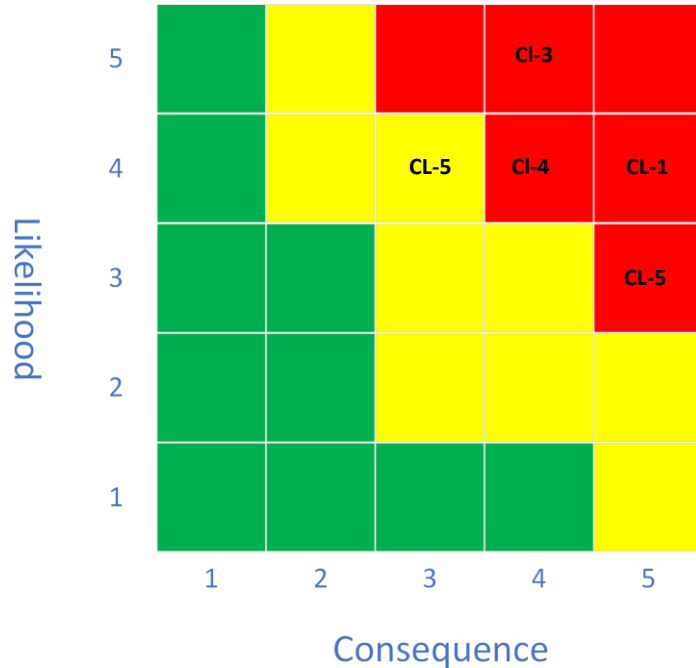
High, medium, and low are relative to one another.

Communications and Data Handling

Computer Characteristic	Initial Checkouts	Initial Deployment	Science Operations	Data Uplink and Downlink	Spacecraft Sun Pointing / Charging	Orbit Maintenance	Safe Mode
Throughput Estimations	Low	--	High	High	Medium	Medium	low
CPU Utilization	Medium	Low	Medium	Medium	Medium	Medium	Low
Non-volatile Memory Usage	Low	--	High	High	Low	Low	Low
Volatile Processor Memory Usage	Low	--	Medium	High	Medium	Medium	Low
Update Frequency	Low	--	Medium	--	High	Medium	High
Timing Accuracy	Low	Strict	Low	Strict	Medium	Medium	Low

High, medium, low, and strict are relative to one another.

Component-Level Risk Slide (Level 4 Requirements)



ID	Risk	Mitigation Strategy	Approach
CL-1	Detectors may exceed thermal limits during high-load operation, causing noise or hardware failure.	Increase radiator efficiency, verify heater control logic, and perform thermal vacuum testing for worst-case conditions.	M
CL-2	Power distribution board may be unable to reliably provide the 5V rail required for all four detectors.	Validate rail loading with engineering models, add margin to regulator performance, test under max-draw scenarios.	M
CL-3	DSC-Domino C&DH interface may experience intermittent communication failures through USB-C under vibration or thermal cycling.	Perform vibration/thermal stress testing on connectors, consider strain-relief bracket, evaluate direct-solder alternative.	M
CL-4	Heater elements may not supply sufficient heat fast enough to prevent detectors from dipping below -20 °C.	Model heater sizing, improve mounting conductivity, run cold-start thermal TVAC tests to verify response speed.	M
CL-5	Some payload fastening points (Domino + moderator interface) may not maintain mechanical stability under launch loads.	Re-evaluate bolt selection, verify torque specs, and perform modal analysis + bench-level vibration testing.	M

Mission Advisory Board

- **Dr. Tim Steffensmeier**
 - Assistant Vice President and Director of Engagements and Outreach, Kansas State University
- **Bill Swartz**
 - Atmospheric and Remote Sensing Scientist, John Hopkins University Applied Physics Laboratory
- **Dr. Amir Bahadori**
 - Professor and Program Director, Kansas State University Nuclear Engineering
 - Former NASA civil servant in the Space Radiation Analysis Group (SRAG)
- **Dr. Matt O'Keefe**
 - Professor, Kansas State University Chemical Engineering

Mission Advisory Board

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 - Former Industry Advisor, Mike Wiegers Department of Electrical and Computer Engineering, Kansas State University

CubeSat Project Donor

- Kevin Needham
 - President, Kiewit Power Engineering
 - Senior Vice President, Power Markets & Strategy

Acronyms (A-L)

- ADC(S) – Attitude Determination and Control (Subsystem)
- ATID – Analysis, Test, Inspect, Demonstrate
- B-dot/S-dot – Method of angular velocity determination using change in Earth or Sun vector orientation over time
- BMS – Battery Management System
- bps – Bits per Second
- CAD – Computer Aided Design
- CAN – Controller Area Network
- CD(&)H – Communications and Data Handling
- ConOps – Concept of Operations
- CSS – Coarse Sun Sensor
- EGSE – Electrical Ground Station Equipment
- EM – Electromagnetic
- EMI – Electromagnetic Interference
- EPD – Energy and Power Distribution
- FSS – Fine Sun Sensor
- FSW – Flight Software
- GEVS – General Environmental Verification Standard
- GMSK – Gaussian Minimum Shift Keying
- GNSS – Global Navigation Satellite System
- GPS – Global Positioning System
- GS – Ground Station
- GSE – Ground Support Equipment
- HDPE – High Density Polyethylene
- HIL – Hardware In The Loop
- H&S – Health and Status
- I2C – Inter-Integrated Circuit
- KSU – Kansas State University
- KB – Kilobyte(s)
- LEO – Low Earth Orbit
- LV – Launch Vehicle

Acronyms (M-Z)

- MAB – Mission Advisory Board
- MCX – Micro Coaxial Connector
- MDR – Mission Definition Review
- MGSE – Mechanical Ground Station Equipment
- mrg - Margin
- MS – Microsoft
- MSN – Main Subsystem Network
- MSND – Microstructured Semiconductor Neutron Detector
- MTM – Magnetometer
- MTQ – Magnetorquer (either air or solid core)
- NASA – National Aeronautics and Space Administration
- NOAA – National Oceanic and Atmospheric Administration
- OBC – Onboard Computer
- PCDU – Power Control Distribution Unit
- PDR – Preliminary Design Review
- PID – Proportional-Integral-Derivative (controller)
- PLD – Payload
- PV - Photovoltaic
- RDT – Radiation Detection Technologies
- RF-SCOE – Radio Frequency Special Check-Out Equipment
- RS-422 – Recommended Standard 422
- RS-485 – Recommended Standard 485
- RTC – Real Time Clock
- RTD – Resistance Temperature Detector
- RVM – Requirement Verification Matrix
- Rx – Receive
- SAA – South Atlantic Anomaly
- SIL – Software In The Loop
- SMP - Small Push-On Connector
- SR – System Risk
- TBD – To Be Determined
- TRIAD – Tri-Axial Attitude Determination
- Tx – Transmit
- UHF – Ultra High Frequency